

Buyer Market Power and Exchange Rate Pass-through*

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Abstract

Large foreign buyers dominate export markets and use their market power to impose markdowns on sellers—and this shapes how exchange-rate shocks transmit to export prices. Using the universe of Colombian customs transactions for 2007–2020, which link every exporter to its foreign buyers, I document three facts: (i) a few importers dominate each product–destination market and pay different prices for identical goods; (ii) a 10% peso depreciation raises seller-currency prices by only 1% for sellers tied to the largest buyers, versus 15% for those selling to small buyers, after controlling for invoice currency and seller-time fixed effects; and (iii) markets with higher buyer concentration exhibit lower aggregate pass-through. I rationalize these facts with a structurally estimated open-economy oligopsony model in which buyers internalize their influence over an upward-sloping supply curve. The model generates a markdown channel—whereby larger buyers increase their markdowns following a depreciation, absorbing the exchange-rate gain at the seller’s expense—and shows that aggregating the firm-level equilibrium conditions yields the Herfindahl–Hirschman Index as the sufficient statistic for market-level pass-through. Using simulated method of moments, I estimate the within-product cross-buyer and across-product supply elasticities, recovering average markdowns of 16% of the marginal revenue product. In a counterfactual with no buyer market power, seller revenues rise but pass-through nearly doubles, revealing a trade-off between price levels and exchange-rate insulation that is quantitatively significant for exporters in concentrated markets in developing countries.

JEL Codes: D43, E31, F31, F41, F42, L1

Keywords: Market power, Exchange rate, Oligopsony, Market structure, Markdown, Exchange-rate pass-through

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1 Introduction

Large firms dominate many sectors of the global economy. This growing concentration has been shown to carry important macroeconomic consequences (Autor et al., 2020; Gutiérrez and Philippon, 2019; De Loecker, Eeckhout and Unger, 2020). In the context of international markets, a vast group of small exporting firms often sell their goods to just a handful of large, multinational buyers. For example, the top one percent of importers account for 83.5% of U.S. imports (Bernard et al., 2018). This raises the question of how the presence of large buyers affects prices and price dynamics in export markets. In particular, do large firms exert buyer market power by imposing markdowns? If so, when there is an exchange rate shock, do large firms leverage this buyer market power to increase their profits? What are the consequences for smaller connected firms? And, at the sector level, how does buyer concentration shape aggregate price dynamics in international markets?

This paper studies buyer market power in international markets and its impact on exchange rate pass-through. Exchange-rate pass-through (ERPT) corresponds to the change in international prices in the seller’s currency as a response to a change in the exchange rate. I combine a novel transaction-level dataset covering the universe of Colombian exports with an oligopsony model of buyer market power in international trade. The conclusion is that buyer market power moderates the response of international prices to exchange rate shocks, implying a trade-off for sellers: lower prices due to markdowns, but less volatile price adjustments. The main mechanism behind this effect is that large firms have *more* variable markdowns and can use this as a tool to maintain more stable prices. For example, when the Colombian currency depreciates, foreign buyers absorb the shock by *increasing* their markdowns—extracting a larger share of the exchange-rate gain and preventing Colombian exporters’ prices from rising proportionally with the exchange rate. I directly estimate this effect between firms within a market.

I begin by documenting stylized facts on Colombian export markets. This paper uses data on exports from Colombia to the rest of the world from 2007 to 2020. I exploit the granularity of my data, containing identifiers of buyer, seller, product, destination country, and year in each transaction. Export data are matched to data on bilateral exchange rates for each year and destination country. I define a *market* as a product-destination-year combination. First, I find that sales are concentrated among a few large foreign buyers in each market, and that a given seller faces different prices for different buyers of the same products and destination country.

Second, I differentiate the exchange rate pass-through for larger and smaller buyers. I control by a variety of fixed effects including seller-time to account for sellers’ marginal cost, and year-product-country fixed effects to isolate the differences between markets, comparing across buyers with different market shares.¹ I find that larger buyers face a lower exchange rate pass-through to prices in the seller’s currency, ranging from 1% for the largest buyers to 15% for the smaller ones. Thus, when the currency of the seller’s country depreciates, sellers in concentrated markets face attenuated price increases (in the seller’s currency) relative to exporters that sell to smaller buyers.

Third, I aggregate at the market level. Starting from the firm-level regression, I calculate a weighted average of price changes—with weights corresponding to buyer shares—to obtain an

¹Note that even though this might drop the exchange rate coefficient, the coefficient of interest—the one for the interaction term—does not change.

expression for the average price change in a market as a function of the Herfindahl concentration index. I run the same regression at the market level and include market-level controls to isolate the effect of market concentration on the ERPT. I find that a higher concentration of sales among buyers displays more moderate changes in market average prices after an exchange rate shock (i.e., lower exchange rate pass-through).

To further ensure the validity of the main results, I conduct a series of robustness tests. A large empirical literature highlights the prevalence of U.S. dollar invoicing and the role of sticky prices in shaping exchange-rate pass-through ([Gopinath et al., 2020](#)), raising the question of whether the patterns documented above simply reflect dominant-currency pricing rather than buyer market power. To address this concern, I implement a set of targeted exercises: I restrict the sample to exports destined for the United States, where the bilateral and dollar rates coincide, estimate the baseline specification only for dollar-invoiced transactions, and include the dollar exchange rate—replicating the specification in [Gopinath et al. \(2020\)](#)—as an alternative exchange-rate definition. Across all cases, the key result holds. This indicates that the mechanism uncovered in the data² is distinct from, and robust to, the dominant-currency pricing channel emphasized in prior work.

Beyond the analysis of sticky prices and dollar invoicing, I perform several additional robustness checks to ensure the empirical results are not driven by modeling choices or sample composition. First, I incorporate import intensity, following [Amiti, Itskhoki and Konings \(2014\)](#), to account for potential heterogeneity in buyers’ cost structures that might correlate with markdown behavior. Second, I consider alternative definitions of buyer market share to confirm that the results do not hinge on a specific measure of concentration. Third, I re-estimate the main specifications across alternative subsamples, varying the set of destinations, products, and firms included in the analysis. Fourth, I rule out standard quantity discounts as an alternative mechanism by showing that price dispersion across buyers persists after controlling for quantities and that the pass-through gradient is present even in goods with limited variation in shipment size. I also show that these results are not driven by seller market power: controlling for seller heterogeneity and interacting exchange-rate changes with seller-size measures leaves the main coefficients unchanged. Across all exercises, the findings remain consistent.

Motivated by these stylized facts, I propose an open economy model of oligopsony that accounts for buyer market power in international markets and shows its consequences for price determination in international transactions. In my model, sellers are located in the home country and buyers are in foreign countries. On the supply side, buyers choose quantities they want to purchase (strategically, accounting for how their demand affects the market). The price is then determined by the market supply curve. Each buyer knows that buying more pushes the price up along the upward-sloping supply curve. The supply curve is microfounded with a discrete-choice problem, where sellers are price takers and choose which product to produce and which buyer to supply. On the demand side, buyers observe the quantities supplied and choose the price they are willing to pay for a product. Given a finite number of buyers, they act strategically, internalizing their influence over prices. In equilibrium, buyers pay sellers a price marked down from the marginal revenue of the product.

²The theoretical mechanism is formalized in the next section. It differs from [Gopinath et al. \(2020\)](#) in focusing on markdown adjustments arising from buyer market power rather than on price stickiness or currency of invoicing as the primary drivers of incomplete pass-through. The model assumes flexible prices and expresses revenues and costs in the seller’s currency for tractability; this does not imply an assumption of producer-currency pricing. Instead, it provides a consistent framework to analyze how buyer market power affects prices and markdowns.

In my model, I hold the seller side fixed to isolate the mechanism that links buyer market power to the trade-off between prices (markdowns) and exposure to exchange-rate volatility. Endogenizing both sides simultaneously would introduce additional strategic margins, such as exporters' pricing, quality upgrading, or assortative matching responses, that would interact with buyer behavior and confound the measurement of the mechanism I aim to quantify.

The first theoretical result is that markdowns are increasing in the buyer's market share—that is, larger buyers have greater markdowns (μ). Aggregating the firm-level markdowns across all firms in a market, I find market-level average markdowns are increasing in buyer market concentration. Additionally, markdowns depend on sellers' within-product cross-buyer elasticity of substitution and the cross-product elasticity of substitution. Lower elasticities correspond to greater markdowns. Intuitively, if substitution across buyers and products is costly for sellers, buyers have more market power and higher markdowns.

The second theoretical result is that the price response to exchange rate shocks varies with buyer market share. This is a novel source of exchange rate pass-through dispersion that, to my knowledge, has not been previously studied in the literature. The overall effect is driven by two mechanisms that partially offset each other in how pass-through varies with buyer size: a markdown channel and a marginal-revenue channel. Both channels reduce pass-through, but the markdown channel grows with buyer market share while the marginal-revenue channel shrinks, so the net gradient—larger buyers face lower pass-through—reflects the dominance of the markdown channel, as confirmed empirically.

On the one hand, the markdown channel implies that following a depreciation of the seller's currency, buyers *increase* their markdowns, extracting a larger share of the exchange-rate gain and keeping prices more stable in the seller's currency relative to what they would be absent buyer market power. Larger buyers tend to have more variable markdowns and increase them more elastically in response to exchange-rate shocks.

On the other hand, the marginal-revenue channel implies that, following a change in the exchange rate, a standard price effect induces sellers to change their quantity supplied, which in turn affects marginal revenue. Because sellers have a lower supply elasticity in concentrated markets—intuitively, the costs of finding another buyer are higher for these sellers—larger buyers face smaller changes in marginal revenue. In contrast to the markdown channel, prices in the seller's currency are more volatile.

Aggregating the firm-level equilibrium conditions across all buyers in a market — weighting each buyer's price response by its revenue share — yields a market-level pass-through expression in which the Herfindahl–Hirschman Index of buyer concentration emerges as the sufficient statistic for aggregate pass-through. This is a derived result of the model: the HHI is not imposed as a measure of concentration but falls out of the aggregation, providing a tight structural bridge between the micro-level mechanism and the market-level empirical specification.

I then map the structural equation in the model for the exchange rate pass-through to the estimates in my data. The model shows that buyer market share serves as a sufficient statistic for pass-through heterogeneity across buyers: conditional on product-destination-time, all relevant information about a buyer's markdown is summarized by its market share, so no direct measure of

markdowns is required for estimation. The results thus reveal that the markdown channel is more empirically relevant than the marginal-revenue channel. Intuitively, larger buyers internalize the upward-sloping supply curve and are aware that each additional unit they buy increases the price of every other unit.

I proceed using indirect inference to quantify the markdowns for large firms and estimate how they change in response to an exchange rate shock. In the model, two elasticities govern the magnitude of this effect: the cross-product elasticity of supply and the within-product cross-buyer elasticity of supply. I propose an approach that integrates (i) empirical estimates of the exchange rate pass-through elasticities, (ii) moments from the cross-section of prices and (iii) a simulated method of moments to estimate these elasticities by indirect inference. I find that the average markdown is approximately 16% of the marginal revenue product.

Finally, I use the model to simulate a counterfactual economy with no buyer market power. In a perfectly competitive economy, sellers' revenues are higher due to a price effect (i.e., the absence of markdowns) as well as a quantity effect (i.e., they adjust quantities in response to higher prices). However, revenues in the seller currency are more elastic in response to international shocks, potentially generating greater volatility.

I illustrate my findings with an example. Starbucks, a large U.S. buyer of Colombian coffee, receives a higher markdown (i.e., a price discount) than smaller U.S. firms buying coffee from Colombia. All else equal, Starbucks is thus able to pay lower prices for coffee. Moreover, following a COP depreciation, Starbucks—exploiting its buyer market power—increases its markdown, capturing a larger share of the exchange-rate gain. As a result, COP prices received by Colombian coffee exporters are less responsive to exchange rate shocks when selling to Starbucks than when selling to smaller buyers. In the aggregate, if the U.S.–Colombia coffee market is dominated by large buyers like Starbucks, the average market price for coffee is also reduced and less responsive to shocks. In a counterfactual world where Starbucks and other large firms did not have such market power, the sellers in developing countries would increase their revenues because they would sell at a higher price.

This paper contributes to four strands of the literature. First, it contributes to the extensive literature on the exchange rate disconnect (e.g., [Engel \(2000\)](#); [Obstfeld and Rogoff \(2000\)](#); [Itskhoki and Mukhin \(2021\)](#); [Gopinath et al. \(2020\)](#)), with particular attention to the incomplete pass-through of exchange rate movements into international prices. The earlier literature highlights three main mechanisms behind incomplete pass-through: nominal rigidities under local currency pricing ([Devereux and Engel, 2003](#); [Gopinath and Rigobon, 2008](#)), variable markups leading to pricing-to-market ([Atkeson and Burstein, 2008](#)), and local distribution costs ([Burstein, Neves and Rebelo, 2003](#); [Goldberg and Campa, 2010](#)). More recent work has emphasized additional factors, including the role of imported inputs ([Amiti, Itskhoki and Konings, 2014](#); [Wang and Yu, 2021](#)), product quality ([Freitag and Lein, 2023](#); [Chen and Juvenal, 2016](#)),³ firm balance sheet conditions such as debt or trade credit ([Hardy, Saffie and Simonovska, 2023](#); [Cui, Guo and Juarez, 2024](#); [Kim and Lee, 2023](#)), quantity discounts ([Ganapati and Hottman, 2026](#)) and currency invoicing ([Devereux, Dong and Tomlin, 2017](#); [Chen, Chung and Novy, 2022](#); [Bonadio, Fischer and Sauré, 2020](#)).⁴ In this literature, the

³See also [Rodríguez-Lopez \(2011\)](#) who show that aggregate import prices may be subject to composition bias due to changes along the extensive margin of trade.

⁴[Bonadio, Fischer and Sauré \(2020\)](#) focuses on how invoicing affects the speed of pass-through, while the other papers

emphasis has largely been on the seller side characteristics (Berman, Martin and Mayer, 2012; Burstein and Gopinath, 2014),⁵ whereas a theoretical contribution of this paper is to introduce buyer market power and a buyer-concentration effect. Empirically, the detailed buyer–seller data I use allow me to quantify the role of buyer–seller relationships in shaping exchange rate pass-through and in capturing the markdown response. More closely related to my work, Auer and Schoenle (2016) incorporate market structure into the analysis of exchange rate pass-through and Amiti, Itskhoki and Konings (2019) highlight strategic complementarities in firms’ pricing responses to competitors. I complement their contributions by focusing on buyer market power, analyzing its implications for exporters, and the potential stability gains from selling to large buyers.

Second, similar to my paper, a growing body of work on buyer market power in labor markets uses oligopsony and monopsony models to explain why workers’ wages are marked down from their marginal products (Berger, Herkenhoff and Mongey, 2022; Azar, Marinescu and Steinbaum, 2019; Lamadon, Mogstad and Setzler, 2022). My theoretical approach most closely resembles Tortarolo and Zarate (2018); Felix (2022) in labor markets.⁶ Lehr (2026) documents a similar size gradient in the market for corporate inventors, where larger employers face less elastic labor supply and greater monopsony power. Closest to my empirical question is Chan, Salgado and Xu (2026), who estimate the passthrough of firm-level productivity shocks into wages in Danish matched employer–employee data and read its incompleteness as evidence of labor market power—the same logic by which I attribute incomplete exchange-rate passthrough to buyer markdowns. Both papers work within labor markets, where the marked-down seller is a worker and the object of interest is the wage. I carry the logic into international goods markets, where the seller is an exporting firm, and ask how much of an exchange-rate movement that firm retains in the price it receives and whether the markdowns that depress that price also dampen its exposure to volatility.

Third, this paper relates to the literature on market power (De Loecker, Eeckhout and Unger, 2020; De Loecker et al., 2016) and imperfect competition in firm-to-firm trade (Atkeson and Burstein, 2008; Burstein, Cravino and Rojas, 2024; Kikkawa, Magerman and Dhyne, 2019; Morlacco, 2019). Contemporaneous work includes Alviarez et al. (2025, 2026), who study pricing under tariff-induced cost shocks in a two-sided bargaining setting. This paper complements theirs in three ways: I study exchange-rate shocks (volatile and frequent, unlike permanent tariffs), focus on seller-currency prices (the relevant object for exporters in developing countries), and use an oligopsony framework that allows me to structurally estimate the markdown channel and quantify its implications for seller revenue volatility—a welfare question they do not address. My approach is closest to Zavala (2022), who model farmer crop choice and exporter–buyer strategic interaction to measure markdowns in Ecuador’s agricultural value chains. I also rely on an oligopsony structure, but in an open-economy setting and with a different objective: rather than identifying the level of markdowns, I focus on the pass-through of exchange-rate shocks. This shift in focus allows me to highlight a trade-off for sellers—between lower prices and lower exposure to exchange-rate volatility—that has not been explored in the existing literature. I also complement work documenting different prices for the same products, such as Ignatenko (2024), where the mechanism is oligopolistic price discrimination. In my

examine its extent depending on the invoicing currency.

⁵These papers show that seller concentration reduces pass-through into domestic prices. The seller side is analyzed throughout the literature and further discussed in the Appendix.

⁶These papers focus on employees rather than intermediate inputs, as well as Berger, Herkenhoff and Mongey (2022) when looking at U.S. labor markets.

case, the mechanism is oligopsonistic, with variable markdowns, which allows me to structurally estimate new parameters such as the substitution elasticities across products and across buyers within a product.

Fourth, I contribute to a nascent literature on buyer–seller links, global value chains, and shock transmissions (Devereux, Dong and Tomlin, 2017; Huneus, 2018; Lim, 2018; Hottman and Monarch, 2020; Dhyne et al., 2021). Because of data availability, most of these papers focus on firm-to-firm transactions in the domestic context while my paper and a few others (Adão et al., 2022; Bernard et al., 2019) analyze the international markets. I contribute to this literature by documenting the existence of price dispersion for the same seller, product and destination in the international setting. Additionally, I estimate the cross-buyer elasticity of substitution, a key parameter that had not been previously estimated.

The rest of the paper proceeds as follows. Section 2 presents my data and empirical setting. Section 3 documents some key stylized facts on buyer–seller relationships in Colombia and their consequences for exchange rate pass-through. Section 4 presents the model that links buyer market concentration to export prices, yielding a specification for estimating the effect of buyer market power on exchange rate pass-through. Section 5 uses the empirical estimates to calibrate the model and, through indirect inference, recover the key elasticities needed to quantify the markdown channel. Section 6 proposes a counterfactual scenario with no buyer market power. Section 7 concludes.

2 Data

This paper combines buyer–seller transaction data for Colombia in international markets with data on bilateral exchange rate shocks. In this section, I describe the data and present summary statistics relevant for the analysis.

One of the challenges of studying buyer market power in international markets is the lack of detailed information on bilateral transactions between buyers and sellers. I use novel data on the universe of cross-border trade transactions between Colombian exporters and foreign firms during 2007–2020.⁷ The data come from the Colombian National Directorate of Taxes and Customs (DIAN; Dirección de Impuestos y Aduanas Nacionales de Colombia).⁸ For each transaction, DIAN reports the value and quantity shipped (in USD and in COP), the shipment date, the 10-digit Harmonized System (HS10) code of the product traded, the country of destination, the weight, the port through which this transaction occurred and the transportation mode. The key element of the dataset is that I am able to uniquely identify the foreign firm interacting with the Colombian firms and, in this way, I can conduct a buyer–seller analysis.

I combine this administrative microdata with data on bilateral exchange rates from International Financial Statistics of the International Monetary Fund (IMF). In particular, I use the monthly nominal bilateral exchange rate expressed as local currency per USD.

The bilateral exchange rate is the theoretically appropriate object in this setting. In the model,

⁷In the appendix, I include robustness checks conducted with data at the transaction level for importing firms.

⁸This dataset was accessed through Datamyne.

buyers' profit functions are expressed in terms of the bilateral rate between the seller's and buyer's countries, and the markdown adjustment mechanism operates through this channel regardless of invoicing currency. At the same time, given that 98% of Colombian export transactions are dollar-invoiced, I also run all main specifications using the COP/USD dollar exchange rate directly, following [Gopinath et al. \(2020\)](#).

The second key variable is the price. Constructing price and volume indices using customs data presents a significant hurdle, primarily due to the unit value bias. This bias arises because unit values, derived by dividing observed values by quantities, do not reflect actual prices. Even in the absence of price fluctuations, unit values can vary due to shifts in composition. To address this bias, I eliminate 8-digit products with a unit value variance higher than a threshold of 200% as those observations are more likely to be biased.⁹

The data consists of all exports from 50,869 Colombian firms producing 6,941 different HS10-level goods¹⁰ exported to 54 different countries during 2007–2020. Table 1 summarizes the main descriptive statistics relevant for my analysis. In each year of data, an average of 13,382 sellers trade with 39,028 buyers each year.¹¹ Each combination of destination and HS10 product includes, on average, 4.55 buyers, suggesting only a few buyers for a large number of sellers. Each of these buyers buys on average 3.68 products from Colombia.

Table 1: Annual Summary Statistics

Variable	Mean	SD
# Products	6,941	91
# Sellers	13,382	5,479
# Buyers	39,028	2,914
# Buyers by destination $\times$ product	4.6	23
# Products by buyer	3.7	9

Notes: In this table products are at the 10-digit Harmonized System level. Source: Colombian Customs Data.

3 Stylized Facts

In Colombian export markets, a small number of large foreign buyers account for the bulk of purchases from each seller. In this section, I document three stylized facts that together indicate significant buyer market power in Colombian export markets, and I highlight that the second and third facts represent the novel contribution of this paper.

First, as established in other contexts, a handful of large buyers purchase the majority of Colombian exports in each destination market. These buyers pay different prices for the same seller in international markets. Both of these patterns have been demonstrated in other contexts ([Morlacco, 2019](#); [Bernard et al., 2019](#); [Ignatenko, 2024](#); [Fontaine, Martin and Mejean, 2020](#); [Zavala, 2022](#)) and

⁹This approach aligns with the methodology suggested by [Boz, Cerutti and Pugacheva \(2019\)](#).

¹⁰Each product is identified with a 10-digit code, which corresponds to the Harmonized Commodity Description and Coding System at the highest level of disaggregation. An example for this could be women's or girls' cotton panties versus knitted or crocheted panties.

¹¹Note that these buyers correspond to all possible destinations.

confirm that buyer concentration and differential pricing are pervasive features of international trade. Second, and new to the literature, I find that larger buyers face a lower exchange-rate pass-through to prices in the seller’s currency. This finding not only reinforces the presence of buyer market power but also highlights its crucial role in shaping firms’ price-adjustment behavior when exchange rates fluctuate.¹² Third, I provide the first market-level evidence that exchange-rate pass-through systematically declines with buyer concentration: in markets dominated by a small number of large buyers, exporters adjust their prices significantly less in response to exchange-rate shocks.

3.1 Fact I: Larger buyers dominate international markets in Colombian exports and they pay different prices

Fact I.A: Most Colombian exports are sold to the largest foreign buyers in each market I document the well-known dominance of large firms in export markets. I define a market as a destination $\times$ product $\times$ year, where a product is at the HS10 level.

First, I identify the top buyers (top 3, top 5, top 10) of exports in each market and calculate how much they contribute to the total value bought in each market. Figure 1, Panel A shows that the value of the exports bought by the top three buyers in each market accounts for 78% of exports from Colombia, suggesting the high degree of buyer concentration in Colombia’s export market. For example, for the coffee market into the U.S. for a certain year, this would mean Starbucks, Peet’s Coffee and Dunkin Donuts buy most of Colombia’s coffee sold to the U.S., by value. Although, in most cases, these buyers appear to be specific manufacturing firms, as suggested by the names in the dataset, the results would remain unchanged even if they were intermediaries or retail firms.

Second, I calculate the degree of concentration of sales by using a standard measure of concentration, the Herfindahl-Hirschman Concentration Index (HHI). Before defining this index, I define S_{bjkt} as Buyer b ’s share of the nominal value of all exports of Product j to Country k in period t .

$$S_{bjkt} = \frac{p_{bjkt}q_{bjkt}}{\sum_{b' \in \mathcal{B}_{jkt}} p_{b'jkt}q_{b'jkt}}, \quad (3.1)$$

I then define the HHI.

$$HHI_{jkt} = \sum_{b \in \mathcal{B}_{jkt}} S_{bjkt}^2, \quad (3.2)$$

where $\mathcal{B}_{jkt}$ is the set of buyers active in market (j, k, t) .

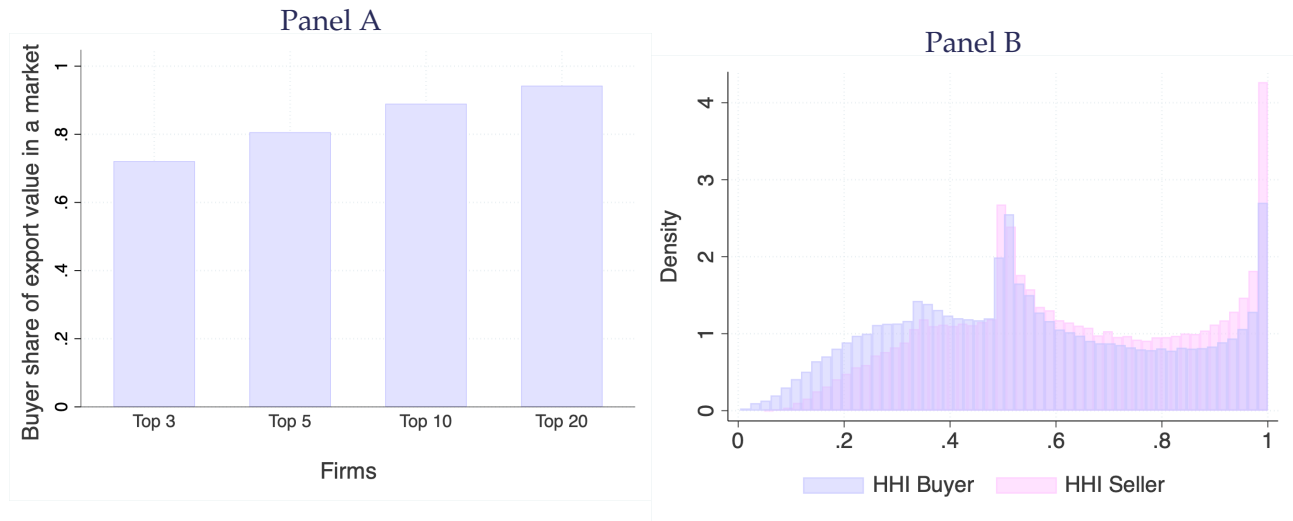
Figure 1, Panel B plots the distribution of the HHI. Note that in a market with only one buyer the HHI would be 1, while in a market with two buyers where each of them accounts for half of the market share, the HHI is 0.5. The figure shows a considerable number of markets with a high HHI, implying a high degree of concentration of sales among buyers.

I benchmark the observed level of concentration against the HHI for sellers, comparing the concentration of buyers in Colombian markets with that concentration of sellers. Figure 1, Panel B

¹²There is some contemporaneous evidence, such as [Alviarez et al. \(2025\)](#), which focuses on cost shocks—in particular only at the firm level, on tariffs in the United States. In contrast, this fact examines exchange rate pass-through in the context of currency depreciations at the market level, addressing the challenges faced by small sellers in developing countries.

indicates that the concentration of export flows among buyers is as important as the concentration among sellers, and, therefore, it could have important economic implications—precisely the focus of the analysis in this paper.

Figure 1: Buyer market concentration



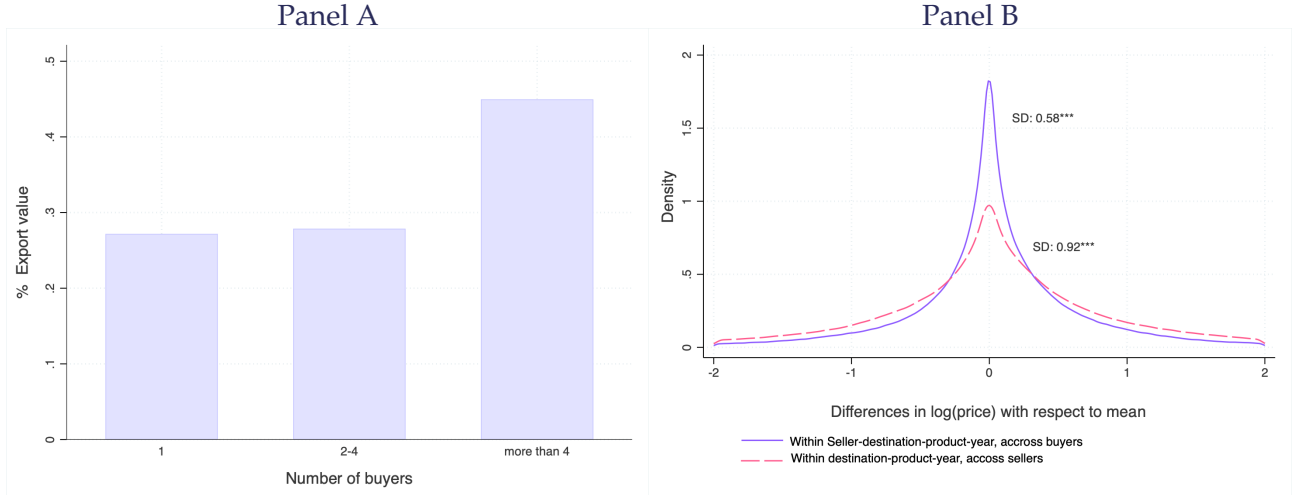
Notes: This figure shows the concentration of Colombian exports among foreign buyers. Panel A shows how much of the export value in a market, where the market is defined as destination country $\times$ HS10 product $\times$ year, corresponds to the top buyers. Top buyers are ranked by their purchases in the given market. Panel B shows the distribution of the HHI using equation 3.2 for the buyer market share (blue) and the seller market share (pink). [Source](#): Colombian Customs Data.

In this paper, I focus on buyer-side variation while keeping seller concentration constant. Seller-side market power in exchange-rate pass-through is already well documented ([Atkeson and Burstein, 2008](#); [Amiti, Itskhoki and Konings, 2014](#); [Alviarez et al., 2025](#)); focusing on the buyer side allows me to isolate a new mechanism — large foreign buyers’ ability to compress margins and buffer small exporters against exchange-rate shocks — without the confounding influence of dual concentrations.¹³ I acknowledge that Figure 1 Panel B shows seller concentration is comparable to buyer concentration in the data, and Figure 9 documents assortative matching between large buyers and large sellers. The model’s assumption of a continuum of sellers is therefore a deliberate simplification rather than a literal description of the market. Crucially, it does not bias the empirical results: in Section A.2.5 I explicitly control for seller market share and seller HHI alongside buyer concentration, and the coefficient on buyer market share remains significant and unchanged.

Fact I.B: Buyers pay different prices for the same seller and product in a given year I document the existence of multi-buyer firms in a market and that these firms receive different prices for the same product among their buyers.

¹³Model parameters can be readily adjusted to incorporate seller concentration, so extending the framework to jointly study both sides is feasible, either within this structure or in a bargaining framework as in [Alviarez et al. \(2025\)](#).

Figure 2: Export value explained by multibuyer sellers and top buyers and price dispersion



Notes: This figure shows characteristics of multibuyer sellers. Panel A highlights that sellers with more than one buyer account for half of the export value on average per market. Panel B illustrates the price dispersion after including product fixed effects, country of destination fixed effects and year fixed effect. The blue line includes also seller fixed effects. That is, for a given seller, product, year and country of destination, prices have a standard deviation of 0.58%. Source: Colombian Customs Data.

Figure 2, Panel A shows many multibuyer firms in Colombian export markets. In my sample, these firms account for roughly 80% of the exports value of the country. I document this fact for sellers (exporters) in Colombia. As documented in Figure 2, Panel B, the same firm, exporting the same product to the same destination in the same year, receives different prices from different buyers. This is true even controlling for sector $\times$ destination $\times$ year fixed effects to compare similar destination markets (i.e., controlling for size of the market, as well as growth of a particular sector). The standard deviation from the mean of prices received by one firm for the same product to the same destination across similar buyers is around 0.58%. This suggests specific buyers have characteristics that considerably affect a firm's price. More in detail, in Appendix A.2.2, I explore empirically the relationship between price dispersion and buyer size. The theoretical model in Section 4 provides a structural explanation for this dispersion: buyers with larger market shares face a less elastic supply curve (equation 4.7), which allows them to impose larger markdowns μ_{bjk} on sellers. Since markdowns vary across buyers within the same product-destination market, prices received by the same seller differ across buyers even for identical goods. This is oligopsonistic price discrimination driven by buyer market power, distinct from quantity discounts (ruled out in Section 3.4), quality differences, or long-term contracts (both addressed in the appendix).

To ensure robustness, I consider several potential mechanisms that could explain observed price differences, including product differentiation, exchange rate volatility, long-term contracts, price rigidity, and directed search. In Appendix A.2.3, I examine whether these price differences can be attributed to product differentiation or exchange rate volatility. By analyzing price dispersion at the monthly level for both differentiated products and commodities, I show that price discrimination persists even for identical goods, indicating that these price differences cannot be fully explained by these factors. Additionally, in Appendix A.2.9, I address the potential influence of long-term contracts by incorporating the length of buyer-seller relationships as a control variable. To account for price rigidity, I include fixed effects at the firm or product level to capture potential price stickiness. Finally, I use buyer-seller fixed effects to account for the possibility of directed search. In Section 3.4, I also

address the possibility of quantity discounts driving results.

3.2 Fact II: Larger buyers face a lower exchange rate pass-through to prices in the seller's currency

In this subsection, I estimate the relationship between buyer market power and exchange-rate pass-through by including the buyer's market share interacted with exchange-rate shocks in a regression where the outcome variable is the log change in price.¹⁴

As the structural model in Section 4 will show, buyer market share serves as a *sufficient statistic* for pass-through heterogeneity across buyers within a product-destination market.¹⁵, justifying its direct use in estimation without requiring a separate measure of markdowns. Formally, the exchange-rate pass-through into seller-currency prices to Buyer b , in Product j and Destination k is

$$\Delta p_{s,b,j,k,t} = \underbrace{[\alpha + \beta S_{b,j,k,t-1}]}_{\text{Exchange rate pass through}} \Delta e_{kt} + \delta S_{b,j,k,t-1} + \underbrace{\tau_{s,j,k} + \tau_{s,t}}_{\text{Fixed Effects}} + \epsilon_{s,b,j,k,t} \quad (3.3)$$

where $\Delta p_{s,b,j,k,t}$ is the log change in price of Good j from Seller s to Buyer b in Country k at Time t , Δe_{kt} is the log bilateral exchange rate change (COP seller currency per 1 unit buyer currency—Destination k). That is, an increase in e_k corresponds to the bilateral depreciation of seller currency, COP, relative to the Destination k buyer currency. $\tau_{s,j,k}$, $\tau_{s,t}$ are destination-product-seller fixed effect, year-seller fixed effect.¹⁶

I estimate parameters α and β with values averaged across seller, product, destination, and period. $S_{b,j,k,t-1}$ corresponds to my measure of buyer market share defined in equation 3.1.¹⁷ Note that $\alpha + \beta S_{b,j,k,t-1}$ corresponds to the exchange rate pass-through coefficient. That is, if this term is zero, a shock to the exchange rate produces no change in the seller-currency prices (COP), and a proportional change (to the change in the exchange rate) in the buyer currency (rest of the world currency).¹⁸

The main empirical contribution of this paper corresponds to the coefficient β , which determines how the market share of the buyer affects the exchange rate pass-through. If this coefficient is negative, larger buyers experience a lower change in price in the seller currency in response to exchange rate changes. For example, if Colombia depreciates its currency by 1%, this translates to a $\alpha + \beta\%$ change for the cases where a buyer is the only buyer in that destination for that product in a given year. However, when there is more than one buyer, the effect of the exchange rate shock is $\alpha + \beta S_{bjk}\%$. I summarize the distribution of this variable in my data in the appendix.

I propose different specifications including the fixed effects indicated by parameters in the

¹⁴Following the literature on exchange-rate pass-through such as [Gopinath, Itskhoki and Rigobon \(2010\)](#); [Amiti, Itskhoki and Konings \(2014\)](#); [Gopinath et al. \(2020\)](#).

¹⁵This parallels the role of import intensity in [Amiti, Itskhoki and Konings \(2014\)](#), where import intensity summarizes all relevant seller-side information about pass-through heterogeneity across exporters. Here, buyer market share plays the analogous role on the buyer side.

¹⁶In the data, I test directly for nonlinearity in this relationship and find no statistically significant evidence.

¹⁷In the appendix, I discuss the assumption that Δe_{kt} is uncorrelated with $S_{b,j,k,t-1}$ and so the OLS estimates of α and β from this regression are the theoretical coefficients in the pass-through relationship.

¹⁸This would correspond to a complete exchange rate pass-through as defined throughout the literature ([Amiti, Itskhoki and Konings, 2014](#); [Gopinath et al., 2020](#)).

theoretical model. In the most demanding specification, I include a year-HS-country fixed effect to isolate differences between markets and compare across buyers with different market shares. Note that this fixed effect absorbs the bilateral exchange rate included on its own, so the level $\Delta(\ln e_{kt})$ term appears in Columns (1)–(3) but is absent from Column (4) of Table 2. Its inclusion is also consistent with the assumption in Section 4 that both the quantities other buyers purchase and the prices other sellers charge to Buyer b do not change in response to the bilateral shock. The remaining columns include progressively less saturated sets of fixed effects, as described in the table notes.¹⁹

Second, I include several fixed effects accounting for the seller dimension, such as a year-seller fixed effect to control for shocks to the marginal cost, quality and characteristics of the buyer–seller relationship such as tenure and product mix. Given year-seller fixed effect controls for seller-level marginal cost variation common across a seller’s products in a given year, it does not absorb product-varying cost shocks within a seller (e.g., a productivity shock specific to coffee relative to cocoa for a multi-product exporter). As a robustness check I replace the year-seller fixed effect with a year-seller-product fixed effect in Appendix A.2.5; the main coefficients are unchanged, confirming the results are not driven by product-specific cost variation.

Table 2 reports the results. As the equation includes different sets of fixed effects, I go from the least-saturated regression to the more-demanding fixed effects. First, in Column (1), I find that, at the annual horizon, the unweighted average exchange rate pass-through elasticity into seller prices in the sample is 0.13, or, equivalently, 0.87 ($= 1 - 0.13$) into destination prices. I include product–destination-specific effects (where industry is defined at the HS 8-digit level) to be consistent with the theory, and year effects to control for common marginal-cost variation. In Column (2), I include an interaction between exchange rates and buyer market share. I show that the simple average coefficient reported in Column (1) masks a considerable amount of heterogeneity, as buyers (for the same seller) with different market share have very different pass-through rates. Buyers with a high market share exhibit a lower exchange rate pass-through into seller-currency export prices. By looking at Table 2, Column (2) pass-through coefficients can be calculated. The median buyer in the sample, a buyer with 13% market share, has a pass-through of 12% in the currency of the seller ($0.148 - 0.164 \cdot 12\% = 12\%$). As the market share increases, the pass-through declines. For example, a buyer with a 50% market share has only a 6% pass-through ($0.148 - 0.164 \cdot 50\% = 6\%$), and the pass-through of a buyer that is a monopsonist, around 100% market share, is approximately 0% ($0.148 - 0.164 \cdot 100\% \approx 0\%$).²⁰

¹⁹In particular, for less-saturated versions of the same regression, I also construct specific market-level controls in the data and include them in the regression (e.g., market price index, inflation, GDP).

²⁰For near-monopsonistic buyers ($S \approx 1$), the linear approximation yields pass-through near zero or very slightly negative. In the data, the mass of observations at $S \approx 1$ is small, and values marginally below zero are not statistically distinguishable from zero.

Table 2: Effect of Buyer's Market Share on Exchange Rate Pass-through

	(1)	(2)	(3)	(4)
	$\Delta \text{Log}(\text{Price})$	$\Delta \text{Log}(\text{Price})$	$\Delta \text{Log}(\text{Price})$	$\Delta \text{Log}(\text{Price})$
$\Delta \ln ER$	0.132*** (0.0509)	0.148*** (0.0547)	0.0692* (0.0355)	
S_{t-1}		-0.0481*** (0.00684)	-0.0565*** (0.00778)	-0.0185*** (0.00659)
$\Delta \ln ER \times S_{t-1}$		-0.164** (0.0789)	-0.117** (0.0494)	-0.155*** (0.0445)
Period-Seller FE	✓	✓		
Period-HS FE			✓	
Country-HS-Seller FE	✓	✓	✓	
Country-Period-HS FE				✓
N	517100	517100	512577	460477

Notes: Results from equation 3.3. The dependent variable corresponds to the log change of prices. ΔER and S_{t-1} are the bilateral exchange rate and the buyer market share, respectively. Products are defined at the HS10 level. Standard errors are clustered at the country-time level and are shown in parentheses. Country-level clustering leaves the significance levels unchanged. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

To better understand the results from my regression, Table 3 shows the number of firms with different levels of exchange rate pass-through and buyer share. The largest buyers have on average between 0% and 5% pass-through while the smallest buyers have an exchange rate pass-through of around 15%.

Table 3: Firms with Different Levels of Exchange Rate Pass-through

EPRT	Number of firms	Mean S
$0 < \alpha + \beta S_t < 0.05$	68,548	0.91
$0.05 < \alpha + \beta S_t < 0.10$	57,011	0.66
$0.10 < \alpha + \beta S_t < 0.15$	82,935	0.44
$0.15 < \alpha + \beta S_t < 0.20$	169,764	0.21
$0.20 < \alpha + \beta S_t$	1,144,500	0.02

Notes: The table shows the number of firms and the mean value for buyer market share S_t for the different categories of exchange rate pass-through coefficients. α and β correspond to the estimates from Table 2.

These results reflect that the dominant mechanism is the markdown channel: larger firms have lower exchange rate pass-through. That is, given larger buyers have market power, they internalize the upward-sloping supply curve for inputs, which implies that each additional unit they buy raises the price of every other unit.²¹ Following a depreciation, large buyers—operating farther from the competitive benchmark—increase their markdown by more absorbing a greater share of the exchange-rate gain at the seller's expense, which prevents seller-currency prices from rising proportionally. For a given buyer, the higher the market power, the steeper the supply curve faced, and so the lower the pass-through of an exchange rate shock to the seller's price.

Appendix A.2.11 addresses the concern that buyer shares may be mechanically related to prices. The negative pass-through gradient remains when buyer size is measured leave-one-seller-out, using

²¹This is analogous to a monopoly case where the only seller internalizes the downward-sloping demand curve.

only other Colombian sellers' transactions to construct the buyer share, and when buyer size is measured with the bilateral seller-buyer share. The appendix provides additional evidence on the economic interpretation of the buyer-size gradient. Moreover, Appendix A.2.12 documents that domestic sales account for about 84 percent of Colombian manufacturing sales during 2007–2020 and shows that high domestic-inclusive foreign-buyer exposure is associated with lower pass-through. Appendix A.2.8 shows that the buyer-size gradient is not absorbed by shipment scale: the interaction remains negative after controlling for log quantity and its interaction with the exchange-rate shock.

3.3 Fact III: Markets with High Concentration of Sales Among Buyers Display Low Exchange Rate Pass-through

I explore the market level exchange rate pass-through. I start from the equation 3.3, and obtain the following regression at the market level:

$$\Delta p_{jkt} = [\alpha + \beta HHI_{jkt}] \Delta e_{kt} + \delta HHI_{jkt} + \mathcal{F}_{jkt} + \epsilon_{jkt}, \quad (3.4)$$

where Δp_{jkt} is the log change in the average price in market (j, k, t) , and $\mathcal{F}_{jkt}$ denotes fixed effects: HS (product), period–HS, country–HS, and country–period.²²

While calculating the exchange rate pass-through at the market level, I can no longer include seller fixed effects to control for specific seller characteristics, such as the seller market share. Thus, the coefficient of this regression could be reflecting either buyer or seller market power. To address this potential issue, I aggregate the information I have at the seller level, and calculate the concentration index also for the sellers. This allows me to disentangle the effects, and I can account accurately for the effect of buyer market concentration. Results for this regression are shown in Table 4. When exports are more concentrated among a few buyers, the exchange rate pass-through for the average market price is lower.

²²I summarize the distribution of the HHI and the exchange rate pass-through at the market level in the appendix.

Table 4: Effect of Market Concentration on Average Exchange Rate Pass-through

	(1)	(2)	(3)	(4)
	$\Delta \ln(\text{Price})$	$\Delta \ln(\text{Price})$	$\Delta \ln(\text{Price})$	$\Delta \ln(\text{Price})$
$\Delta \ln ER$	0.0657*** (0.0243)		0.0289 (0.0302)	
HHI _{buyer}	-0.00909* (0.00519)	-0.000965 (0.00479)	-0.00878* (0.00515)	-0.000303 (0.00475)
$\Delta \ln ER \times \text{HHI}_{\text{buyer}}$	-0.166*** (0.0420)	-0.187*** (0.0395)	-0.159*** (0.0412)	-0.176*** (0.0392)
HHI _{seller}			0.00295 (0.00319)	0.00538** (0.00234)
$\Delta \ln ER \times \text{HHI}_{\text{seller}}$			0.0580** (0.0256)	0.0735*** (0.0191)
HS FE		✓		✓
Period-HS FE	✓		✓	
Country-HS FE	✓		✓	
Country-Period FE		✓		✓
N	334468	376582	334468	376582

Notes: The table shows results for equation 3.4. HHI_{buyer} and HHI_{seller} correspond to the HHI for sales concentrations among buyers and sellers, respectively. They are calculated by using equation 3.2 with the market share of the buyers and sellers. Standard errors are clustered at the country-time level and are shown in parentheses. Country-level clustering leaves the significance levels unchanged. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Column (1) shows that, even without controlling for seller HHI, buyer concentration has a significant relationship with exchange rate pass-through. Columns (2), (3), and (4) include information of the distribution of sellers' market share while controlling for period, HS-country and HS-period fixed effects. My preferred specification is Column (4) because it contains the most restrictive fixed effects. It shows that the concentration of the buyers strongly influences the exchange rate pass-through.

3.4 Robustness

In this section, I consider six sets of robustness tests: discussing price stickiness and currency of invoicing, discussing the presence of seller market power, considering alternative definitions of buyer market share, including import intensity, and using alternative samples in terms of included destinations, firms, and products. I conclude the section by a discussion of the possible selection, alternative mechanisms and measurement issues.

Currency of invoicing and sticky prices. In my model, there are flexible prices and strategic complementarities in pricing. I now briefly comment on the interpretation of my results in an environment with sticky prices, where exporters choose to fix their prices temporarily either in local or in producer currency. I examine the case where the U.S. dollar acts as the dominant currency: firms set export prices in dollars and change them infrequently, as in [Gopinath et al. \(2020\)](#).

In this setting, the results in this paper would confound together the change in the desired markdown with the mechanical changes in markdown induced by the exchange rate movements when prices are sticky in a given currency. To address this concern, I estimate three alternative

specifications: (i) baseline regressions with transactions only to the US, (ii) baseline regressions with transactions only invoiced in US dollars, (iii) baseline regressions including the exchange rate between the Colombian peso and the US dollar.

Table 5: Sticky Prices and Dominant Currency

	(1)	(2)	(3)	(4)	(5)
	$\Delta \ln(\text{Price})$	$\Delta \ln(\text{Price})$	$\Delta \ln(\text{Price})$	$\Delta \ln(\text{Price})$	$\Delta \ln(\text{Price})$
$\Delta \ln ER$	0.143*** (0.0225)		0.128** (0.0591)	0.145** (0.0590)	
S_{t-1}	-0.141*** (0.0255)	-0.0836*** (0.0141)	-0.0635*** (0.00870)	-0.0624*** (0.00856)	-0.0570*** (0.00836)
$\Delta \ln ER \times S_{t-1}$	-0.683*** (0.128)	-0.487*** (0.104)	-0.182** (0.0878)	-0.186** (0.0871)	
$\Delta \ln ER_{US-COP} \times S_{t-1}$					-0.216*** (0.0796)
Buyer FE	✓				
HS-Seller FE	✓	✓			
Period-Seller FE		✓	✓	✓	✓
Country-HS-Seller FE			✓	✓	✓
Sample: US only	✓	✓			
Sample: Full				✓	✓
Sample: \$-invoiced			✓		
ER: Dollar (US-COP)					✓
N	122648	127531	466688	471536	517100

Notes: The table shows results for equation 3.3 but including controls for the sticky prices and the dominant currency paradigm hypothesis. Standard errors are clustered at the country-time level and are shown in parentheses. Country-level clustering leaves the significance levels unchanged. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

First, I restrict the sample to exports destined for the United States. In this case, the bilateral exchange rate and the dollar exchange rate coincide, so any variation in the exchange rate is mechanically tied to movements in the U.S. dollar. This setup effectively absorbs concerns related to which exchange rate is the “relevant” one—whether the global dollar or the bilateral rate—and rules out the possibility that my results are driven by episodes of broad dollar fluctuations. Even in this setting, buyer market power remains strongly relevant, showing that the central mechanism of the paper does not hinge on exchange-rate identification choices but rather on the role of buyers in shaping pass-through. In Table 5, Columns (1) and (2), I present these results using two alternative sets of fixed effects. In both cases, higher buyer market power is associated with lower exchange-rate pass-through, and the estimated effect is even stronger than in the baseline specification.

In Columns (3) and (4), I compare the main specification estimated on the full sample (Column 4) with the same specification restricted to transactions invoiced in U.S. dollars (Column 3). In the Colombian customs data, 98% of transactions are dollar-invoiced, so the restricted sample is nearly identical to the universe of observations. The results are very similar across the two samples—if anything, slightly larger in magnitude when not conditioning on the invoicing currency—which suggests that currency of invoicing is unlikely to bias my estimates.

Finally, in Column (5) of Table 5, I estimate a specification that uses the dollar exchange rate and obtain results consistent with [Gopinath et al. \(2020\)](#) regarding the exchange-rate pass-through component that is independent of buyer market power. Note that the bilateral and dollar exchange

rates cannot be included jointly due to collinearity. In Appendix A.2.10, I report additional specifications designed to facilitate comparison with the aggregate results in Gopinath et al. (2020), restricting the sample to manufactures and adopting a similar set of controls and fixed effects.

Taken together, these results indicate that the role of buyer market power in shaping exchange-rate pass-through is not an artifact of sticky prices or dollar invoicing. The mechanism emphasized in this paper — buyers with greater market power adjusting their markdowns in response to exchange-rate movements — operates even within transactions destined to the United States and within those invoiced in dollars, where sticky-price and dominant-currency channels are either absorbed or held fixed. At the same time, buyer market power may additionally affect the probability that prices are set in the buyer’s currency, reinforcing low short-run pass-through through a complementary sticky-price channel. Disentangling the relative contribution of these two channels is beyond the scope of this paper, though existing work suggests they share common underlying primitives²³

Seller Market Power Although the focus of this paper is on the buyer side, seller market power has been shown to matter for exchange-rate pass-through (e.g., Atkeson and Burstein (2008); Amiti, Itskhoki and Konings (2014); Alvarez et al. (2025)). In this section, I verify that my buyer-side results are not driven by seller concentration. The baseline specifications already absorb seller heterogeneity through fixed effects (e.g., seller $\times$ year and country $\times$ HS $\times$ seller); I now complement this with a series of checks that explicitly control for seller market power, both at the firm and market level.

In the baseline specifications, fixed effects absorb seller heterogeneity (e.g., seller $\times$ year and country $\times$ HS $\times$ seller). In Appendix A.2.5, I report robustness checks that interact exchange-rate changes with multiple seller-size measures (and alternative definitions) as well as with seller market share; across these specifications, the main coefficients remain statistically significant and have the expected signs. At the market level, including seller HHI alongside buyer HHI leaves the negative interaction between pass-through and buyer concentration unchanged.

The model also embeds a seller-side footprint through the marginal-revenue channel: pass-through depends on (i) σ , the elasticity of substitution in the buyer’s CES aggregator, and (ii) x_{bjk} , the CES input share for Product j , which provides a production-based analogue of Buyer b ’s cost share on the sellers supplying that product.²⁴ The term x_{bjk} captures how important the associated seller–product relationship is to the buyer—i.e., a product-level analogue of seller market power—and scales how strongly the seller’s quantity response shifts the buyer’s marginal revenue. Intuitively, a larger x_{bjk} gives that seller–product relationship more weight in the marginal-revenue term, strengthening this seller-side channel.

On the Definition of Buyer Market Share One concern might be that, since buyer market shares are computed only within the set of Colombian exports, the measure could miss the fact that international buyers potentially source similar products from other countries. This could be problematic if goods from different origins were perfect substitutes. However, defining market shares narrowly—treating

²³Gopinath, Itskhoki and Rigobon (2010) show that the sticky price determinants of incomplete pass-through are largely shaped by the same underlying primitives as the flexible price determinants, and they reinforce each other in the cross-section of firms.

²⁴The formal definition of x_{bjk} is given in Equation (4.15).

Colombian exports as distinct markets—is justified. First, Colombian products often exhibit unique quality attributes, branding, and taste profiles, making them imperfect substitutes for similar goods from other origins (e.g., coffee from Ecuador or Uganda). Second, international buyers commonly differentiate by geographic origin, validating this market definition. Hence, the buyer market share computed as a buyer’s share of total international purchases of a specific Colombian product accurately captures a buyer’s relative importance within the narrowly defined market for each Colombian exported product.

In the rest of this section, I explore alternative definitions of buyer market share to ensure that the results are not sensitive to my definition of it. In table 11, in Appendix A.2.7 I include two new definitions of buyer market share. First, in Columns (1)-(2), I define a market as a Product HS - year combination. In this case, the buyer market share corresponds to the share of the exports in a Product HS-year combination bought by a given buyer. Second, I define the buyer market share as the share of a given seller s exports in a Product HS - year that corresponds to a given buyer b . In other words, how relevant is a specific buyer in the exports of a seller in a market, independently of how big is the buyer as a whole in that market. In all cases, the signs of my results are essentially unchanged. The magnitude of the effect fluctuates from a smaller effect when only looking at the within-seller size measure to a bigger effect when considering buyers in all destinations. These results are not surprising since the average buyer shares are much lower when dropping a large share of exports from the denominator in the measure.

As a final exercise, I explore the relationship between these alternative definitions and the seller side. To do this, in the last two Columns of Table 9 in Appendix A.2.5 I include an interaction of the bilateral exchange rate with the seller market share for alternative measures of buyer market share. Results remain significant and with the same sign. Notably, once I expand the market definition to the product-year level, seller market share becomes relevant for pass-through.

Import Intensity Since it has been documented that more import-intensive exporters have significantly lower exchange rate pass-through, as a function of offsetting exchange rate effects on their marginal cost (Amiti, Itskhoki and Konings, 2014), I discuss this topic in this subsection. For import intensity to create a bias in the estimates, it has to be the case that there is a selection of larger buyers into exporters that, in turn, source their intermediate inputs internationally. Specifically these exporters should import more than other exporters.²⁵

To account for this potential effect, I include a measure of import intensity of the exporter, defined as the ratio between the total value of imports for the seller s at time t and the total trade of that seller at time t . Appendix A.2.6 shows the results of this alternative specification. First, in Column (1), I verify my results are unchanged when using the contemporaneous import intensity measure. Next, in Column (2), I include the lagged time-varying import intensity and arrive at the same result.

²⁵Ideally one would also control for buyers’ import intensity, since buyers who source more intermediate inputs internationally may face cost offsets that affect their markdown behavior. However, buyer import intensity is not observed in my data. Instead, I control for seller import intensity, which remains informative: if large buyers systematically match with more import-intensive sellers (as the assortative matching in Figure 9 suggests), then controlling for seller import intensity partially proxies for the buyer-side channel.

Ruling Out Quantity Discounts as an Alternative Mechanism One alternative explanation for the lower pass-through observed for large buyers is standard quantity discounts: high-volume buyers may receive lower unit prices, and these discounts, if largely static, could mechanically attenuate measured pass-through without reflecting strategic markdown adjustments. This is a relevant concern because transaction-level quantity discounts can shape both price variation and measured pass-through (Ganapati and Hottman, 2026). Several features of the data and results point against this interpretation. The price dispersion documented for identical products sold by the same seller to different buyers (Fact I) persists after controlling for quantities and product-destination-year fixed effects, suggesting that price variation is not mechanically tied to shipment size. The pass-through–buyer-size gradient also appears across both homogeneous and differentiated goods and remains when focusing on products where per-unit quantities are relatively uniform. Moreover, Appendix A.2.8, Table 12, shows that the buyer-size interaction remains negative after controlling directly for log shipment quantity and its interaction with exchange-rate changes.

Alternative Samples I further check the robustness of my results within alternative subsamples of the dataset, both in the coverage of export destinations and in the types of products and firms. It reveals the same qualitative and quantitative patterns found in the benchmark sample.

Columns (1)–(2) of Table 18 in Appendix A.2.13 report the results for an alternative set of export destinations—OECD countries.²⁶ It is noteworthy that for the OECD subsample I estimate both a higher baseline pass-through (for firms with zero buyer market share) and a stronger effect of buyer market share on pass-through, than for other countries.

The remaining columns in Table 18 consider different sets of products and firms. I follow the ISIC Rev. 3.1 classification to define which products are commodities. As a robustness check, I also use the subsample of differentiated products only, referenced products only and homogeneous products only (instead of the full set of products presented) constructed using the classification of goods by Rauch (1999). This is available in Appendix A.2.13.

Finally, the sample has included all of the firm’s manufacturing exports rather than restricting it only to HS major products. In Column (7), I include only the firm’s major export products, based on its largest HS code, in order to address the issue of multiproduct firms, identifying a firm’s major products, using the HS 10-digit category. I show that the results are not sensitive to this choice of products.

4 The Model

I develop an oligopsony model in international markets with infinitely many sellers located in the home country and a few large buyers in each foreign market. This concentration of demand gives the buyers market power and allows them to choose the prices they pay.²⁷ The concentration of buyers, and hence their market power, differs across and within products. Given these prices, sellers

²⁶Note the United States only case is shown in Table 5.

²⁷In my baseline model, buyers compete à la Bertrand. However, in Appendix A.1.5, I additionally solve for Cournot competition.

choose which products they produce, and to which buyer they sell. I model the seller’s choice of sector and buyer as a discrete-choice problem, which yields a nested CES supply curve.

The equilibrium price is a function of the relative buyer market share.²⁸ The shape of this function is determined by two key elasticities, the cross-product supply elasticity and within-product cross-buyer supply elasticity, which govern the heterogeneity of costs in the seller’s choice problem. Intuitively, more heterogeneous sellers’ costs lead to greater consequences of buyers’ market power.

While the existing literature has already established that seller-side concentration influences exchange rate pass-through, this paper abstracts from seller concentration to focus exclusively on buyer market power. The model’s assumption of a continuum of sellers is a simplification that isolates the buyer-side mechanism²⁹ and shows buyer markdowns act as a buffer against exchange-rate shocks. This abstraction does not drive the core results: although pass-through elasticities may vary across exporters of different sizes, the main contribution here is to show that large buyers’ ability to compress margins insulates even the smallest sellers from exchange-rate fluctuations. This focus on buyer power is particularly relevant for understanding price-adjustment dynamics in markets where exporters are numerous and fragmented, whereas buyers are few and powerful.

I follow the structure of labor markets in [Berger, Herkenhoff and Mongey \(2022\)](#) and [Felix \(2022\)](#) but applied to international trade, more similar in spirit to [Zavala \(2022\)](#), as it provides a tractable framework to focus squarely on the buyer side of the market. This setup is particularly well-suited to disentangling the effects of shocks that operate through their impact on markups and marginal revenues—mechanisms that, to my knowledge, have not been studied in the context of exchange rate pass-through. It also permits a direct comparison of the mean and variance of shocks in an aggregate setting. For this purpose, unlike [Alviarez et al. \(2025\)](#) and [Ignatenko \(2024\)](#), I only require modeling the buyer side to capture the effects of interest. Moreover, unlike [Huang et al. \(2024\)](#), I do not endogenize the formation of buyer–supplier links; instead, I take these relationships as exogenous and focus on quantifying their effects.

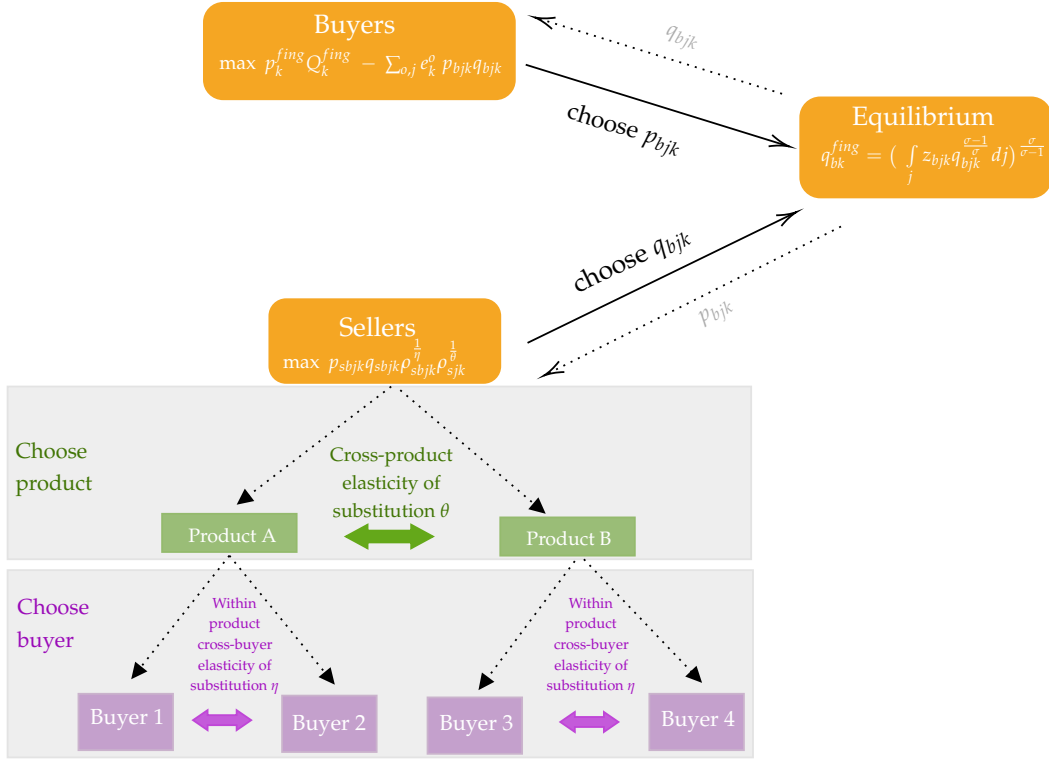
4.1 Timeline and Model Structure

The timeline of the model is as follows: (i) productivity shocks are realized, (ii) buyers choose the price they want to pay for their inputs, and (iii) sellers choose the quantity they are going to supply of each input. I solve this by backward induction, starting with the seller’s problem then moving to the buyer’s problem. Figure 3 summarizes the model structure with notation explained in the text.

²⁸In this sense, the model also connects to the work of [Alviarez et al. \(2025\)](#) although I do not assume a bargaining game. Sellers in my setting correspond to small producers.

²⁹The data show that seller concentration is comparable to buyer concentration (Figure 1 Panel B) and that large buyers tend to match with large sellers (Figure 9). The model is therefore not a literal description of the market. However, controlling for seller market share and seller HHI alongside buyer concentration in Section A.2.5 leaves the buyer-side coefficients unchanged, confirming that omitting seller concentration does not bias the core results.

Figure 3: Model Structure



Notes: This figure displays a diagram of the structure of the model. The upper part shows how quantities and prices are determined in equilibrium. The lower part illustrates the seller input supply decision according to the discrete choice framework.

4.2 Seller Supply Function

An infinite mass of potential sellers in a home country indexed by $s \in [0, 1]$ sell products indexed by $j \in \mathcal{J} \equiv \{1, \dots, M\}$ to buyers b in destination countries $k \in \mathcal{K}$. Each seller makes two decisions: (i) which product to produce and (ii) which buyer to supply. This decision will depend on the sellers' initial endowment, some productivity shocks and the prices offered by the buyers.

Let $\mathcal{B}_{jk}$ denote the set of buyers active in product–destination market (j, k) and define $\mathcal{A} \equiv \{(b, j, k) : (j, k) \in \mathcal{J} \times \mathcal{K}, b \in \mathcal{B}_{jk}\}$. Each Seller s has an endowment, $q_s \sim \psi$, and can allocate it across buyer–product–destination triples in $\mathcal{A}$. As the seller supplies more to one triple, less of the endowment remains for the others: $\sum_{(b,j,k) \in \mathcal{A}} q_{sbjk} = q_s$. Also, sellers with more of the endowment, q_s , can produce more.

Second, apart from their initial endowment, each Seller s for Product j for Buyer b in Destination k receives an idiosyncratic productivity drawn iid from a nested Frechet distribution: He receives an idiosyncratic shock, ρ_{sjk} , for producing each Product j (product-specific shock) and an idiosyncratic shock, ρ_{sbjk} , for supplying each Buyer b within Product j (within-product buyer-specific shock). Therefore, the idiosyncratic shocks determine the supply. A higher shock for Buyer b and Product j mean the seller can supply more if he chooses that buyer and product. Intuitively, ρ_{sjk} corresponds to the availability of inputs and technology for the seller to produce Product j , and ρ_{sbjk} corresponds to search costs and frictions for the seller to connect with Buyer b of Product j .

The shocks capture heterogeneity in their comparative advantage across products and buyer relationships. They can be interpreted as reduced-form representations of factors such as search frictions, relationship specificity, or informational barriers. While this structure loosely resembles search or random utility models, the analogy is not exact: sellers are heterogeneous and choose the single product–buyer combination that maximizes their return, allocating their full quantity to that match. This is a modeling choice that simplifies the supply side while reflecting the idea that sellers may face capacity or relational constraints that limit diversification across buyers. Importantly, even though sellers only match with one buyer in the model, aggregating across many sellers yields a smooth distribution of sales shares across buyers, which is what matters for market-level outcomes and estimation. This approach allows for tractable aggregation while capturing rich heterogeneity and selection in buyer–seller links.

Third, sellers observe the prices offered by different buyers for different products and destinations. Each seller chooses the buyer–product–destination triple that yields the highest return, given its productivity shocks and the prices set by buyers.³⁰

$$\max_{\{q_{sbjk}\}_{(b,j,k) \in \mathcal{A}}} \sum_{(b,j,k) \in \mathcal{A}} p_{bjk} q_{sbjk} \rho_{sbjk}^{\frac{1}{\eta}} \rho_{sjk}^{\frac{1}{\theta}} \quad \text{s.t.} \quad \sum_{(b,j,k) \in \mathcal{A}} q_{sbjk} = q_s.$$

where p_{bjk} is the price of Product j in Destination k when it is purchased as an input by Buyer b . Note that this price varies by Buyer b since buyers have market power. As there are no diminishing returns to selling to a given buyer–product–destination triple, in equilibrium each seller picks one triple and sells its entire endowment there, if there are no ties.³¹ However, buyers will source from different sellers.

For intuition, consider the problem of a seller who has an initial endowment of q_s square feet of land to be cultivated. He could use it for either growing coffee or cocoa beans depending on his technology, ρ_{sjk} —reflecting, for instance, differential access to machinery suited for each crop. If he produces coffee, he could sell it to Starbucks or Peet’s Coffee, with the choice governed by search costs ρ_{sbjk} —capturing, for example, the lower frictions of supplying a buyer with whom he has an established relationship or compatible logistics. Finally, the seller takes into account the prices offered by each buyer before allocating his endowment. There could be a trade-off between producing lower quantities and higher prices offered by the buyers.

The probability that Seller s chooses Product j and Buyer b within Destination k , $\Pr(sbjk)$, is independent of the seller’s endowment q_s . Due to the Frechet distribution of productivity shocks, for a given seller, that is fixing q_s , the probability of choosing Buyer b and Product j is the same as the probability that $\Pr(\text{revenue}_{b'j'k} < \text{revenue}_{bjk})$ for every $j' \in \mathcal{J}$ and $b' \in \mathcal{B}_{j'k}$ such that $(b', j') \neq (b, j)$. Following [Eaton and Kortum \(2002\)](#), this probability is then equal to how much of the production supplied to Destination k goes to each buyer and product. Formally, I define λ_{bjk} as the share of

³⁰Note that there are no costs in this maximization given all the sellers have an endowment. One way some types of costs are included is through the different shocks ρ_{sbjk} and ρ_{sjk} , but not input costs.

³¹The discrete choice framework used here provides a clear structural interpretation of the elasticities of substitution within and across product categories. In particular, it explicitly captures sellers’ substitution patterns across different buyers and product lines, thus highlighting the role of buyer concentration in determining market outcomes. This setting is analogous to one where small sellers transact exclusively with an intermediary, who subsequently resells to multiple buyers.

production supplied to Destination k that is sold to Buyer b of Product j :³²

$$\lambda_{bjk} = \underbrace{\frac{P_{jk}^{1+\theta}}{P_k^{1+\theta}}}_{\Pr(\text{chooses Product } j \mid k)} \underbrace{\frac{p_{bjk}^{1+\eta}}{P_{jk}^{1+\eta}}}_{\Pr(\text{chooses Buyer } b \mid j, k)}, \quad (4.1)$$

where $P_{jk} = \left(\sum_{b \in \mathcal{B}_{jk}} p_{bjk}^{1+\eta} \right)^{\frac{1}{1+\eta}}$ and $P_k = \left(\sum_{j \in \mathcal{J}} P_{jk}^{1+\theta} \right)^{\frac{1}{1+\theta}}$. I derive this in Appendix A.1.3.

Aggregating across sellers yields a nested CES upward-sloping supply curve for Buyer b in Product j , Country k :³³

$$q_{bjk} = \left(\frac{p_{bjk}^\eta}{P_{jk}^{1+\eta}} \right) \left(\frac{P_{jk}^{1+\theta}}{P_k^{1+\theta}} \right) Y_k \quad (4.2)$$

where $Y_k = \sum_{j \in \mathcal{J}} \sum_{b \in \mathcal{B}_{jk}} p_{bjk} q_{bjk}$.

4.2.1 Interpreting Elasticities

There are intuitive interpretations of the parameters θ and η .³⁴ First, θ governs the correlation of product-specific shocks. This means that the higher θ , the more correlated are the seller's productivity draws across sectors. This means that, if the idiosyncratic productivity for the different product is more likely to be similar, the prices in the product will be more relevant to determine the quantity choice. Intuitively, θ will be high if the availability of inputs needed for many different sectors and technology is similar so that there is little heterogeneity in productivity. Finally, θ is the elasticity of substitution across products in the CES supply function. If θ is relatively high, then it is easy to substitute products from the point of view of the seller. Higher substitutability would correspond to higher rates of seller switching between products, in a dynamic setting.

Analogously, η governs the correlation of buyer-specific shocks. The higher η , the more correlated are the seller's draws across buyers within a product. Then, since search costs to connect with one buyer or another are similar, the price each buyer offers will be more important. If η is high then sellers are able to actually connect with many buyers, and there will be low heterogeneity in the cost of finding a buyer.

Following the literature on the topic, I expect $\eta > \theta$, which has different interpretations: (i) Idiosyncratic cost shocks are more strongly correlated across buyers than across products, (ii) there is more heterogeneity in the productivity of producing different products than in the costs of connecting with two different buyers,³⁵ and (iii) sellers are more substitutable within products than across products from the buyer's point of view.

³²All destinations here will differ on the exchange rate, and they might also have different elasticities. More details on this in Section 4.2.1.

³³See Appendix A.1.4 for derivations and intuitions on how prices relative to the price index relate to quantity. Equation 4.2 is equivalent to $q_{bjk} = \left(\frac{p_{bjk}}{P_{jk}} \right)^\eta \left(\frac{P_{jk}}{P_k} \right)^\theta \frac{Y_k}{P_k}$

³⁴The interpretation of elasticities is inspired by Berger, Herkenhoff and Mongey (2022) and Zavala (2022).

³⁵Note that, in the empirical analysis, this condition holds for the same destination and in the same period.

4.3 Buyer's Profit Maximization

There is a finite number of buyers in Foreign Country k . Each buyer purchases her inputs to produce a final good to sell in her home country. A buyer can buy different inputs j from different origins.³⁶ Her production function is CES:

$$Q_{bk}^{finalg} = \left(\sum_{j \in \mathcal{J}} z_{bjk} q_{bjk}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (4.3)$$

where z_{bjk} is an idiosyncratic productivity term for Buyer b , Product j , and Destination k , and is the source of ex-ante heterogeneity across buyers.

Buyers of Product j exert market power over sellers, which I model as Bertrand competition. When deciding the price to pay, buyers form expectations about how sellers will respond. In doing so, they internalize the upward-sloping supply curve: each additional unit they purchase increases the price of every other unit. Note that, as I assume that the market structure is oligopsonistic, a buyer can affect the price index P_{jk} ; however, product markets are negligible in the destination aggregate, so the buyer takes P_k and Y_k as given.

Therefore, the problem of Buyer b , located in Country k that buys Product j consists of choosing the prices they will offer to sellers, p_{bjk} . Buyers maximize the following profit function subject to a production function, Equation 4.3, and the quantity supplied by the seller, Equation 4.2.³⁷

$$\max_{\{p_{bjk}\}_{j \in \mathcal{J}}} p_k^{finalg} Q_{bk}^{finalg} - \sum_{j \in \mathcal{J}} \frac{p_{bjk} q_{bjk}}{e_k} \quad \text{s.t.} \quad Q_{bk}^{finalg} = \left(\sum_{j \in \mathcal{J}} z_{bjk} q_{bjk}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad q_{bjk} = \frac{p_{bjk}^\eta P_{jk}^\theta Y_k}{P_{jk}^\eta P_k^\theta P_k}. \quad (4.4)$$

The first term of the profit function corresponds to the revenue the buyer obtains after selling quantity Q_{bk}^{finalg} of the final good she produces at price p_k^{finalg} . The domestic price of output is modeled as exogenous. The second term corresponds to the cost of inputs from the origin under consideration: quantities q_{bjk} purchased at prices p_{bjk} .

The profit function is expressed in local currency. Buyers sell their final products in the home country so the revenue term is expressed in local currency. As buyers buy these inputs in international markets (costs term), they pay for them in the currency of the producer. I introduce the term e_k , which corresponds to the nominal exchange rate, to convert the costs to local currency. This modeling choice does not assume that trade is necessarily invoiced in the producer's currency, nor does it take a stand on the dominant currency paradigm (DCP). Rather, it is a reduced-form way to capture buyers' exposure to international cost variation driven by exchange rate fluctuations, regardless of the invoicing currency. Importantly, the exchange rate term ensures that all components in the profit function are expressed in the same currency, allowing for a consistent and tractable formulation of firm-level revenues and costs.³⁸ The subindex k indicates the country of the buyer. Because the buyer's

³⁶She can also buy the same input j from different origins. Because the buyer's problem is the same for each origin, I suppress the origin subscript and write the problem for one origin at a time.

³⁷In the baseline specification, I do not include a fixed cost of importing; however, Appendix A.1.17 addresses this, suggesting some selection into importing, similar to the selection into exporting commonly discussed in the literature.

³⁸While prices may be sticky in a dominant currency like the USD, cost-side exposure from inputs and imported goods still introduces exchange rate sensitivity, which the model aims to capture in reduced form.

problem is the same for each origin, I suppress the origin subscript and write the problem for one origin at a time.³⁹ The exchange rate is defined as foreign currency per unit of home currency.

The marginal revenue product of Product j for Buyer b in Destination k is defined as

$$MRP_{bjk} \equiv \frac{\partial(p_k^{finalg} Q_{bk}^{finalg})}{\partial q_{bjk}} = p_k^{finalg} z_{bjk} \left(\frac{q_{bjk}}{Q_{bk}^{finalg}} \right)^{-\frac{1}{\sigma}}. \quad (4.5)$$

The first-order condition (FOC) can then be written as

$$p_{bjk} = \underbrace{\frac{1}{\mu_{bjk}}}_{\text{markdown: } \mu_{bjk} > 1} \times \underbrace{MRP_{bjk}}_{\text{marginal revenue product}} \times e_k, \quad (4.6)$$

where the markdown is $\mu_{bjk} = 1 + \epsilon_{bjk}^{-1}$ and $\epsilon_{bjk} = \frac{\partial \ln q_{bjk}}{\partial \ln p_{bjk}}$ is the supply elasticity faced by Buyer b of Product j in Destination k . Note that a higher μ_{bjk} implies a lower price—the markdown is larger, meaning the buyer pays further below the marginal revenue product.

Equation 4.6 shows that the price of the input in the producer's currency (seller's currency) depends on the markdown, the exchange rate and the marginal value of the input, that is the value the input adds to the final product. In other words, the buyers, who are the ones that have market power, will pay for an input an amount equal to how much this input adds to their revenues, reduced by how much market power they have.⁴⁰

Some relevant intuitions can be derived from equation 4.6. First, MRP_{bjk} is expressed in buyer's currency and the markdown has no unit so, for the price to be in the currency of the seller, I need to multiply by the exchange rate e_k . If all transactions happened in the domestic market (that is, if there is no difference in currency, so $e_k = 1$), then price is equal to the marginal revenue product divided by the markdown. Second, under perfect competition, $\frac{1}{\epsilon_{bjk}} = 0$ and $\mu_{bjk} = 1$, so the price equals the marginal value of the input. When the buyer has market power, he internalizes the upward-sloping supply of inputs, $\frac{1}{\epsilon_{bjk}} > 0$, so $\mu_{bjk} > 1$ and the input price is marked down from the perfectly competitive level. The steeper the supply curve faced by the buyer (higher $\frac{1}{\epsilon_{bjk}}$), the more market power she has, the higher the markdown μ_{bjk} , and the lower the price, ceteris paribus. Alternatively, the more value the input adds to the final good (higher MRP), the higher the price.

4.4 Buyer Market Power and Supply Elasticity

The elasticity of supply characterizes how prices are determined. Given Bertrand competition,⁴¹ the elasticity of supply has the following closed form.

$$\epsilon_{bjk} = \eta(1 - S_{bjk}) + \theta S_{bjk}, \quad (4.7)$$

³⁹In the empirical application, Colombia is the single observed origin.

⁴⁰Note that this is equivalent to [Berger, Herkenhoff and Mongey \(2022\)](#) and [Felix \(2022\)](#) on labor-market power where the wage is equal to the markdown times the marginal productivity of labor. The intuition is that these buyers avoid purchasing the last few units of a good whose value to them is greater than their marginal cost, just to hold down the price paid for prior units.

⁴¹I focus on Bertrand competition and present results under Cournot competition in Appendix A.1.5.

where

$$S_{bjk} = \frac{p_{bjk}q_{bjk}}{\sum_{b' \in \mathcal{B}_{jk}} p_{b'jk}q_{b'jk}} = \frac{p_{bjk}^{1+\eta}}{\sum_{b' \in \mathcal{B}_{jk}} p_{b'jk}^{1+\eta}} \quad (4.8)$$

is the relative size of Buyer b and Product j in Destination k . This variable is key given that, together with the elasticities, it determines the buyer's market power.

Focusing on equation 4.7, the supply elasticity, ϵ_{bjk} , is the weighted average of the elasticity of substitution across products, j , and across buyers, b , where the relative size of the buyers governs these weights.⁴² Note that the smaller the buyer share, which could relate to a higher level of competition (more buyers per market), the more weight on the substitutability across buyers within a product, η . With many buyers, they exert less influence, and sellers can always switch to other buyers of the same product or input. However, as the number of buyers decreases, the relevance relies on the potential substitution between products, θ .

This yields my first theoretical result. Rearranging equation 4.7, and assuming $\eta > \theta$, I find the elasticity of supply is decreasing in buyer market share, and so the markdown μ_{bjk} is increasing in buyer market share. Therefore, larger buyers have larger markdowns (i.e., they pay prices further below the marginal revenue product).

Intuitively, a higher buyer market share implies fewer alternative buyers available for sellers, limiting their ability to substitute across buyers and thereby reducing sellers' bargaining power. When a buyer accounts for a large share of total demand for a given product, sellers face fewer alternative opportunities and become increasingly dependent on this dominant buyer. As a result, sellers become less responsive to price changes (lower elasticity of supply), enabling the buyer to negotiate greater markdowns. In contrast, when buyers have smaller market shares, sellers retain more flexibility to shift sales across alternative buyers, making supply more elastic and markdowns smaller. Thus, I summarize this result in the following proposition:

Proposition 1 1. *The markdown of Buyer b for Product j in Destination k is increasing in that buyer's market share in the market:*

$$\mu_{bjk} = 1 + [\eta(1 - S_{bjk}) + \theta S_{bjk}]^{-1}; \quad \frac{\partial \ln \mu_{bjk}}{\partial \ln S_{bjk}} > 0. \quad (4.9)$$

2. *The marginal revenue product defined in Equation (4.5) is increasing in the buyer's market share:*

$$MRP_{bjk} = p_k^{finalg} z_{bjk} \left(\frac{q_{bjk}}{Q_{bk}^{finalg}} \right)^{-\frac{1}{\sigma}}; \quad \frac{\partial \ln MRP_{bjk}}{\partial \ln S_{bjk}} > 0.$$

Proof See Appendix A.1.7.

⁴²See Appendix A.1.6 for proof.

4.5 Concentration

In this section, I aggregate my previous results at the market level. Aggregating the right-hand side of equation 4.7 across all firms in a local market, weighting each firm by its buyer market share, gives the key relationship between the degree of buyer market power in the inputs market and its concentration level.

Proposition 2 *Suppose input supply follows a nested CES, and buyers compete for sellers à la Bertrand. The average price markdown in market jk is approximated by*

$$\mu_{jk} \equiv 1 + \epsilon_{jk}^{-1} = 1 + [\eta(1 - HHI_{jk}) + \theta HHI_{jk}]^{-1} \approx \frac{\overline{MRP}_{jk}}{\overline{P}_{jk}} e_k. \quad (4.10)$$

where $\overline{MRP}_{jk}$ and $\overline{P}_{jk}$ are Market jk 's quantity-weighted average marginal revenue product of the input and average price, respectively, $\epsilon_{jk} = \sum_{b \in \mathcal{B}_{jk}} S_{bjk} \epsilon_{bjk}$ is the buyer-share-weighted average market supply elasticity, and $HHI_{jk} = \sum_{b \in \mathcal{B}_{jk}} S_{bjk}^2$ is the market's HHI.

Proof See Appendix A.1.8.

The intuition behind Proposition 2 is straightforward. It demonstrates how the aggregate degree of buyer market power in an input market relates directly to the market's buyer concentration, captured by the Herfindahl-Hirschman Index (HHI). Specifically, when buyer concentration (HHI) is higher, a few dominant buyers control larger shares of the market. This concentration reduces sellers' flexibility in substituting across buyers, decreasing the aggregate elasticity of market supply. A lower supply elasticity allows dominant buyers to extract larger markdowns. Conversely, when the market is less concentrated (lower HHI), sellers have greater flexibility to switch buyers, increasing supply elasticity and thus reducing average markdowns. This aggregate relationship neatly captures how buyer concentration shapes market-wide markdowns through its impact on supply elasticity.

After obtaining an equilibrium price equation and showing how it depends on the markdown,⁴³ I turn now to the relationship between markdowns and price adjustments caused by exchange rate shocks.

4.6 Exchange Rate Pass-through

In this section, I investigate the role of buyer market power in determining the export price response to exchange rate shocks (exchange rate pass-through elasticity). I consider a generic exchange rate shock at the country-pair level, Δe_k , between the home country and destination Country k .

By definition, a bilateral exchange rate shock affects the prices and quantities for all exports in the home country. This means that, after an exchange rate shock, when Buyer b chooses the new price, full efficiency would require considering how the shock affects the prices chosen by all the other buyers of Seller s .⁴⁴ Consistent with my assumption in the buyer profit-maximization problem, I

⁴³In Appendix A.2.2, I show the relationship between size and price level.

⁴⁴Intuitively, by affecting the price and quantities for other buyer-seller pairs, a given shock may affect the price p_{bjk}

assume that when Buyer b chooses the new bilateral price, she takes as given both prices and quantities of all other pairs. The analysis therefore focuses on the direct effect of the shock on the price, p_{bjk} .⁴⁵

The Bertrand assumption—buyers simultaneously choose prices taking competitors' prices as fixed—is standard in oligopsony models and natural here because buyers in different firms or countries face independent procurement decisions and cannot coordinate on quantities. Under Nash conjectures in prices, each buyer's best response is computed holding rivals' strategies fixed, which is precisely the assumption embedded in the firm-level FOC (equation 4.6). The robustness of this assumption is confirmed in Appendix A.1.5, where I solve the Cournot alternative and show that all qualitative results are preserved.

Log-differentiating equation 4.6, and using the result in Proposition 1, I rewrite the log change in price, $d\ln p_{bjk}$, as⁴⁶

$$d\ln p_{bjk} = -d\ln \mu_{bjk} + d\ln MRP_{bjk} + d\ln e_k. \quad (4.11)$$

4.6.1 Direct Effect

In the event of a bilateral exchange rate shock, the resulting change in price implied by expression 4.11 can be decomposed into a direct effect, holding rival buyers' prices fixed, and an indirect effect induced by their responses. In this section, I focus on the direct effect.

The direct effect can be thought of as the overall effect when Seller s is very small relative to the Product j industry in Country k , such that rival buyers' prices remain fixed, and changes in the bilateral exchange rate do not affect aggregate outcomes.

Holding rival buyers' prices fixed and solving for each term, I derive

$$d\ln \mu_{bjk} = \frac{d\ln \mu_{bjk}}{d\ln S_{bjk}} \frac{d\ln S_{bjk}}{d\ln p_{bjk}} d\ln p_{bjk} \quad (4.12)$$

$$= Y_{bjk}(1 + \eta)(1 - S_{bjk}) d\ln p_{bjk} \quad (4.13)$$

$$= \Gamma_{bjk} d\ln p_{bjk}, \quad (4.14)$$

where I define $Y_{bjk} = \frac{\partial \ln \mu_{bjk}}{\partial \ln S_{bjk}} > 0$ as the partial elasticity of the markdown with respect to buyer share S_{bjk} (positive by Proposition 1), and $\Gamma_{bjk} = \frac{d\ln \mu_{bjk}}{d\ln p_{bjk}} > 0$ captures the equilibrium feedback from prices to markdowns. In the case of constant markdowns, $\Gamma_{bjk} = 0$.

through changes in buyer market share. Appendix A.1.11 considers how the pass-through formula would change once these indirect effects are considered. The more general pass-through formula can be derived by solving a complex system of equations for each Seller s .

⁴⁵I validate this assumption in the empirical section, where I show that the effect of the country-pair-level shock to the bilateral price is unchanged regardless of whether or not the quantities or prices of other buyers in the same Product j and Destination k are controlled for in the estimation.

⁴⁶Note that I am dropping the subindex s ; I will assume sellers are homogeneous in the prices they receive from the buyers so I can isolate the buyer effects. In my empirical part, I control for differences in two different sellers connected with the same buyer.

For the marginal-revenue term, define the CES input share as

$$x_{bjk} \equiv \frac{z_{bjk} q_{bjk}^{\frac{\sigma-1}{\sigma}}}{\sum_{j' \in \mathcal{J}} z_{bj'k} q_{bj'k}^{\frac{\sigma-1}{\sigma}}}. \quad (4.15)$$

Using this input share, we can rewrite the marginal revenue product as $MRP_{bjk} = p_k^{final} z_{bjk}^{\frac{\sigma}{\sigma-1}} x_{bjk}^{-\frac{1}{\sigma-1}}$. We can then derive the log change in the marginal revenue product as

$$d \ln MRP_{bjk} = \frac{d \ln MRP_{bjk}}{d \ln x_{bjk}} \frac{d \ln x_{bjk}}{d \ln q_{bjk}} \frac{d \ln q_{bjk}}{d \ln p_{bjk}} d \ln p_{bjk} \quad (4.16)$$

$$= -\frac{1}{\sigma-1} \frac{\sigma-1}{\sigma} (1-x_{bjk}) \epsilon_{bjk} d \ln p_{bjk} \quad (4.17)$$

$$= -\frac{1}{\sigma} (1-x_{bjk}) \epsilon_{bjk} d \ln p_{bjk} \quad (4.18)$$

$$= \Phi_{bjk} d \ln p_{bjk}, \quad (4.19)$$

where σ is the elasticity of substitution in the CES production function and ϵ_{bjk} is given by Equation (4.7). When the buyers' production technology has constant returns to scale and all inputs adjust proportionally, $d \ln MRP_{bjk} = 0$.

Substituting equations 4.12–4.16 into 4.11, it is possible to write the log change in the price, p_{bjk} , for each Buyer b in Product j and Destination k as a function of the buyer's market share, S_{bjk} , and fundamentals.

$$d \ln p_{bjk} = -\Gamma_{bjk} d \ln p_{bjk} + \Phi_{bjk} d \ln p_{bjk} + d \ln e_k \quad (4.20)$$

Proposition 3 characterizes the direct component of the pass-through of an exchange rate shock into the price, p_{bjk} .

Proposition 3 *The pass-through of a bilateral exchange-rate shock to p_{bjk} , holding rival buyers' prices fixed ($d \ln p_{b'jk} = 0$ for all $b' \in \mathcal{B}_{jk} \setminus \{b\}$), is*

$$\frac{d \ln p_{bjk}}{d \ln e_k} = \frac{1}{1 + \underbrace{\Gamma_{bjk}(\eta, \theta, S_{bjk})}_{\text{Markdown channel}} - \underbrace{\Phi_{bjk}(\varphi_{bjk}, S_{bjk})}_{\text{Marginal Revenue Channel}}}, \quad (4.21)$$

where $\Gamma_{bjk} = Y_{bjk}(1+\eta)(1-S_{bjk}) > 0$ and $\Phi_{bjk} = \varphi_{bjk}\epsilon_{bjk} < 0$, with $\varphi_{bjk} = -(1-x_{bjk})/\sigma$.

Since $\Gamma_{bjk} > 0$ and increases with buyer market share (Corollary 1), while $-\Phi_{bjk} > 0$, the denominator exceeds one, yielding pass-through below one. Both channels reduce pass-through, but the markdown channel—which grows with buyer market share—is empirically dominant.

Proof See Appendix A.1.9.

Equation 4.21 shows that pass-through can be written as a function of buyer share and the parameter vector $v = \{\eta, \theta, \sigma\}$. In Appendix A.1.11, I explore the indirect effect.⁴⁷

⁴⁷I also explore a numerical simulation with two firms in Appendix A.1.15.

4.6.2 Aggregate Level

In this section, I derive market-level exchange-rate pass-through. Using Proposition 6 in Appendix A.1.11, I calculate the average exchange-rate pass-through by sector and destination in terms of the HHI.

Proposition 4 *The average exchange-rate pass-through is given by*

$$\psi_{jk} = \frac{d\ln P_{jk}}{d\ln e_k} = \frac{1}{1 + \tilde{\Gamma}_{jk}(\eta, \theta, HHI_{jk}) - \tilde{\Phi}_{jk}(\varphi_j, HHI_{jk})} + \frac{\tilde{\Gamma}_{jk}(\eta, \theta, HHI_{jk}) - \tilde{\Phi}_{jk}(\varphi_j, HHI_{jk})}{1 + \tilde{\Gamma}_{jk}(\eta, \theta, HHI_{jk}) - \tilde{\Phi}_{jk}(\varphi_j, HHI_{jk})} \frac{d\ln P_k}{d\ln e_k} + \frac{\widetilde{MRP}_q}{1 + \tilde{\Gamma}_{jk}(\eta, \theta, HHI_{jk}) - \tilde{\Phi}_{jk}(\varphi_j, HHI_{jk})} \frac{d\ln q_k}{d\ln e_k}. \quad (4.22)$$

where $\tilde{\Gamma}_{jk} = d\ln \mu_{jk} / d\ln e_k$ and $\tilde{\Phi}_{jk} = d\ln MRP_{jk} / d\ln e_k$.

Proof See Appendix A.1.12.

4.7 Channels

In this section, I decompose the overall exchange rate pass-through effect into markdown and marginal-revenue channels. From my theoretical model, I derive an expression to quantify each of these channels in the empirical part in Section 5.2.

4.7.1 Markdown Channel

The markdown channel is driven by the endogenous response of buyer market share—and therefore markdowns—to the exchange-rate shock. Following a positive exchange-rate shock ($\uparrow e_k$, a depreciation of the seller's currency), the seller-currency price p_{bjk} begins to rise. As p_{bjk} rises relative to the product-level price index P_{jk} , the buyer's market share S_{bjk} increases (more sellers are attracted to the buyer offering a higher relative price). By Proposition 1, $\frac{\partial \ln \mu_{bjk}}{\partial \ln S_{bjk}} > 0$, so the higher market share raises the markdown μ_{bjk} . This higher markdown exerts downward pressure on p_{bjk} —captured by the term $-\Gamma_{bjk} d\ln p_{bjk}$ in Equation (4.20)—keeping it from rising proportionally with e_k . In this way, buyers *increase* their markdowns following a depreciation, absorbing part of the exchange-rate gain at the seller's expense and stabilizing seller-currency prices.

The key theoretical result of the model is that, at the firm level, the markdown channel Γ_{bjk} is an increasing function of buyer market share. Therefore, the markdown channel operates differently for buyers with different market shares: higher market share leads to more-variable markdowns. Intuitively, buyers with higher market share already have higher markdowns—they pay prices farther below the marginal revenue product. Given this, they have more *scope* to further increase their markdown in response to an exchange-rate shock. After a depreciation, larger buyers increase μ_{bjk} by more (larger Γ_{bjk}), absorbing a greater share of the exchange-rate gain, which is why they exhibit lower pass-through into seller-currency prices.

To identify the magnitude of this effect and formally analyze each component of this channel, I focus on a direct implication of Proposition 1.

Corollary 1 *The markdown channel under fixed rival prices is*

$$\Gamma_{bjk} = \frac{\partial \ln \mu_{bjk}}{\partial \ln p_{bjk}} = \frac{(\eta - \theta)(1 + \eta)S_{bjk}(1 - S_{bjk})}{[1 + \eta(1 - S_{bjk}) + \theta S_{bjk}][\eta(1 - S_{bjk}) + \theta S_{bjk}]} > 0.$$

If the cross-product elasticity of substitution is lower than the within-product cross-buyer elasticity, $\eta > \theta > 1$, then firms with higher S_{bjk} have more-variable markdowns:

$$\frac{\partial \Gamma_{bjk}}{\partial S_{bjk}} > 0.$$

Proof See Appendix A.1.10 for details.

The chain of events implied by Propositions 1–3 and Corollary 1 can be understood as follows. A positive exchange-rate shock (a depreciation) raises the buyer’s valuation of the input in the seller’s currency, placing upward pressure on p_{bjk} , as required by Equation 4.6. As p_{bjk} rises relative to the market benchmark, the buyer attracts more inputs and its share increases ($\partial \ln S_{bjk} / \partial \ln p_{bjk} > 0$). The higher share lowers the supply elasticity it faces ($\partial \epsilon_{bjk} / \partial S_{bjk} = \theta - \eta < 0$) and raises its markdown ($\partial \ln \mu_{bjk} / \partial \ln S_{bjk} > 0$). The buyer therefore retains part of the exchange-rate gain, restraining the increase in p_{bjk} and generating incomplete pass-through ($d \ln p_{bjk} / d \ln e_k < 1$). Moreover, because the markdown response increases with buyer share ($\partial \Gamma_{bjk} / \partial S_{bjk} > 0$), higher-share buyers exhibit lower pass-through through this channel.⁴⁸

The exchange-rate shock also operates through the marginal-revenue channel. As the higher price attracts more inputs, the quantity purchased by the buyer increases. Because inputs have diminishing marginal returns, this lowers their marginal revenue product ($\Phi_{bjk} < 0$) and reduces the buyer’s willingness to raise p_{bjk} . Thus, both channels attenuate pass-through: the markdown channel operates through buyer market power, whereas the marginal-revenue channel operates through diminishing marginal returns. This second channel is explored in detail in the following subsection.

4.7.2 Marginal-Revenue Channel

The marginal-revenue channel captures the price response due to changes in the buyer’s average revenue. When the bilateral price increases due to a positive exchange rate shock, a standard supply effect leads the seller to supply more of that variety. Higher supply decreases the marginal revenue, decreasing the price.

Rearranging equation 4.16, The marginal-revenue channel takes the following form.

$$\frac{d \ln MRP_{bjk}}{d \ln p_{bjk}} = \frac{-1}{\sigma} (1 - x_{bjk}) \epsilon_{bjk} = \Phi_{bjk} < 0$$

The marginal-revenue channel depends on (i) σ , the parameter for elasticity of substitution in

⁴⁸This feedback requires $\eta > \theta$. If $\eta = \theta$, supply elasticity is independent of buyer share and the variable-markdown channel disappears ($\Gamma_{bjk} = 0$). If the common elasticity also approaches infinity, the markdown converges to one and buyer market power disappears. These limiting cases are analogous to the labor-market settings in Berger, Herkenhoff and Mongey (2022) and Felix (2022).

the buyer's CES production function, (ii) x_{bjk} , the share of input j in the buyer's production costs, and (iii) the elasticity of supply.⁴⁹ I interpret how each parameter contributes to this channel.

First, the higher the σ , the more substitutability between products in the production function and the less relevant the marginal-revenue channel. In the extreme, if $\sigma \rightarrow \infty$, then every input has a close substitute either from another seller in that same country or in another country and there is no differential marginal-revenue effect for larger buyers, because there is no marginal-revenue effect at all.⁵⁰

Second, a higher x_{bjk} yields a more-relevant product for the buyers' production. If $x_{bjk} = 1$, input j is the only input and the marginal revenue will be constant, where increasing one unit of the input will increase the marginal revenue the same amount. If that were the case, then the buyer's market share would be irrelevant for this channel because, again, this channel would be shut down.

Third, a higher elasticity of supply yields a bigger marginal-revenue channel. Note that this is the only term in the marginal-revenue channel that depends on the buyer market share. As the elasticity of supply is smaller for bigger buyers, the bigger the buyer, the less substantial the revenue (and price) decrease.

5 Estimation

In this section, I map the empirical moments from Facts II and III into the model and estimate the parameters governing the markdown and marginal-revenue channels using indirect inference. I then recover the key elasticities—across products and across buyers within products—that discipline the strength of the markdown channel.

5.1 Exchange Rate Pass-through

Consider a sudden change in the bilateral exchange rate between the home country and Destination k . Below, I analyze the degree to which the exchange rate shock is passed on to international prices depending on buyer market power.

I derive the pass-through regression by linearizing the theoretical relationship between buyer market share and pass-through in equation 4.21, following the approach of [Amiti, Itskhoki and Konings \(2014\)](#), but here focusing on buyer market power instead of import intensity. This provides tractable coefficients that can be mapped into the data and guides the empirical specification.⁵¹

⁴⁹As shown in equation 4.16, the marginal-revenue channel depends on the buyer's production function because it is related to how the product bought is used for production. In my baseline model, I propose a CES production function, but I solve for alternative specifications in Appendix A.1.16.

⁵⁰If the production function were Cobb Douglas, then $\sigma = 1$. This case is explored in Appendix A.1.16.

⁵¹The first-order approximation omits terms of order S_{bjk}^2 and higher. Whether the linear approximation is adequate depends on whether the pass-through function $\frac{1}{1+\Gamma_{bjk}-\Phi_{bjk}}$ exhibits low curvature over the empirically relevant range of buyer shares—not on whether shares themselves are small. Indeed, a central finding of this paper is that buyer shares are large and economically significant. As a validation, the SMM estimation in Section 5 recovers structural parameters by matching the empirical $\hat{\alpha}$ and $\hat{\beta}$ using the full nonlinear model; the close fit between model and data moments (Table on parameters) confirms that the linear specification adequately captures the pass-through variation across the observed distribution of buyer shares.

5.1.1 Linearization

To estimate the effect of an exchange rate shock on prices for buyers with different market shares, after linearizing on the buyer market share, S_{bjk} , I calculate a first-order approximation, replace the differential d with a time difference Δ ⁵² and rearrange to derive Proposition 5.⁵³

Proposition 5 1. *The first-order approximation to the exchange rate pass-through elasticity into prices in seller currency for Buyer b in Product j and Destination k is given by*

$$\psi^*_{bjkt} \approx \mathbb{E} \left[\frac{\Delta p_{bjkt}}{\Delta e_{kt}} \right] = \alpha_{bjkt} + \beta_{bjkt} S_{bjkt}. \quad (5.1)$$

2. *The first-order approximation to the exchange rate pass-through elasticity into producer currency export prices for Product j and Destination k is given by*

$$\psi^*_{jkt} \approx \mathbb{E} \left[\frac{\Delta P_{jkt}}{\Delta e_{kt}} \right] = \alpha_{jkt} + \beta_{jkt} HHI_{jkt}, \quad (5.2)$$

where HHI_{jkt} corresponds to the concentration of that sector in that destination.

Proof See Appendix A.1.13.

The pass-through elasticity, ψ^*_{bjkt} , measures the buyer-product-destination price's equilibrium log change relative to the log change in the bilateral exchange rate, averaged across all possible states of the world and economic shocks. Proposition 5 relates firm-level pass-through to buyer market share, which forms a sufficient statistic for cross-sectional variation in pass-through within the product-destination universe. Furthermore, Proposition 5 provides closed-form expressions for coefficients α_{jk} and β_{jk} .

The values of the coefficients in this relationship (α_{jk} and β_{jk}) can be estimated in the data. Starting from Proposition 5, I arrive at my main empirical specification at the firm level (Fact II), where I regress the annual change in the log export price on the change in the log exchange rate, interacted with the buyer market share in equation 5.1. I also arrive at the regression at the market-level (Fact III), that corresponds to equation 5.2.

5.2 Estimation of the Markdown Channel

In the model, two key elasticities govern market power and so, the magnitude of the markdown channel: the elasticity of substitution across products, θ , and the elasticity of substitution within product, across buyers, η . In this section, I describe an approach that integrates (i) new empirical

⁵²I define $\Delta A_t = \ln A_t - \ln A_{t-1}$

⁵³While Fact 3 documents the raw reduced-form correlation between buyer size and pass-through, equation 5.1 arises from the theoretical framework and delivers an estimable regression at the firm-buyer-product level. This specification imposes the structure implied by the model and includes the fixed effects required by theory (e.g., product-destination and time-seller effects), so that the coefficients α and β can be interpreted as the empirical counterparts of the model parameters. In this sense, equation 3.3 is not simply a re-statement of Fact 3, but the theory-consistent regression that links the micro data to the model and provides the sufficient statistics used in the quantitative analysis.

estimates using bilateral exchange rate shocks (see Section 3.2) and (ii) new moments from the cross-section, into (iii) a simulated method of moments routine in which all unknown parameters are estimated jointly.

5.2.1 Challenges for Estimation

Equation 4.21 shows that the pass-through term, $\frac{dp_{bjk}}{de_k}$ is a function of three parameters— η , θ , σ —and S_{bjk} . Once I linearize on buyer market shares, S_{bjk} , I have two coefficients (equation 5.1) which I obtain from running the regression in the data. The sizes of the coefficients β and α are informative on the magnitudes of the elasticities θ and η . However, I cannot disentangle them from the effect of the marginal revenue, φ . This is a well-known issue in the markup literature (De Loecker and Warzynski, 2012), which is usually addressed by estimating the production function and backing out market power.⁵⁴ Instead, I combine the elasticities from the empirical part with moments from the cross section and use the structure of the model to estimate η and θ directly, along with other parameters.

5.2.2 Indirect Inference

I recover the parameters of the model through indirect inference implemented as simulated method of moments (SMM). I estimate all parameters jointly, but outline the estimation procedure separately for each group of parameters. Appendix A.3 provides further details. For this exercise, I follow a top-down approach done for labor markets in Berger, Herkenhoff and Mongey (2022) and agriculture intermediaries in Zavala (2022).

Estimates for η and θ To estimate η and θ , I proceed in three steps: (1) Estimate equation 5.1 in the actual data. (2) Simulate equation 5.1 in the model. (3) Pick η and θ so that the coefficients α and β from the model match their counterparts in the data.

I estimate equation 5.1 in the actual data already in section 3.2 and obtain $\hat{\alpha}$ and $\hat{\beta}$. To simulate equation 5.1, I use the following procedure. First, I draw z_{bjk} from an exogenous distribution.⁵⁵ For each guess of η and θ , I solve the model. Next, I shock the prices by drawing from the distribution of bilateral exchange rate shocks. I solve the model again to create a simulated panel, treating the outcomes across these two model economies as panel data. The resulting exchange rate pass-through coefficients, denoted $\beta(\eta, \theta)$ and $\alpha(\eta, \theta)$, are functions of η and θ .

I pick η and θ so that the pass-through coefficients estimated from the simulated data match the coefficients I estimated from the actual data (coefficients from Table 2) such that

$$(\hat{\eta}, \hat{\theta}) = \operatorname{argmin}_{\eta, \theta} \{ \|\hat{\alpha} - \alpha(\eta, \theta)\| + \|\hat{\beta} - \beta(\eta, \theta)\| \}.$$

⁵⁴Another typical problem for the estimation of the elasticity of supply (and so the markdown) is that, when firms behave strategically, the structural elasticity cannot be measured using how prices respond to a well-identified shock. The structural elasticity is a partial equilibrium concept answering the counterfactual: How much firms change supply, holding their competitors' q_{sbjk} constant. The reduced-form elasticity includes all other firms' responses.

⁵⁵This is needed to have nonsymmetric buyer market shares.

Estimate for σ I take advantage of the data available for Colombia and use cross-sectional moments to estimate parameters that govern the marginal revenue product. Holding η and θ fixed, I use the ratio between prices of different products from the same buyer to derive an expression that can be estimated through regression using the available data, and its coefficient becomes a function of σ , specifically $\frac{1}{\sigma}$.⁵⁶

$$\frac{p_{bjk}}{p_{bj'k}} = \frac{\varepsilon_{bjk} (1 + \varepsilon_{bj'k})}{(1 + \varepsilon_{bjk}) \varepsilon_{bj'k}} \frac{z_{bjk}}{z_{bj'k}} \left(\frac{q_{bj'k}}{q_{bjk}} \right)^{\frac{1}{\sigma}}$$

$$\ln \left(\frac{p_{bjk}}{p_{bj'k}} \right) = \underbrace{\ln \left(\frac{\varepsilon_{bjk} (1 + \varepsilon_{bj'k})}{(1 + \varepsilon_{bjk}) \varepsilon_{bj'k}} \right)}_{\omega_0} + \ln \left(\frac{z_{bjk}}{z_{bj'k}} \right) + \underbrace{\frac{1}{\sigma}}_{\omega_1} \ln \left(\frac{q_{bj'k}}{q_{bjk}} \right)$$

External Parameters: ε_k , z_{bjk} and Others I assume that (log) buyer productivity, $\log z_{bjk}$, and log changes in exchange rate shocks, Δe_k , follow normal distributions.

$$\log z_{bjk} \sim N(\mu_z, \sigma_z^2) \text{ and } \ln \Delta e_k \sim N(\mu_e, \sigma_e^2)$$

For buyer productivity, I choose (μ_z, σ_z^2) to match the distribution of buyers' market shares.⁵⁷ For bilateral exchange rate shocks, I choose (μ_e, σ_e^2) to match the distribution of log changes in the bilateral exchange rate in the data. Finally, the numbers of products and buyers are also chosen to match the data from Colombia.

5.2.3 Parameter Estimates

Table 6: Summary of parameters

Parameter	Description	Value	Moment	Model	Data
A. Assigned					
N	Number of products	6,983			
M_j	Number of buyers per product	17			
μ_{e_k}	Mean of $\ln ER$ changes	0.03			
$\sigma_{e_k}^2$	SD of $\ln ER$ changes	0.1			
B. Estimates					
θ	Across-product substitutability	5.14	Baseline pass-through $\hat{\alpha}$	0.181	0.175
η	Across-buyer substitutability	7.17	Interaction buyer share $\hat{\beta}$	-0.243	-0.241
σ	Input substitutability	0.3	Relative price level	0.012	0.013
z_{bjk}	Productivity shifter	0.05	Average firm size	0.21	0.23

Notes: Panel A reports externally assigned parameters. Panel B reports jointly estimated SMM parameters. "Model" and "Data" report simulated and empirical target moments, respectively.

⁵⁶In Appendix A.3.1, I explain the detailed procedure for this.

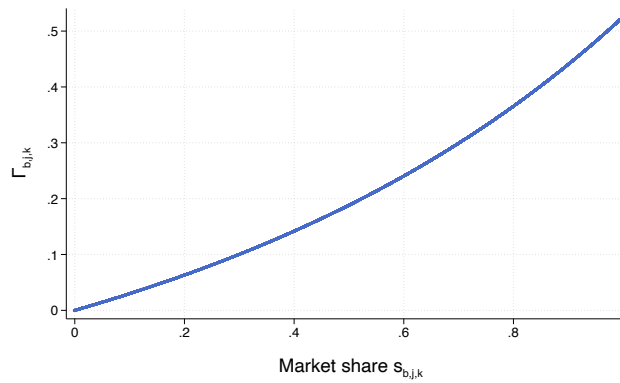
⁵⁷For the productivity distribution, I follow Zavala (2022). However, results remain very similar when using a Pareto distribution. While the Pareto specification can better capture firm selection dynamics, such as entry and exit decisions, it often produces a highly skewed distribution, with a few dominant buyers holding disproportionately large market shares. In contrast, the normal distribution provides a better, more conservative fit by smoothing the extremes and avoiding excessive concentration of market power. This leads to more balanced market share distributions and makes the model more robust to outliers, while still preserving the key relationships in buyer market power and pass-through.

In Appendix A.3, I include the procedure for calculating the standard errors, sensitivity analysis, and proof of uniqueness.

5.2.4 Quantifying the Markdown Channel

Using the estimates for η and θ together with the corresponding S_{bjk} from my data, I calculate the implied markdowns Colombian exporters face. I find the average markdown to be 16%. Then, using the structural equation from the model, I quantify the markdown channel. Figure 4 shows the markdown channel Γ_{bjk} is bigger for buyers with larger market shares after substituting the estimated elasticities.

Figure 4: Markdown Channel and ERPT varies with size



Notes: Figure plots buyer market share on the x-axis and the markdown channel Γ_{bjk} on the y-axis. A higher Γ_{bjk} means that following a depreciation, the buyer increases its markdown by more, absorbing a larger share of the exchange-rate gain. Buyer market share is defined as the share of the market, defined as destination country $\times$ product $\times$ year, purchased by a given buyer.

Figure 4 reveals a positive relationship between the responsiveness of the markdown and the market share. That is, firms with larger market shares make larger adjustments to their markdowns when faced by a price shock—they increase μ_{bjk} by more after a depreciation.

6 Counterfactual: Eliminating Buyer Market Power

To explore the aggregate implications of buyer market power for the sellers in Colombia, I propose a counterfactual where I eliminate buyer market power. Moving from an oligopsony structure to perfect competition with no strategic interactions implies not only changes in the level of revenues but also in the volatility of these revenues.

6.1 Level Effect

I simulate an alternative model in which buyers lack market power. Under perfect competition, buyers still face upward-sloping supply curves, whose shapes are determined by the cross-product elasticities of substitution (η) and within-product cross-buyer elasticity of substitution (θ). However, they do not internalize their influence over the price. Rather, they perceive a perfectly elastic supply

curve ($\epsilon_{bjk} = \infty$), so $\mu_{bjk} = 1$. Input prices are no longer marked down from their marginal-revenue product.

By comparing the sellers' revenues in both versions of the economy, I aim to estimate the welfare losses caused by buyers' markdowns. The total impact of buyer market power is the log difference in sellers' revenues between the two scenarios:⁵⁸

$$\text{Total Effect} = \log \sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^C q_{bjk}^C - \log \sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^M q_{bjk}^M, \quad (6.1)$$

where $q_{bjk} \equiv \int_0^1 q_{sbjk} ds$ is the aggregate quantity supplied to Buyer b in market (j, k) .

To perform the welfare analysis, I first aggregate the economy's revenues with and without market power. Then, I calculate the percentage change by dividing the change in aggregated revenues by the revenues with market power. To estimate prices without market power, I re-scale the simulated prices by the markdown, obtaining hypothetical prices in the absence of monopsony, such that $p_{bjk}^C = p_{bjk}^M \mu_{bjk}$. Using these prices, I determine the corresponding quantities to calculate the revenues. The welfare effect is thus described by the following equations and results:

$$\text{Total Effect} = \frac{\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^C q_{bjk}^C - \sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^M q_{bjk}^M}{\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^M q_{bjk}^M} \quad (6.2)$$

The change in the sellers' revenues can be decomposed into a quantity effect and a price effect.⁵⁹ I first simulate the model with and without market power to quantify these effects. To differentiate the total effect between the price and quantity effects, I utilize the following expression:

$$\underbrace{\log \left(\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^{\text{PerfComp}} \hat{q}_{bjk}^{\text{Olig}} \right) - \log \left(\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^{\text{Olig}} \hat{q}_{bjk}^{\text{Olig}} \right)}_{\text{Price effect}} + \underbrace{\log \left(\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^{\text{PerfComp}} \hat{q}_{bjk}^{\text{PerfComp}} \right) - \log \left(\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^{\text{PerfComp}} \hat{q}_{bjk}^{\text{Olig}} \right)}_{\text{Quantity effect}} \quad (6.3)$$

6.1.1 Price Effect

The price effect corresponds to the increase in price when removing markdowns; sellers earn higher revenues for supplying the same product to the same buyer. This effect can be thought of as a redistribution from buyers to sellers. To measure this effect, I calculate sellers' revenue using

⁵⁸The model simulation's approach to scaling quantities by dividing them by Q_k limits my ability to estimate the effects as previously described fully. Consequently, all revenues are adjusted by Q_k , where Q_k represents the equilibrium quantity for total import payments P_k in economy k for an additional unit of aggregate import price. I use $\hat{q}_{bjk} \equiv \frac{q_{bjk}}{Q_k}$.

⁵⁹My analysis shares the spirit of [Edmond, Midrigan and Xu \(2023\)](#), which accounts for the costs of markups.

quantities from the oligopsony model-baseline and prices from the perfect-competition counterfactual.

$$PriceEffect = \frac{\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^C q_{bjk}^C - \sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^M q_{bjk}^C}{\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^M q_{bjk}^M} \quad (6.4)$$

6.1.2 Quantity Effect

The quantity effect corresponds to efficiency gains. In the model, sellers trade off the price of a given buyer and a given product with their idiosyncratic shock for producing that product and supplying that buyer. This leads to misallocation: Some sellers do not produce the product in which they are most productive simply because its price index is very low. Conditional on a product, some sellers do not supply the buyers with lower information frictions to connect with them, simply because their prices are too low. Once buyer market power is removed, the trade-off lessens and allows sellers to produce the product they are most productive on and supply their buyer with lower search costs/frictions.

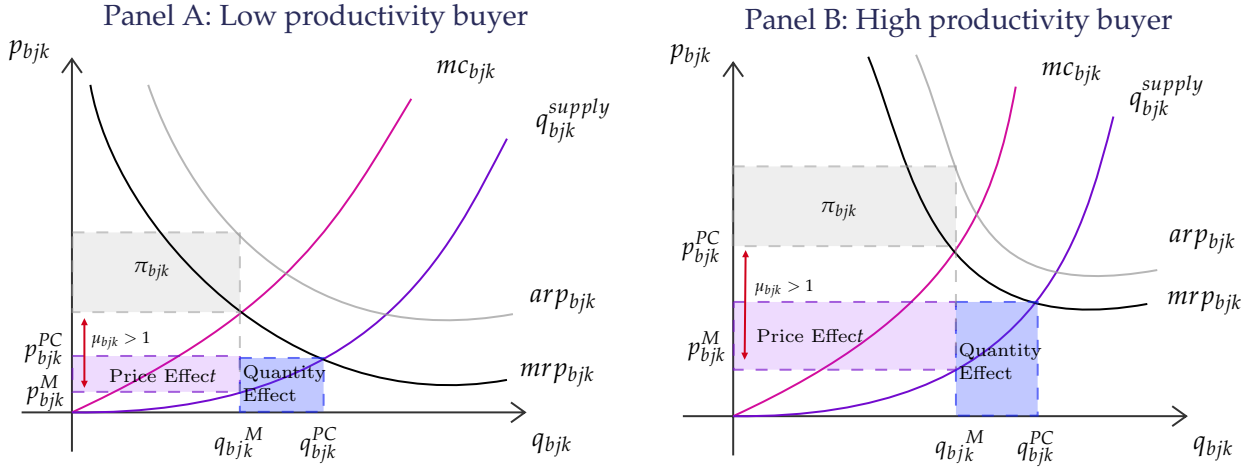
$$QuantityEffect = \frac{\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^M q_{bjk}^C - \sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^M q_{bjk}^M}{\sum_{(b,j,k) \in \mathcal{A}} p_{bjk}^M q_{bjk}^M} \quad (6.5)$$

6.1.3 Quantifying the Total Effect, Price Effect and Quantity Effect

The magnitude of the quantity effect and price effect varies by the market shares of the buyer. Figure 5 shows graphically both the price and quantity effect for a large buyer relative to a small one. Panel A shows a low-productivity buyer scenario, where the price markdown from buyer market power mainly redistributes revenue from sellers to the buyer, without substantially affecting the equilibrium quantity. The quantity effect, reflecting allocative inefficiency, remains relatively minor compared to the price effect. In contrast, Panel B depicts a high-productivity buyer with a larger market share, resulting in more pronounced markdowns and significant reductions in equilibrium quantity. Here, buyer market power not only redistributes revenue but also creates meaningful allocative inefficiencies, evident from a larger quantity effect.⁶⁰

⁶⁰This highlights that productivity differences among buyers, which determine market shares, shape the welfare implications of market power predominantly through price effects. Although the overall efficiency loss may appear modest, it remains economically significant, consistent with magnitudes commonly discussed in macroeconomic welfare analyses.

Figure 5: Price and Quantity Effect



Notes: This figure shows the price and the quantity effect from comparing with a case of perfect competition depending on the productivity of the buyer, which in turn determines the buyer market share.

I find that sellers' revenues would be 0.2% higher in the absence of market power. Redistribution from buyers to sellers increases income by 0% and efficiency gains would increase sellers' revenues by 0.199%.⁶¹ Therefore, the markdowns do not stop sellers from picking the efficient product-seller combination but they still lose a share of their revenues as a result of the lower prices they face. These revenues are weighted; they represent the value of the buyer's share in the aggregated Q increase given an increase in the aggregated price. Since productivity is the main driver of shares' differences, they remain more or less constant with or without market power, which explains why the effect comes from the price and not so much from the quantities.

6.2 Effect on Δ in Revenues with Exchange Rate Shocks

In the rest of this section, I quantify the welfare effects of buyer market power by comparing the volatility of the revenues faced by the sellers in an oligopsony structure to those in perfect competition.

To this end, consider a single consumer, a seller in the home country, whose income is equal to the revenues y_t obtained from exporting their products. Revenues from exports are the only source of income such that $c_t = y_t$. At the same time, revenues in each period follow a random walk:

$$Y_t = Y_{t-1} e^{\mu} e^{-1/2\sigma^2} \epsilon_t,$$

where $\ln(\epsilon_t)$ is a normally distributed random variable with mean 0 and variance σ^2 . I assume a random walk structure for firm revenues primarily for analytical simplicity and transparency. This assumption allows the analysis to focus directly on volatility stemming from exchange rate shocks rather than predictable trends or cycles. While alternative processes (e.g., AR(1) or structural VARs) might capture some observed regularities more realistically, the random walk assumption is standard (Lucas, 2003) and enables straightforward quantification of welfare implications. Under

⁶¹While all sellers benefit from perfect competition, the gains are not distributed equally: markets with higher baseline levels of buyer market concentration experience greater increases.

these assumptions $E(e^{-(1/2)\sigma^2}\epsilon_t) = 1$. Preferences over such consumption paths are assumed to be

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \frac{((1+\lambda)C_t^{PerfComp})^{1-\gamma}}{1-\gamma} \right], \quad (6.6)$$

where β is a subjective discount rate, γ is the coefficient of risk aversion, and the expectation is taken with respect to the distribution of shocks ϵ_t .

I compare the utility difference for a seller in an oligopsony structure with one in perfect competition. An extremely risk-averse consumer would prefer the case in oligopsonistic competition. I quantify this utility difference by multiplying the perfect competition path by a constant factor $1 + \lambda$ in all dates and states, choosing λ so the seller is indifferent between the oligopsony and the compensated perfect-competition path. Therefore, λ is chosen to solve

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \frac{((1+\lambda)C_t^{PerfComp})^{1-\gamma}}{1-\gamma} \right] = \mathbb{E} \left[\sum_{t=0}^{\infty} \frac{\beta^t}{1-\gamma} C_t^{Olig(1-\gamma)} \right] \quad (6.7)$$

Canceling, taking logs and collecting terms gives

$$\lambda \approx \frac{1}{2} \gamma (\sigma_{PerfComp}^2 - \sigma_{Oligop}^2) \quad (6.8)$$

Note that the compensation parameter λ —the welfare gain from eliminating volatility from buyer market power—depends on three terms: the risk-aversion parameter γ , the amount of risk present in each case $\sigma_{PerfComp}^2$ and σ_{Oligop}^2 . The last two terms correspond to variance of the ϵ_t for each case.

To estimate λ , estimates of these parameters are needed. As both scenarios have a different variance for their change in revenues, I use the structural model to connect pass-through to revenue volatility. From Proposition 5, the exchange-rate component of revenue variance equals the squared pass-through elasticity times the variance of exchange-rate shocks. Specifically,

$$\text{var}[\Delta Y_t] = (\hat{\alpha} + \hat{\beta} S_{b,j,k,t})^2 \text{var}[\Delta e_{k,t}] \quad (6.9)$$

I substitute $S_{b,j,k,t} = 0$ for the perfect competition case and $S_{b,j,k,t} = 1$ for the oligopsony case, obtaining 0.0008 and 0.000003, respectively. Seeing this agent as a representative seller in a developing country, I use estimates of the coefficient of risk aversion, γ . In macroeconomics and finance, this coefficient ranges from 1 (lowest risk aversion) to 5 (highest risk aversion). I pick a coefficient of 1 and calculate λ :

$$\lambda \approx 0.0004.$$

From the literature, these welfare losses of a monetary-policy regime are low, but they are on the order of magnitude of Lucas (2003). Comparing this lambda with Lucas (2003) (0.0005), I can conclude buyer market power accounts for 80% of the costs of welfare related to eliminating the whole business cycle in the U.S. when evaluating the volatility of the sellers' income in Colombia.⁶²

Taking into account both effects, the level effect on prices and the volatility of revenues, I can

⁶²This will be quantifying only one of the sources of risk that compose the whole cost of business cycles.

quantify how the key parameters—the elasticities, risk aversion, and utility function—shape the welfare effects of buyer market power on sellers.

7 Conclusion

This paper studies buyer market power in international markets and its impact on the exchange rate pass-through. I combine a novel transaction-level dataset covering the universe of Colombian exports that crucially contains information on the identity of the foreign buyer for the period 2007–2020 with an oligopsony model of buyer market power in international trade. The main conclusion is that buyer market power significantly shapes exchange rate pass-through, introducing a novel source of pass-through dispersion across buyers and a quantitatively important trade-off between price levels and exchange-rate insulation.

Theoretically, buyer market power has implications for price determination and for how these prices react to exchange rates. First, buyers with higher market share have a higher markdown μ , and so a lower price, all else equal. Second, buyer market power has consequences for the exchange rate pass-through. The overall effect is driven by two mechanisms that partially offset each other in how pass-through varies with buyer size: a markdown channel and a marginal-revenue channel. Both reduce pass-through, but the markdown channel grows with buyer size while the marginal-revenue channel shrinks, making the markdown channel empirically dominant.

My empirical strategy focuses on estimating the Colombian exports' pass-through elasticity to the rest of the world. At the firm level, my findings suggest that bigger buyers pay lower prices, and have a lower exchange rate pass-through to sellers' currency, ranging from 1% for the largest to 15% for the smaller buyers. The mechanism behind this is that large buyers' greater market power leads to *more* variable markdowns: following a depreciation, large buyers increase their markdown μ_{bjk} by more (larger Γ_{bjk}), absorbing a greater share of the exchange-rate gain at the seller's expense, which prevents seller-currency prices from rising proportionally. At the market level, in markets where buyers are more concentrated, prices have higher markdowns and exchange rate pass-through in seller currency is lower.

Finally, I calibrate the model and obtain key elasticities that allow me to simulate a counterfactual scenario where buyers have no market power. Under this scenario, sellers receive higher prices but their revenues are more responsive to exchange rate shocks. In this setting, seller currency depreciations are much more common than appreciations. On balance, sellers are less likely to benefit from reduced volatility than to be disadvantaged by attenuated revenue gains during depreciation episodes.

This paper has important policy implications for sellers from developing countries who sell their products to large firms. Even though when selling to a large firm they might receive marked-down prices, these prices are more stable during exchange rate shocks—because large buyers absorb the shock by increasing their markdowns. Countries in Latin America frequently experience depreciations, so this mechanism prevents sellers from capturing the full gains of favorable exchange rate movements. On the other hand, multinationals abroad might find it less appealing to connect with small sellers.

The findings of this paper carry a broader macroeconomic implication. Buyer concentration in

global trade has risen substantially over recent decades ([Autor et al., 2020](#); [De Loecker, Eeckhout and Unger, 2020](#)), and the mechanism documented here suggests that this trend may have contributed to the secular decline in exchange-rate pass-through observed across countries. As import markets become increasingly dominated by large buyers, the markdown channel identified in this paper grows stronger: larger buyers have more variable markdowns, absorb more of each exchange-rate shock, and transmit less of it into seller-currency prices. To the extent that this buyer-side force has been strengthening alongside rising concentration, exchange rates may be becoming less effective as a macroeconomic stabilization tool for small open economies—independently of changes in invoicing practices or monetary policy credibility emphasized in prior work. Quantifying this channel at the aggregate level represents a promising direction for future research.

The growing concentration of sales among large, multinational firms opens several directions for future research. At the firm level, future work could focus on exploring which kinds of buyers are the best investment for small sellers in developing countries in the long run. At the same time, which markets contribute more to the growth of these small firms. Relevant policy questions at the market level remain unanswered: How does the market structure in terms of concentration of two different countries affect when they engage in trade?

A Appendix

A.1 Appendix: Theoretical Model

A.1.1 Notation

- $\mathcal{J} = \{1, \dots, M\}$: finite product set
- $\mathcal{K}$: destination set
- $\mathcal{B}_{jk}$: finite set of buyers active in product–destination market (j, k)
- $\mathcal{A} = \{(b, j, k) : k \in \mathcal{K}, j \in \mathcal{J}, b \in \mathcal{B}_{jk}\}$: seller choice set
- $\rho_{s,j,k}$: s 's idiosyncratic shock for producing j
- $\rho_{b,s,j,k}$: s 's idiosyncratic shock for supplying b with j
- z_{bjk} : CES productivity weight for Buyer b , Product j , and Destination k
- q_s : Seller's endowment
- $q_{s,b,j,k}$: Seller s 's production of good j for buyer b in Destination k
- $Q_{b,k}^{finalg}$: Buyer b 's production of the final good in Destination k
- $p_{s,b,j,k}$: input j 's price at destination k if it is bought by b from s
- $p_{b,j,k}$: input j 's aggregate price for b in k
- $P_{j,k}$: input j 's price index in destination k
- $\overline{P}_{j,k}$: Market j 's average price in destination k
- P_j : input j 's aggregate price index
- P_k : aggregate price index in destination k
- p_k^{finalg} : final good's price in Destination k
- $\lambda_{b,j,k}$: share of production supplied to Destination k that is sold to Buyer b of Product j
- $S_{b,j,k}$: Relative size of buyer b in input j 's market in destination k
- $\chi_{b,j,k}$: j 's share of b 's expenditure in k
- $MRP_{b,j,k}$: marginal revenue of input j in buyer b 's production
- $\overline{MRP}_{j,k}$: market of input j 's average marginal revenue product in destination k
- $\mu_{s,b,j,k}$: buyer b 's markdown buying input j from seller s in destination k ($\mu > 1$, higher μ means lower price)
- $\mu_{b,j,k}$: buyer b 's markdown in input j 's market in destination k

- $\mu_{j,k}$: markdown of input j in destination k
- $HHI_{j,k}$: market j 's concentration index in destination k
- θ : Elasticity of substitution across products in the seller's CES supply function
- η : Elasticity of substitution across buyers within a product
- $\varepsilon_{b,j,k}$: Supply elasticity faced by buyer b of input j in destination k
- $\varepsilon_{j,k}$: Market j 's average elasticity
- $Y_{b,j,k}$: $\frac{\partial \ln \mu_{b,j,k}}{\partial \ln S_{b,j,k}} > 0$ (positive by Proposition 1)
- $\Gamma_{b,j,k}$: $\frac{d \ln \mu_{b,j,k}}{d \ln p_{b,j,k}} > 0$ (the markdown channel, positive)
- $\phi_{b,j,k}$: $MRP_{b,j,k}$'s elasticity with respect to $p_{b,j,k}$ (negative)
- σ : Elasticity of substitution in the final good's CES
- e_k : nominal exchange rate between the origin under consideration and Destination k , $\left(\frac{\text{origin currency}}{\text{destination } k \text{ currency}}\right)$; the origin subscript is suppressed because the buyer's problem is the same for each origin
- Δe_k : exchange-rate shock for the origin–destination pair under consideration
- $\Psi_{b,j,k}$: Exchange rate pass-through to $p_{b,j,k}$
- $\Psi_{j,k}$: Exchange rate pass-through to $P_{j,k}$
- α : Estimated pass through independent of $S_{b,j,k}$
- β : Estimated pass through that depends on $S_{b,j,k}$

A.1.2 Supply Side: Frechet Shocks

I assume that the shock is drawn from a nested Frechet distribution. Then,

$$H(\vec{\rho}) = \exp \left[- \sum_{j \in \mathcal{J}} B_{jk} \left(\sum_{b \in \mathcal{B}_{jk}} B_{bjk} \rho_{bjk}^{-(1+\eta)} \right)^{\frac{1+\theta}{1+\eta}} \right], \text{ with } \theta < \eta,$$

Here $B_{jk} > 0$ is the Fréchet scale for product–destination nest (j,k) , and $B_{bjk} > 0$ is the buyer scale within (j,k) . Both are normalized to one thereafter.

The seller chooses the buyer that is going to yield the maximum profits. I will do this for buyer b in product j . The density function of choosing buyer b and product j is:

$$H_{bjk}(\vec{\rho}) = -(1+\theta) \tilde{P}_{jk}^{(1+\theta)-(1+\eta)} B_{bjk} \rho_{bjk}^{-(1+\eta)-1} \exp \left(- \sum_{j' \in \mathcal{J}} \tilde{P}_{j'k}^{1+\theta} \right) d\rho_{bjk}$$

where $\tilde{P}_{j'k'} = B_{j'k'} \left(\sum_{b' \in \mathcal{B}_{j'k'}} B_{b'j'k'} \rho_{b'j'k'}^{-(1+\eta)} \right)^{\frac{1}{1+\eta}}$.

A.1.3 Supply Side: Share

I need to integrate two things, first the probability of choosing Buyer b and Product j , and second the total quantities. For a given seller, that is fixing q_s and Destination k , the probability of choosing Buyer b and Product j is the same as the probability that $\rho_{b'j'k} \leq \frac{p_{bjk}}{p_{b'j'k}} \rho_{bjk}$ for every other buyer-product pair (b', j') in Destination k .⁶³ Then⁶⁴

$$\begin{aligned} \lambda_{bjk} &= P \left(\rho_{b'j'k} \leq \frac{p_{bjk}}{p_{b'j'k}} \rho_{bjk} \text{ for every } j' \in \mathcal{J} \text{ and } b' \in \mathcal{B}_{j'k} \text{ such that } (b', j') \neq (b, j) \right) \\ &= \int_0^\infty \exp \left[- \underbrace{\sum_{j' \in \mathcal{J}} B_{j'k} \left(\sum_{b' \in \mathcal{B}_{j'k}} B_{b'j'k} \left(\frac{p_{bjk}}{p_{b'j'k}} \right)^{-(1+\eta)} \rho^{-(1+\eta)} \right)^{\frac{1+\theta}{1+\eta}}}_{\Pr(\rho_{b'j'k} \leq (p_{bjk}/p_{b'j'k})\rho)} \right] \underbrace{dH_{bjk}(\rho)}_{\text{density of } \rho_{bjk}} \\ &= \underbrace{\frac{P_{jk}^{1+\theta}}{\sum_{j' \in \mathcal{J}} P_{j'k}^{1+\theta}}}_{\Pr(\text{chooses } j|k)} \underbrace{\frac{B_{bjk} p_{bjk}^{1+\eta}}{P_{jk}^{1+\eta}}}_{\Pr(\text{chooses } b|j,k)} \underbrace{\int_0^\infty -\exp(-u) du}_{=1} \end{aligned}$$

This expression has an intuitive interpretation: conditional on choosing Product j in Destination k , the probability of choosing Buyer b , $\Pr(b | j, k)$, depends on how large the price of Buyer b (numerator) is relative to the price index of Product j (denominator), which is a CES aggregate of prices across buyers within a product. The probability of choosing Product j , conditional on Destination k , $\Pr(j | k)$, then depends on how large the price index of Product j (numerator) is relative to the overall price index (denominator) in Destination k , which is a CES aggregate of price indexes across products.

Therefore, the share of sellers' production purchased by Buyer b in Product j and Destination k is:

$$\lambda_{bjk} = \frac{P_{jk}^{1+\theta} B_{bjk} p_{bjk}^{1+\eta}}{P_k^{1+\theta} P_{jk}^{1+\eta}} \quad (\text{A.1})$$

where $P_{jk} = B_{jk} \left(\sum_{b' \in \mathcal{B}_{jk}} B_{b'jk} p_{b'jk}^{1+\eta} \right)^{\frac{1}{1+\eta}}$. In the rest of the paper, I will simplify B_{bjk} and B_{jk} to 1.

⁶³This means that revenue is higher for Buyer b and Product j , conditional on Destination k .

⁶⁴This is the probability that the shock is higher than any other shock. Specifically by looking at the equations I can see it is the probability that e_{sb} is higher than another shock (cdf of e on point $\frac{p_{sb}}{p_{b'j'}} \rho$) throughout the whole distribution of shocks e (integral part). Also, $\lambda = \int_0^\infty H_{sb}(\rho, \frac{p_{sb}}{p_{b'j'}} \rho, \dots)$.

A.1.4 Supply Curve: Choice of Quantity

Aggregating across sellers yields a nested CES supply curve for Buyer b in Product j . I note that $p_{bjk}q_{bjk} = \lambda_{bjk}P_kQ_k$.

The expected quantity supplied by Seller s to Buyer b in Product j is $q_{sbjk} = q_s \times \Pr(sbjk)$. Integrating over sellers yields the total quantity in Product j supplied to Buyer b :

$$q_{bjk} = \int_0^1 \Pr(sbjk) q_{sk} dR = \frac{p_{bjk}^\eta}{\sum_a P_{bjk}^{1+\eta}} \frac{\sum_s (p_{bjk}^{1+\eta})^{\frac{1+\theta}{1+\eta}}}{\sum_{s'} (\sum P_{b'jk}^{1+\eta})^{\frac{1+\theta}{1+\eta}}} \underbrace{\int_0^1 p_{bjk} q_{sk} dR}_{Y_k}$$

So, the quantity supplied to Buyer b of Product j , destination k is:

$$q_{bjk} = \left(\frac{p_{bjk}^\eta}{P_{jk}^{1+\eta}} \right) \left(\frac{P_{jk}^{1+\theta}}{P_k^{1+\theta}} \right) Y_k \quad (\text{A.2})$$

where $P_k = \left(\sum_{j \in \mathcal{J}} P_{jk}^{1+\theta} \right)^{\frac{1}{1+\theta}}$.

Defining $x \equiv Y_k / P_k^{1+\theta}$, equation A.2 becomes

$$q_{bjk} = p_{bjk}^\eta \cdot \underbrace{P_{jk}^{\theta-\eta}}_x \cdot \left(P_k^{1+\theta} \right)^{-1} Y_k. \quad (\text{A.3})$$

Using the CES aggregator $q_{jk} \equiv \left(\sum_{b \in \mathcal{B}_{jk}} q_{bjk}^{(1+\eta)/\eta} \right)^{\eta/(1+\eta)}$ and substituting (A.3):

$$\begin{aligned} q_{jk} &= \left(\sum_{b \in \mathcal{B}_{jk}} \left(p_{bjk}^\eta P_{jk}^{\theta-\eta} x \right)^{\frac{1+\eta}{\eta}} \right)^{\frac{\eta}{1+\eta}} \\ q_{jk}^{\frac{1+\eta}{\eta}} &= \sum_{b \in \mathcal{B}_{jk}} p_{bjk}^{1+\eta} \cdot P_{jk}^{(\theta-\eta)\frac{1+\eta}{\eta}} \cdot x^{\frac{1+\eta}{\eta}} \\ q_{jk} &= \left(\sum_{b \in \mathcal{B}_{jk}} p_{bjk}^{1+\eta} \right)^{\frac{\eta}{1+\eta}} \cdot P_{jk}^{\theta-\eta} \cdot x = P_{jk}^\eta \cdot P_{jk}^{\theta-\eta} \cdot x \\ q_{jk} &= P_{jk}^\theta \cdot x \end{aligned} \quad (\text{A.4})$$

From (A.4) and the definition of x :

$$q_{jk} = P_{jk}^\theta \cdot x \implies q_{jk} = P_{jk}^\theta \cdot \left(P_k^{1+\theta} \right)^{-1} Y_k \implies \frac{q_{jk}}{P_{jk}^\theta} = \frac{Y_k}{P_k^{1+\theta}},$$

By (A.4):

$$x = \frac{Y_k}{P_k^{1+\theta}}. \quad (\text{A.5})$$

Substituting (A.5) into (A.3):

$$q_{bjk} = \left(\frac{p_{bjk}^\eta}{P_{jk}^\eta} \right) \left(\frac{P_{jk}^\theta}{P_k^\theta} \right) \frac{Y_k}{P_k}, \quad (\text{A.6})$$

Aggregating at the country level, the price and quantity indices are $P_k = \left(\sum_{j \in \mathcal{J}} P_{jk}^{1+\theta} \right)^{1/(1+\theta)}$ and $Q_k = \left(\sum_{j \in \mathcal{J}} q_{jk}^{(1+\theta)/\theta} \right)^{\theta/(1+\theta)}$. It follows that $p_{bjk} q_{bjk} = \lambda_{bjk} P_k Q_k$, where $P_k Q_k = Y_k$.

A.1.5 Cournot Competition

In the baseline model, buyers compete à la Bertrand: each buyer b chooses the price p_{bjk} to offer sellers, taking other buyers' prices as given. Here I solve the Cournot alternative, where each buyer instead chooses quantity q_{bjk} , taking other buyers' quantities $\{q_{b'jk}\}_{b' \neq b}$ as fixed. All qualitative results are preserved.

First-Order Condition Under Cournot, buyer b solves:

$$\max_{q_{bjk}} p_k^{finalg} Q_{bk}^{finalg} - \frac{1}{e_k} p_{bjk}(q_{bjk}) q_{bjk} \quad \text{s.t.} \quad Q_{bk}^{finalg} = \left(\sum_{j \in \mathcal{J}} z_{bjk} q_{bjk}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad q_{bjk} = \frac{p_{bjk}^\eta P_{jk}^\theta Y_k}{P_{jk}^\eta P_k^\theta P_k} \quad (\text{A.7})$$

The first-order condition with respect to q_{bjk} is:

$$MRP_{bjk} \cdot e_k = \frac{\partial(p_{bjk} q_{bjk})}{\partial q_{bjk}} = p_{bjk} \left(1 + \frac{1}{\epsilon_{bjk}^C} \right) \quad (\text{A.8})$$

where ϵ_{bjk}^C is the supply elasticity faced by buyer b when other buyers' quantities are held fixed. Rearranging gives the same structural form as Bertrand (equation 4.6):

$$p_{bjk} = \frac{1}{\mu_{bjk}^C} \cdot MRP_{bjk} \cdot e_k, \quad \mu_{bjk}^C = 1 + \frac{1}{\epsilon_{bjk}^C} > 1 \quad (\text{A.9})$$

Cournot Supply Elasticity Under Bertrand, buyer b takes other buyers' prices fixed, giving:

$$\epsilon_{bjk}^B = \eta(1 - S_{bjk}) + \theta S_{bjk}$$

Under Cournot, buyer b takes other buyers' quantities fixed instead. When p_{bjk} rises, P_{jk} rises, which forces other buyers' prices to adjust to keep their quantities unchanged. Imposing $d \ln q_{b'jk} = 0$ on

equation 4.2:

$$d \ln p_{b'jk} = \frac{\eta - \theta}{\eta} d \ln P_{jk}$$

Substituting into $d \ln P_{jk} = S_{bjk} d \ln p_{bjk} + (1 - S_{bjk}) d \ln p_{b'jk}$ and solving for $\frac{d \ln P_{jk}}{d \ln p_{bjk}}$, then using $\epsilon_{bjk}^C = \frac{d \ln q_{bjk}}{d \ln p_{bjk}} \Big|_{q-bjk}$ yields:

$$\epsilon_{bjk}^C = \frac{\eta \theta}{\theta(1 - S_{bjk}) + \eta S_{bjk}} \quad (\text{A.10})$$

Note the elegant symmetry with Bertrand: the roles of η and θ are swapped. Both elasticities share the same limits— $\epsilon^C = \epsilon^B = \eta$ at $S_{bjk} = 0$ and $\epsilon^C = \epsilon^B = \theta$ at $S_{bjk} = 1$ —but for all interior values $S_{bjk} \in (0, 1)$:

$$\epsilon_{bjk}^C \leq \epsilon_{bjk}^B$$

since $\frac{\eta \theta}{\theta(1-S)+\eta S} \leq \eta(1-S) + \theta S$ for $\eta > \theta > 0$. Therefore $\mu_{bjk}^C \geq \mu_{bjk}^B$: Cournot buyers extract weakly larger markdowns than Bertrand buyers.

Qualitative Results Under Cournot Since ϵ_{bjk}^C is strictly decreasing in S_{bjk} (for $\eta > \theta$), μ_{bjk}^C is strictly increasing in buyer market share, preserving Proposition 1. Following the same derivation as Section 4.6 but using ϵ_{bjk}^C :

$$\frac{d \ln p_{bjk}}{d \ln e_k} = \frac{1}{1 + \Gamma_{bjk}^C - \Phi_{bjk}^C} \quad (\text{A.11})$$

where $\Gamma_{bjk}^C = \frac{d \ln \mu_{bjk}^C}{d \ln p_{bjk}} > 0$ is the Cournot markdown channel and $\Phi_{bjk}^C < 0$ is the marginal-revenue channel, both evaluated using ϵ_{bjk}^C . Since Γ_{bjk}^C increases with S_{bjk} , all results of Propositions 3 and Corollary 1 hold under Cournot: larger buyers have higher markdowns, more variable markdowns, and lower exchange-rate pass-through. Quantitatively, Cournot implies larger markdowns and lower pass-through than Bertrand for all $S_{bjk} \in (0, 1)$, since buyers internalize more of the supply response when holding competitors' quantities rather than prices fixed.

A.1.6 Firm-level elasticity of Supply

To solve for the value of ϵ_{bj} , I start from the quantity supplied and solve:

$$\epsilon_{bjk} = \frac{\partial q_{bjk}}{\partial p_{bjk}} \frac{p_{bjk}}{q_{bjk}}$$

$$\frac{\partial q_{bjk}}{\partial p_{bjk}} \frac{p_{bjk}}{q_{bjk}} = \left[\eta \frac{q_{bjk}}{p_{bjk}} + (\theta - \eta) q_{bjk} \frac{p_{bjk}^\eta}{P_{jk}^{1+\eta}} \right] \frac{p_{bjk}}{q_{bjk}}$$

$$\epsilon_{bjk} = \frac{\partial q_{bjk}}{\partial p_{bjk}} \frac{p_{bjk}}{q_{bjk}} = \eta(1 - S_{bjk}) + \theta S_{bjk}$$

A.1.7 Proof of Proposition 1: Markdown and Marginal Revenue Product

Item 1: Markdown. The markdown is

$$\mu_{bjk} = 1 + \epsilon_{bjk}^{-1}, \quad \epsilon_{bjk} = \eta + (\theta - \eta)S_{bjk}.$$

Since $\eta > \theta$,

$$\frac{d\mu_{bjk}}{dS_{bjk}} = \frac{\eta - \theta}{[\eta + (\theta - \eta)S_{bjk}]^2} > 0.$$

Therefore,

$$\frac{d \ln \mu_{bjk}}{d \ln S_{bjk}} > 0.$$

Item 2: Marginal revenue product. Compare equilibria that satisfy the buyer's FOC, holding the exchange rate e_k and rival prices fixed. Rearranging equation 4.6 gives

$$MRP_{bjk} = \frac{p_{bjk}\mu_{bjk}}{e_k}.$$

Therefore,

$$\frac{d \ln MRP_{bjk}}{d \ln S_{bjk}} = \frac{d \ln p_{bjk}}{d \ln S_{bjk}} + \frac{d \ln \mu_{bjk}}{d \ln S_{bjk}}.$$

The second component is precisely the result in Item 1 of Proposition 1, so it is positive. To obtain the first component, start with equation 4.8:

$$S_{bjk} = \frac{p_{bjk}^{1+\eta}}{p_{bjk}^{1+\eta} + \sum_{b' \in \mathcal{B}_{jk} \setminus \{b\}} p_{b'jk}^{1+\eta}}.$$

Rearranging directly,

$$S_{bjk} \sum_{b' \in \mathcal{B}_{jk} \setminus \{b\}} p_{b'jk}^{1+\eta} = (1 - S_{bjk})p_{bjk}^{1+\eta},$$

which implies

$$p_{bjk} = \left(\sum_{b' \in \mathcal{B}_{jk} \setminus \{b\}} p_{b'jk}^{1+\eta} \right)^{\frac{1}{1+\eta}} \left(\frac{S_{bjk}}{1 - S_{bjk}} \right)^{\frac{1}{1+\eta}},$$

Taking logs,

$$\ln p_{bjk} = \frac{1}{1+\eta} \left[\ln \left(\sum_{b' \in \mathcal{B}_{jk} \setminus \{b\}} p_{b'jk}^{1+\eta} \right) + \ln S_{bjk} - \ln(1 - S_{bjk}) \right].$$

Holding rival prices fixed and differentiating with respect to $\ln S_{bjk}$ gives the first component of the marginal revenue product derivative:

$$\frac{d \ln p_{bjk}}{d \ln S_{bjk}} = \frac{1}{(1+\eta)(1 - S_{bjk})} > 0.$$

Combining both terms,

$$\frac{d \ln MRP_{bjk}}{d \ln S_{bjk}} = \frac{1}{(1 + \eta)(1 - S_{bjk})} + \frac{d \ln \mu_{bjk}}{d \ln S_{bjk}} > 0.$$

Thus, along the equilibrium locus, a larger buyer share is associated with both a higher price and a higher markdown. With e_k fixed, the FOC requires a higher marginal revenue product. $\square$

A.1.8 Proof of Proposition 2: Aggregate Markdown

Calculate the share-weighted average of individual supply elasticities:

$$\epsilon_{jk} \equiv \sum_{b \in \mathcal{B}_{jk}} S_{bjk} \epsilon_{bjk} = \eta(1 - HHI_{jk}) + \theta HHI_{jk}.$$

Then define the aggregate markdown by inverting this aggregate elasticity:

$$\mu_{jk} \equiv 1 + (\epsilon_{jk})^{-1} = 1 + [\eta(1 - HHI_{jk}) + \theta HHI_{jk}]^{-1}.$$

Under this definition, HHI is an exact sufficient statistic for the aggregate markdown. We next show how this expression approximates the exact aggregate markdown under Bertrand. Define the quantity-weighted averages

$$\overline{MRP}_{jk} \equiv \frac{\sum_{b \in \mathcal{B}_{jk}} q_{bjk} MRP_{bjk}}{\sum_{b \in \mathcal{B}_{jk}} q_{bjk}}, \quad \bar{P}_{jk} \equiv \frac{\sum_{b \in \mathcal{B}_{jk}} q_{bjk} p_{bjk}}{\sum_{b \in \mathcal{B}_{jk}} q_{bjk}}.$$

Their ratio gives the exact aggregate markdown:

$$\begin{aligned} \frac{e_k \overline{MRP}_{jk}}{\bar{P}_{jk}} &= e_k \frac{\sum_b q_{bjk} MRP_{bjk}}{\sum_b p_{bjk} q_{bjk}} \\ &= \sum_b \frac{p_{bjk} q_{bjk}}{\sum_{b'} p_{b'jk} q_{b'jk}} \frac{e_k MRP_{bjk}}{p_{bjk}} \\ &= \sum_b S_{bjk} \mu_{bjk} = 1 + \sum_b \frac{S_{bjk}}{\epsilon_{bjk}} \equiv \mu_{jk}^*, \end{aligned}$$

where the buyer-level first-order condition implies $e_k MRP_{bjk} / p_{bjk} = \mu_{bjk} = 1 + \epsilon_{bjk}^{-1}$ and $\sum_b S_{bjk} = 1$. To characterize the approximation error, begin with the following nonnegative expression and work

backward:

$$\begin{aligned}
0 &\leq \sum_{b \in \mathcal{B}_{jk}} S_{bjk} \frac{(\epsilon_{bjk} - \epsilon_{jk})^2}{\epsilon_{bjk}(\epsilon_{jk})^2} \\
&= \sum_{b \in \mathcal{B}_{jk}} S_{bjk} \left(\frac{\epsilon_{bjk}}{(\epsilon_{jk})^2} - \frac{2}{\epsilon_{jk}} + \frac{1}{\epsilon_{bjk}} \right) \\
&= \frac{\sum_b S_{bjk} \epsilon_{bjk}}{(\epsilon_{jk})^2} - \frac{2 \sum_b S_{bjk}}{\epsilon_{jk}} + \sum_b \frac{S_{bjk}}{\epsilon_{bjk}} \\
&= \sum_b \frac{S_{bjk}}{\epsilon_{bjk}} - \frac{1}{\epsilon_{jk}} = \mu_{jk}^* - \mu_{jk},
\end{aligned}$$

where we used $\sum_b S_{bjk} = 1$ and $\sum_b S_{bjk} \epsilon_{bjk} = \epsilon_{jk}$. Hence,

$$\mu_{jk}^* - \mu_{jk} = \sum_{b \in \mathcal{B}_{jk}} S_{bjk} \frac{(\epsilon_{bjk} - \epsilon_{jk})^2}{\epsilon_{bjk}(\epsilon_{jk})^2} \geq 0,$$

which is clearly nonnegative. Thus, $\mu_{jk}^* \geq \mu_{jk}$, with equality when individual supply elasticities are identical. This is also the inequality implied by Jensen's inequality. Define the share-weighted variance as

$$\text{Var}_S(\epsilon_{bjk}) \equiv \sum_b S_{bjk} (\epsilon_{bjk} - \epsilon_{jk})^2.$$

In this model, $\epsilon_{bjk} = \eta + (\theta - \eta)S_{bjk}$ and $\epsilon_{jk} = \eta + (\theta - \eta)HHI_{jk}$, so

$$\begin{aligned}
\text{Var}_S(\epsilon_{bjk}) &= (\eta - \theta)^2 \sum_b S_{bjk} (S_{bjk} - HHI_{jk})^2 \\
&= (\eta - \theta)^2 \left(\sum_b S_{bjk}^3 - HHI_{jk}^2 \right).
\end{aligned}$$

When individual elasticities are close to their weighted average, replace ϵ_{bjk} in the denominator of the exact gap by ϵ_{jk} . Combining these results gives

$$\begin{aligned}
\mu_{jk}^* &\simeq \mu_{jk} + \frac{\text{Var}_S(\epsilon_{bjk})}{(\epsilon_{jk})^3} \\
&= 1 + \frac{1}{\eta(1 - HHI_{jk}) + \theta HHI_{jk}} + \frac{(\eta - \theta)^2 \left(\sum_b S_{bjk}^3 - HHI_{jk}^2 \right)}{[\eta(1 - HHI_{jk}) + \theta HHI_{jk}]^3}.
\end{aligned}$$

Thus, the approximation error is governed by the relative dispersion of buyer-level supply elasticities and decreases with the market's average supply elasticity.

A.1.9 Proof of Proposition 3: Direct Pass-through

Substituting Equations 4.12 and 4.16 into Equation 4.11 gives

$$d \ln p_{bjk} = -\Gamma_{bjk} d \ln p_{bjk} + \Phi_{bjk} d \ln p_{bjk} + d \ln e_k.$$

Collecting the price terms,

$$(1 + \Gamma_{bjk} - \Phi_{bjk}) d\ln p_{bjk} = d\ln e_k.$$

Therefore,

$$\frac{d\ln p_{bjk}}{d\ln e_k} = \frac{1}{1 + \Gamma_{bjk} - \Phi_{bjk}}.$$

Since $\Gamma_{bjk} > 0$ and $\Phi_{bjk} < 0$, both channels reduce pass-through below one. $\square$

A.1.10 Proof of Corollary 1: Markdown Channel

Proposition 1 implies

$$Y_{bjk} \equiv \frac{\partial \ln \mu_{bjk}}{\partial \ln S_{bjk}} = \frac{(\eta - \theta)S_{bjk}}{\epsilon_{bjk}(1 + \epsilon_{bjk})} > 0.$$

Under fixed rival prices, the CES price index gives

$$\frac{\partial \ln P_{jk}}{\partial \ln p_{bjk}} = S_{bjk}.$$

Therefore, using $S_{bjk} = (p_{bjk}/P_{jk})^{1+\eta}$,

$$\frac{\partial \ln S_{bjk}}{\partial \ln p_{bjk}} = (1 + \eta)(1 - S_{bjk}).$$

Combining these results,

$$\begin{aligned} \Gamma_{bjk} &= \frac{\partial \ln \mu_{bjk}}{\partial \ln S_{bjk}} \frac{\partial \ln S_{bjk}}{\partial \ln p_{bjk}} \\ &= \frac{(\eta - \theta)(1 + \eta)S_{bjk}(1 - S_{bjk})}{\epsilon_{bjk}(1 + \epsilon_{bjk})} > 0. \end{aligned}$$

To determine how the markdown channel varies with buyer share, differentiate this expression with respect to S_{bjk} :

$$\frac{\partial \Gamma_{bjk}}{\partial S_{bjk}} = \frac{(\eta - \theta)(1 + \eta)}{\epsilon_{bjk}^2(1 + \epsilon_{bjk})^2} \left[\eta(1 + \eta)(1 - 2S_{bjk}) + (\eta - \theta)(1 + \eta + \theta)S_{bjk}^2 \right].$$

The factor outside the brackets is positive. The second term inside the brackets is also positive, while the first can become negative only when $S_{bjk} > 1/2$. Thus, Γ_{bjk} increases throughout the empirical support and may decline only for a buyer controlling a very large share of the market. For such a buyer, an increase in its offered price moves P_{jk} nearly one for one, so its relative price and share respond less. The calibrated pass-through function can therefore turn upward only at very high shares, outside the empirical support. $\square$

A.1.11 Indirect Effect

Proposition 6 includes the additional responses of market prices and quantities.

Proposition 6 Let $\Gamma_{bjk}^R \equiv Y_{bjk}(1 + \eta) > 0$ denote the markdown elasticity with respect to the buyer's relative price, $\varphi_{bjk} \equiv \partial \ln MRP_{bjk} / \partial \ln q_{bjk} < 0$, and $\Phi_{bjk}^R \equiv \varphi_{bjk}\eta < 0$.⁶⁵ Including the additional aggregate responses, pass-through is

$$\frac{d \ln p_{bjk}}{d \ln e_k} = \frac{1}{1 + \Gamma_{bjk}^R - \Phi_{bjk}^R} + \frac{\Gamma_{bjk}^R - \Phi_{bjk}^R}{1 + \Gamma_{bjk}^R - \Phi_{bjk}^R} \frac{d \ln P_{jk}}{d \ln e_k} + \frac{\varphi_{bjk}}{1 + \Gamma_{bjk}^R - \Phi_{bjk}^R} \frac{d \ln q_{jk}}{d \ln e_k}. \quad (\text{A.12})$$

Proof Start from

$$p_{bjk} = \frac{MRP_{bjk}}{\mu_{bjk}} e_k.$$

Taking logs and log changes gives

$$d \ln p_{bjk} = -d \ln \mu_{bjk} + d \ln MRP_{bjk} + d \ln e_k.$$

First, consider the markdown term. Since

$$S_{bjk} = \left(\frac{p_{bjk}}{P_{jk}} \right)^{1+\eta},$$

the markdown depends on the buyer's relative price. Therefore,

$$d \ln \mu_{bjk} = \Gamma_{bjk}^R (d \ln p_{bjk} - d \ln P_{jk}), \quad \Gamma_{bjk}^R \equiv Y_{bjk}(1 + \eta) > 0.$$

Second, consider the marginal-revenue term. From Equation (4.5),

$$d \ln MRP_{bjk} = \varphi_{bjk} d \ln q_{bjk}, \quad \varphi_{bjk} \equiv \frac{\partial \ln MRP_{bjk}}{\partial \ln q_{bjk}} = -\frac{1 - x_{bjk}}{\sigma} < 0.$$

The buyer-level quantity can be written as

$$q_{bjk} = \left(\frac{p_{bjk}}{P_{jk}} \right)^\eta q_{jk},$$

where Appendix A.1.4 shows that $q_{jk} = (P_{jk}/P_k)^\theta Y_k/P_k$. Taking log changes,

$$d \ln q_{bjk} = \eta (d \ln p_{bjk} - d \ln P_{jk}) + d \ln q_{jk}.$$

Hence,

$$d \ln MRP_{bjk} = \Phi_{bjk}^R (d \ln p_{bjk} - d \ln P_{jk}) + \varphi_{bjk} d \ln q_{jk}, \quad \Phi_{bjk}^R \equiv \varphi_{bjk}\eta < 0.$$

Substituting the markdown and marginal-revenue terms into the log change in price gives

$$d \ln p_{bjk} = -\Gamma_{bjk}^R (d \ln p_{bjk} - d \ln P_{jk}) + \Phi_{bjk}^R (d \ln p_{bjk} - d \ln P_{jk}) + \varphi_{bjk} d \ln q_{jk} + d \ln e_k,$$

⁶⁵ Γ_{bjk}^R and Φ_{bjk}^R are the coefficients on the relative-price response $d \ln p_{bjk} - d \ln P_{jk}$ in the markdown and marginal-revenue terms, respectively. Under fixed rival prices, $d \ln P_{jk} = S_{bjk} d \ln p_{bjk}$, so $\Gamma_{bjk}^R = \Gamma_{bjk}/(1 - S_{bjk})$. Moreover, because $\Phi_{bjk} = \varphi_{bjk}\epsilon_{bjk}$ in the direct effect, $\Phi_{bjk}^R = (\eta/\epsilon_{bjk})\Phi_{bjk}$. The proposition could be written using Γ_{bjk} and Φ_{bjk} alone, but the superscript R keeps the relative-price formulation cleaner and separates it from the response of q_{jk} .

or, after collecting terms,

$$(1 + \Gamma_{bjk}^R - \Phi_{bjk}^R) \text{dln} p_{bjk} = (\Gamma_{bjk}^R - \Phi_{bjk}^R) \text{dln} P_{jk} + \varphi_{bjk} \text{dln} q_{jk} + \text{dln} e_k.$$

Dividing by $\text{dln} e_k$ gives

$$(1 + \Gamma_{bjk}^R - \Phi_{bjk}^R) \frac{\text{dln} p_{bjk}}{\text{dln} e_k} = 1 + (\Gamma_{bjk}^R - \Phi_{bjk}^R) \frac{\text{dln} P_{jk}}{\text{dln} e_k} + \varphi_{bjk} \frac{\text{dln} q_{jk}}{\text{dln} e_k}.$$

Solving for pass-through,

$$\begin{aligned} \frac{\text{dln} p_{bjk}}{\text{dln} e_k} &= \frac{1}{1 + \Gamma_{bjk}^R - \Phi_{bjk}^R} \\ &+ \frac{\Gamma_{bjk}^R - \Phi_{bjk}^R}{1 + \Gamma_{bjk}^R - \Phi_{bjk}^R} \frac{\text{dln} P_{jk}}{\text{dln} e_k} \\ &+ \frac{\varphi_{bjk}}{1 + \Gamma_{bjk}^R - \Phi_{bjk}^R} \frac{\text{dln} q_{jk}}{\text{dln} e_k}, \end{aligned}$$

which is Equation (A.12).

Under fixed rival prices, the response of the CES price index is

$$\text{dln} P_{jk} = \frac{\partial \ln P_{jk}}{\partial \ln p_{bjk}} \text{dln} p_{bjk} = S_{bjk} \text{dln} p_{bjk}.$$

Moreover, since $q_{jk} = (P_{jk}/P_k)^\theta Y_k/P_k$ and the direct calculation holds P_k and Y_k fixed,

$$\text{dln} q_{jk} = \theta \text{dln} P_{jk} = \theta S_{bjk} \text{dln} p_{bjk}.$$

The markdown response therefore becomes

$$\begin{aligned} \text{dln} \mu_{bjk} &= \Gamma_{bjk}^R (\text{dln} p_{bjk} - \text{dln} P_{jk}) \\ &= \Gamma_{bjk}^R (1 - S_{bjk}) \text{dln} p_{bjk} \equiv \Gamma_{bjk} \text{dln} p_{bjk}. \end{aligned}$$

Similarly, the buyer-level quantity response is

$$\begin{aligned} \text{dln} q_{bjk} &= \eta (\text{dln} p_{bjk} - \text{dln} P_{jk}) + \text{dln} q_{jk} \\ &= [\eta (1 - S_{bjk}) + \theta S_{bjk}] \text{dln} p_{bjk} \\ &= \epsilon_{bjk} \text{dln} p_{bjk}. \end{aligned}$$

Thus,

$$\text{dln} \text{MRP}_{bjk} = \varphi_{bjk} \epsilon_{bjk} \text{dln} p_{bjk} \equiv \Phi_{bjk} \text{dln} p_{bjk}.$$

Substituting these two responses into the log price equation gives

$$\text{dln} p_{bjk} = -\Gamma_{bjk} \text{dln} p_{bjk} + \Phi_{bjk} \text{dln} p_{bjk} + \text{dln} e_k,$$

so the general expression reduces to Proposition 3.

A.1.12 Proof of Proposition 4: Aggregate Exchange Rate Pass-through

Starting from:

$$\bar{P}_{jk} = \frac{1}{\mu_{jk}} \overline{MRP}_{jk} e_k$$

Following analogous steps to the firm-level derivation (Proof of Proposition 3), but at the market level using HHI_{jk} in place of S_{bjk} , and allowing for aggregate price feedback through P_k and q_k :

$$\frac{d \ln P_{jk}}{d \ln e_k} = \frac{1}{1 + \bar{\Gamma}_{jk} - \bar{\Phi}_{jk}} + \frac{\bar{\Gamma}_{jk} - \bar{\Phi}_{jk}}{1 + \bar{\Gamma}_{jk} - \bar{\Phi}_{jk}} \frac{d \ln P_k}{d \ln e_k} + \frac{\overline{MRP}_q}{1 + \bar{\Gamma}_{jk} - \bar{\Phi}_{jk}} \frac{d \ln q_k}{d \ln e_k}$$

where $\bar{\Gamma}_{jk} > 0$ and $\bar{\Phi}_{jk} < 0$ are the market-level analogues of Γ_{bjk} and Φ_{bjk} , evaluated using HHI_{jk} in place of S_{bjk} . The denominator $1 + \bar{\Gamma}_{jk} - \bar{\Phi}_{jk} > 1$ ensures market-level pass-through is also below one.

I solve further to understand the aggregated markdown channel effect. Let $\kappa \equiv \frac{dHHI_{jk}}{dER}$ denote the effect of an exogenous exchange rate shock on the Herfindahl–Hirschman Index. Plugging $\mu_{jk} = 1 + \epsilon_{jk}^{-1}$ into $\frac{d\mu_{jk}}{dER}$ and differentiating via the chain rule:

$$\frac{d\mu_{jk}}{dER} = \frac{d\mu_{jk}}{dHHI_{jk}} \frac{dHHI_{jk}}{dER} = \left(\frac{1}{\theta} - \frac{1}{\eta} \right) \kappa.$$

I then compute the standard errors for κ under the assumption that the effect on concentration and the input-supply parameters are independent:

$$\begin{aligned} \text{Var}(\kappa) &= \text{Var} \left[\left(\frac{1}{\theta} - \frac{1}{\eta} \right) \kappa \right] \\ &= \text{Var} \left(\frac{1}{\theta} - \frac{1}{\eta} \right) \mathbb{E}[\kappa^2] - \left[\mathbb{E} \left(\frac{1}{\theta} - \frac{1}{\eta} \right) \right]^2 [\mathbb{E}(\kappa)]^2 \\ &= \left[\text{Var} \left(\frac{1}{\theta} - \frac{1}{\eta} \right) + \left(\mathbb{E} \left(\frac{1}{\theta} - \frac{1}{\eta} \right) \right)^2 \right] [\text{Var}(\kappa) + (\mathbb{E}[\kappa])^2] - \left(\frac{1}{\theta} - \frac{1}{\eta} \right)^2 [\mathbb{E}(\kappa)]^2, \end{aligned}$$

whose components can all be evaluated using sample estimates.

A.1.13 Log-Linearization and First-Order Approximation

Starting from Proposition 3 of the main text:

$$d \ln p_{bjk} = \frac{1}{1 + \underbrace{\Gamma_{bjk}}_{\text{markdown channel}} - \underbrace{\Phi_{bjk}}_{\text{marginal revenue channel}}} d \ln e_k,$$

where $\Gamma_{bjk} > 0$ and $-\Phi_{bjk} > 0$, so both channels reduce pass-through below one. Writing out the explicit form and using $\Gamma_{bjk} = Y_{bjk}(1 + \eta)(1 - S_{bjk})$ and $\Phi_{bjk} = \varphi_{bjk}\epsilon_{bjk}$ with $\varphi_{bjk} = -\frac{1}{\sigma}(1 - x_{bjk})$:

$$d\ln p_{bjk} = \frac{1}{1 + Y_{bjk}(1 + \eta)(1 - S_{bjk}) - \varphi_{bjk}\epsilon_{bjk}} d\ln e_k.$$

Taking a first-order approximation in S_{bjk} around $\bar{S}_{bjk}$ and dividing by $d\ln e_k$, letting $D_{bjk} \equiv 1 + \bar{Y}_{bjk}(1 + \eta)(1 - \bar{S}_{bjk}) - \bar{\varphi}_{bjk}\bar{\epsilon}_{bjk}$:

$$\frac{d\ln p_{bjk}}{d\ln e_k} \approx \frac{1}{D_{bjk}} - \frac{\frac{\partial Y_{bjk}}{\partial S_{bjk}} \Big|_{\bar{S}_{bjk}} (1 + \eta)(1 - \bar{S}_{bjk}) - \bar{Y}_{bjk}(1 + \eta) - \bar{\varphi}_{bjk} \frac{\partial \epsilon_{bjk}}{\partial S_{bjk}} \Big|_{\bar{S}_{bjk}}}{D_{bjk}^2} (S_{bjk} - \bar{S}_{bjk}).$$

Rearranging as $\alpha_{bjk} + \beta_{bjk} S_{bjk}$:

$$\frac{d\ln p_{bjk}}{d\ln e_k} \approx \alpha_{bjk} + \beta_{bjk} S_{bjk},$$

where:

$$\alpha_{bjk} = \frac{1}{D_{bjk}} + \frac{\frac{\partial Y_{bjk}}{\partial S_{bjk}} \Big|_{\bar{S}_{bjk}} (1 + \eta)(1 - \bar{S}_{bjk}) - \bar{Y}_{bjk}(1 + \eta) - \bar{\varphi}_{bjk} \frac{\partial \epsilon_{bjk}}{\partial S_{bjk}} \Big|_{\bar{S}_{bjk}}}{D_{bjk}^2} \cdot \bar{S}_{bjk},$$

$$\beta_{bjk} = -\frac{\frac{\partial Y_{bjk}}{\partial S_{bjk}} \Big|_{\bar{S}_{bjk}} (1 + \eta)(1 - \bar{S}_{bjk}) - \bar{Y}_{bjk}(1 + \eta) - \bar{\varphi}_{bjk} \frac{\partial \epsilon_{bjk}}{\partial S_{bjk}} \Big|_{\bar{S}_{bjk}}}{D_{bjk}^2}.$$

β_{bjk} is the coefficient of interest. Note that $\frac{\partial \epsilon_{bjk}}{\partial S_{bjk}} = \theta - \eta < 0$ (since $\eta > \theta$), so the third term in the numerator contributes positively to the numerator of $-\beta_{bjk}$. When the markdown channel dominates—i.e. the first term exceeds the sum of the second and third—the numerator of $-\beta_{bjk}$ is positive and $\beta_{bjk} < 0$: larger buyers exhibit lower exchange rate pass-through, consistent with the empirical estimates in Table 2.

Applying an analogous procedure to Proposition 4 yields the market-level result.

A.1.14 General Equilibrium: Incomplete Pass-through

The direct and indirect effects together account for the general equilibrium effect. In this section I show that, because buyers' market shares respond differently to exchange rate shocks depending on their size, aggregate exchange rate pass-through is incomplete.

I consider the following example focusing only on the markdown effect and leaving the marginal revenue product constant.⁶⁶

⁶⁶The same result holds with variable MRP_{bjk} ; the only difference is that the indirect-effect terms also need to be included, but they too yield complete ERPT.

$$P_{bjk} = \frac{\epsilon_{bjk}}{1 + \epsilon_{bjk}} P_{bjk}^f$$

Pass-through:

$$\begin{aligned} \text{dln} p_{bjk} &= -\text{dln} \mu_{bjk} + \text{dln} p_{bjk}^f \\ \text{dln} p_{bjk} &= -\Gamma_{bjk}(\text{dln} p_{bjk} - \text{dln} P_k) + \text{dln} e_k + \tilde{u}_{bjk} \\ \text{dln} p_{bjk} &= \frac{1}{1 + \Gamma_{bjk}} \text{dln} e_k + \frac{\Gamma_{bjk}}{1 + \Gamma_{bjk}} \text{dln} P_k + \tilde{u}_{bjk} \end{aligned}$$

If $\text{dln} P_k = \sum_{b \in \mathcal{B}_k} S_{bjk} \text{dln} p_{bjk}$, then

$$\text{dln} p_{bjk} = \frac{1}{1 + \Gamma_{bjk}} \text{dln} e_k + \frac{\Gamma_{bjk}}{1 + \Gamma_{bjk}} \sum_{b \in \mathcal{B}_k} S_{bjk} \text{dln} p_{bjk} + \tilde{u}_{bjk}.$$

If $\text{dln} p_{bjk} = \text{dln} p_{zjk}$ for all $b \neq z$, then $\sum_{b \in \mathcal{B}_k} S_{bjk} \text{dln} p_{bjk} = \text{dln} P_k$ and pass-through would be complete:

$$\begin{aligned} \frac{1 + \Gamma_{bjk} - \Gamma_{bjk}}{1 + \Gamma_{bjk}} \text{dln} p_{bjk} &= \frac{1}{1 + \Gamma_{bjk}} \text{dln} e_k + \tilde{u}_{bjk} \\ \text{dln} p_{bjk} &= \text{dln} e_k + (1 + \Gamma_{bjk}) \tilde{u}_{bjk}. \end{aligned}$$

Since markdowns are more sensitive to the relative price $p_{bjk} - P_k$ the larger the buyer's market share S_{bjk} , it follows that $\text{dln} p_{bjk} \neq \text{dln} p_{zjk}$ whenever $S_{bjk} \neq S_{zjk}$, so pass-through is incomplete in general equilibrium. These results are qualitatively unchanged if firms compete in quantities (Cournot).⁶⁷

Finally, note that with a finite number of positive-mass buyers per sector, any change in a product's price p_{bjk} has a non-zero effect on the aggregate sector price P_k . Markdowns are constant if $S_{bjk} = 0$ or $S_{bjk} = 1$. Hence, ERPT (both the direct effect and the sum of the direct and indirect effects) is non-monotonic in buyer size and, at the aggregate level, in the Herfindahl–Hirschman Index for buyer concentration in a market.⁶⁸

More specifically, the sensitivity of P_k to the exchange rate—which determines the indirect effect of ERPT—depends on details such as the source of the exchange rate shock (which shapes the response of costs and prices of domestic producers competing with foreign exporters) and whether the shock is idiosyncratic to a bilateral country pair or is a common shock (e.g., a simultaneous dollar depreciation against all trading partners). [Gopinath, Itskhoki and Rigobon \(2010\)](#); [Auer and Schoenle \(2016\)](#); [Pennings \(2017\)](#) document this.

⁶⁷Results are analogous to a case with variable markups as in [Burstein and Gopinath \(2014\)](#).

⁶⁸Alternative approaches could generate greater firm-level variation in exchange-rate exposure. For example: (i) if markets were defined as product–time combinations so that two different destinations belong to the same market, a bilateral exchange rate shock would affect only firms connected to a given destination; (ii) accounting for a fixed cost of importing on the buyer side would imply some buyers exit or enter following a bilateral shock, altering the market shares of incumbents. In the latter case the market-share equation would include an extra term for the change in the number of firms in the market.

A.1.15 Heterogeneous Price and Share Response

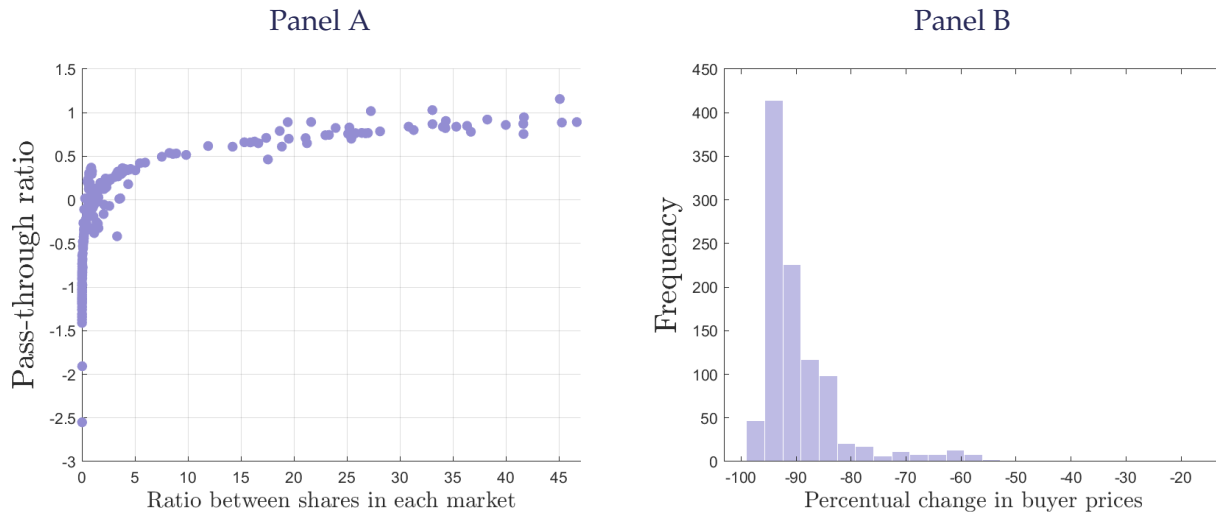
In my model, deviations from relative PPP at the aggregate level arise from buyer market power and buyer market concentration. In this section I present model results on buyers' pricing decisions and the role that heterogeneity in size plays in generating the main findings. I (i) demonstrate this numerically, (ii) provide intuition, and (iii) include a mathematical proof.

Numerical Simulation. Figure 6 presents a simplified economy with two buyers per country, subject to a 1% exchange rate shock in every country. I evaluate the two ERPT terms numerically to understand the dispersion in pricing behaviour.

Panel A emphasises the role of market share in price setting post-shock. The Y-axis measures pass-through as the log ratio of price changes between the two buyers, $\log(\Delta p_{1jk} / \Delta p_{2jk})$; the X-axis shows the ratio of their market shares. Buyers in countries with larger market-share ratios exhibit greater relative pass-throughs, even though every buyer faces an identical shock.

Panel B shows that, despite the identical shock, prices change by different amounts, indicating incomplete pass-through. The histogram of buyers' percentage price changes reveals substantial heterogeneity in pricing-to-market at the firm level. Figure 6 therefore demonstrates that aggregate price-index movements produced by the model are accounted for by the pricing behaviour of large firms.

Figure 6: Heterogeneous Price Response to a 1% Exchange Rate Shock



Notes: Panel A displays the relationship between the ratio of buyers' price changes after a 1% exchange rate increase and the ratio of their pre-shock market shares. Buyers with a larger share increase their price more. Panel B shows the distribution of buyers' percentage price changes after the same shock, illustrating incomplete pass-through.

Intuition. An increase in the exchange rate affects prices through two channels. First, lower input costs (the buyer pays less in its own currency) raise the buyer's willingness to pay—this *direct effect* is identical for every buyer. Second, each buyer's markdown changes by a different amount because buyers differ in productivity and therefore in market share. The supply elasticity a buyer faces depends on its market share, so the *indirect effect*—the feedback from the shock through markdown adjustment—differs across buyers. Adding both effects, the total price change is heterogeneous across

buyers, leading to heterogeneous market-share changes and incomplete aggregate pass-through.

Theoretical Proof. I establish conditions under which pass-through is heterogeneous in general equilibrium. For clarity, I work with a simplified version of the model: one market (a country–product combination), two buyers, and perfectly elastic substitution of inputs within a buyer. This lets me focus on the markdown effect.

Define each buyer's price as a function of p_1 and p_2 :

$$\begin{aligned} MRP_1 &= z_1, \quad S_1 = \frac{p_1^{1+\eta}}{p_1^{1+\eta} + p_2^{1+\eta}}, \quad \varepsilon_1 = \left(\frac{p_1^{1+\eta}}{p_1^{1+\eta} + p_2^{1+\eta}} \right) (\theta - \eta) + \eta, \\ \mu_1 &= \frac{p_1^{1+\eta}(1 + \theta) + p_2^{1+\eta}(1 + \eta)}{p_1^{1+\eta}\theta + p_2^{1+\eta}\eta}. \end{aligned}$$

Substituting into the price equation:

$$p_1 = \frac{p_1^{1+\eta}\theta + p_2^{1+\eta}\eta}{p_1^{1+\eta}(1 + \theta) + p_2^{1+\eta}(1 + \eta)} z_1 e. \quad (\text{A.13})$$

Taking logs and differentiating:

$$d \ln p_1 = \frac{(1 + \eta)p_1^\eta \theta dp_1 + (1 + \eta)p_2^\eta \eta dp_2}{p_1^{1+\eta}\theta + p_2^{1+\eta}\eta} - \frac{(1 + \eta)p_1^\eta(1 + \theta) dp_1 + (1 + \eta)p_2^\eta(1 + \eta) dp_2}{p_1^{1+\eta}(1 + \theta) + p_2^{1+\eta}(1 + \eta)} + d \ln e. \quad (\text{A.14})$$

Rearranging gives

$$d \ln p_1 = d \ln p_2 \frac{\Gamma_1}{1 - (1 + \eta)\phi_1} + d \ln e \frac{1}{1 - (1 + \eta)\phi_1}, \quad (\text{A.15})$$

where

$$\begin{aligned} \Gamma_i &\equiv \frac{\eta p_j^{1+\eta}}{\theta p_i^{1+\eta} + \eta p_j^{1+\eta}} - \frac{(1 + \eta) p_j^{1+\eta}}{(1 + \theta) p_i^{1+\eta} + (1 + \eta) p_j^{1+\eta}}, \\ \phi_i &\equiv \frac{\theta p_i^{1+\eta}}{\theta p_i^{1+\eta} + \eta p_j^{1+\eta}} - \frac{(1 + \theta) p_i^{1+\eta}}{(1 + \theta) p_i^{1+\eta} + (1 + \eta) p_j^{1+\eta}}. \end{aligned}$$

The relative price change is therefore

$$\frac{d \ln p_1}{d \ln p_2} = \frac{1 - (1 + \eta)\phi_2 + \Gamma_1}{1 - (1 + \eta)\phi_1 + \Gamma_2}, \quad (\text{A.16})$$

which equals one only if $p_1 = p_2$. To see this, note that equality requires

$$\frac{(1+\eta)(1+\theta)(p_1^{2(1+\eta)} - p_2^{2(1+\eta)})}{\left((1+\eta)p_1^{1+\eta} + (1+\theta)p_2^{1+\eta}\right)\left((1+\theta)p_1^{1+\eta} + (1+\eta)p_2^{1+\eta}\right)} = \frac{\eta\theta(p_1^{2(1+\eta)} - p_2^{2(1+\eta)})}{\left(\eta p_1^{1+\eta} + \theta p_2^{1+\eta}\right)\left(\theta p_1^{1+\eta} + \eta p_2^{1+\eta}\right)}.$$

The scalar coefficients on each side are not equal for all p_1, p_2 , so the equation holds only when $p_1^{2(1+\eta)} = p_2^{2(1+\eta)}$, i.e. $p_1 = p_2$. Therefore,

$$d \ln p_1 = d \ln p_2 \iff p_1 = p_2.$$

Only buyers with identical prices—hence identical characteristics—have the same pass-through.

A.1.16 Marginal Revenue Effect

I evaluate the marginal revenue channel under three alternative production technologies: (a) one-input, (b) Cobb–Douglas, and (c) CES.

Case I: One Input.

$$Q_{bk}^{finalg} = \left(\frac{q_{bjk}}{\sigma}\right)^\sigma.$$

Assuming the buyer's output price does not change:

$$\begin{aligned} MRP &= \frac{\partial \text{revenues}}{\partial q_{bjk}} = \sigma \frac{Q_{bk}^{finalg}}{q_{bjk}}, \\ \frac{d \ln MRP_{bjk}}{d \ln p_{bjk}} &= \frac{d \ln MRP_{bjk}}{d \ln q_{bjk}} \frac{d \ln q_{bjk}}{d \ln p_{bjk}} = \frac{d MRP}{d q_{bjk}} \frac{q_{bjk}}{MRP_{bjk}} \epsilon_{bjk} = (\sigma - 1) \epsilon_{bjk}. \end{aligned}$$

Case II: Cobb–Douglas.

$$Q_{bk}^{finalg} = \prod_{j \in \mathcal{J}} \left(\frac{q_{bjk}}{\sigma}\right)^\sigma.$$

Assuming the buyer's output price does not change:

$$\begin{aligned} \frac{d \ln MRP_{bjk}}{d \ln p_{bjk}} &= \left[-\frac{Q_{bk}^{finalg}}{q_{bjk}^2} \frac{\partial q_{bjk}}{\partial p_{bjk}} + \frac{\partial Q_{bk}^{finalg}}{\partial q_{bjk}} \frac{\partial q_{bjk}}{\partial p_{bjk}} \frac{1}{q_{bjk}} + \sum_{j' \in \mathcal{J} \setminus \{j\}} \frac{\partial Q_{bk}^{finalg}}{\partial q_{bj'k}} \frac{\partial q_{bj'k}}{\partial p_{bjk}} \frac{1}{q_{bjk}} \right] \frac{p_{bjk}}{MRP_{bjk}} \\ &= \epsilon_{bjk}(\sigma - 1) + \sum_{j' \in \mathcal{J} \setminus \{j\}} \frac{\partial Q_{bk}^{finalg}}{\partial q_{bj'k}} \frac{\partial q_{bj'k}}{\partial p_{bjk}} \frac{1}{q_{bjk}} \frac{p_{bjk}}{MRP_{bjk}} < 0. \end{aligned}$$

Case III: CES.

With a CES production function:

$$Q_{bk}^{finalg} = \left(\sum_{j' \in \mathcal{J}} q_{bj'k}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}.$$

The marginal revenue product satisfies:

$$MRP_{bjk} \propto \left(\frac{Q_{bk}^{finalg}}{q_{bjk}} \right) \frac{q_{bjk}^{\frac{\sigma-1}{\sigma}}}{\sum_{j' \in \mathcal{J}} q_{bj'k}^{\frac{\sigma-1}{\sigma}}} = x_{bjk} \frac{Q_{bk}^{finalg}}{q_{bjk}},$$

where x_{bjk} is Buyer b 's expenditure share on Product j .⁶⁹

Totally differentiating in logs:

$$d \ln MRP_{bjk} = d \ln x_{bjk} + d \ln Q_{bk}^{finalg} - d \ln q_{bjk}.$$

We obtain:

$$\begin{aligned} d \ln x_{bjk} &= \frac{\sigma-1}{\sigma} d \ln q_{bjk} - \frac{\sigma-1}{\sigma} x_{bjk} d \ln q_{bjk} = \frac{\sigma-1}{\sigma} (1 - x_{bjk}) \frac{d \ln q_{bjk}}{d \ln p_{bjk}}, \\ d \ln Q_{bk}^{finalg} &= x_{bjk} \frac{d \ln q_{bjk}}{d \ln p_{bjk}}. \end{aligned}$$

Substituting:

$$d \ln MRP_{bjk} = \left(\frac{\sigma-1}{\sigma} - 1 \right) (1 - x_{bjk}) \frac{d \ln q_{bjk}}{d \ln p_{bjk}} = \frac{-1}{\sigma} (1 - x_{bjk}) \frac{d \ln q_{bjk}}{d \ln p_{bjk}},$$

so that

$$\frac{d \ln MRP_{bjk}}{d \ln p_{bjk}} = \frac{-1}{\sigma} (1 - x_{bjk}) \epsilon_{bjk}.$$

A.1.17 Extensive Margin: Fixed Cost of Importing

I introduce a fixed cost of importing, κ_m , denominated in buyers' currency. A firm can avoid paying κ_m by setting its imports to zero. This extension accounts for changes in buyer market shares arising from entry and exit.

The buyer's problem is essentially identical to the baseline except for an extra fixed cost term. Specifically, a firm imports if⁷⁰

$$z_{bjk} (q_{bjk})^\alpha - \frac{1}{e_k} p_{bjk} q_{bjk} \geq \frac{1}{e_k} \kappa_{bjk}.$$

⁶⁹Equivalently, this product or input index may be denoted by s , as in the earlier version of the derivation.

⁷⁰Following [Atkeson and Burstein \(2008\)](#), I arrange buyers within each market by physical productivity. I focus on equilibria where firms make sequential entry decisions: the most productive firm decides first, followed by the next most productive (assuming no less productive firm is already importing), and so on. The specific ordering has minimal quantitative impact once the model is calibrated to match the high concentration observed in the data.

The primary distinction from the baseline is the expression for the change in buyer market shares following an exchange rate shock. The buyer market share is:

$$S_{bjk} = \frac{p_{bjk}^{1+\eta}}{\sum_{z=1}^{B(e_k)} p_{zjk}^{1+\eta}},$$

where the number of buyers B now depends on the exchange rate e_k . The market-share log-change includes an extra term for buyer entry and exit:

$$d\ln S_{bjk} = (1 + \eta) \left(1 - S_{bjk} - \sum_{z=1}^{B(e_k)} S_{zjk} \frac{d \ln p_{zjk}}{d \ln p_{bjk}} - \underbrace{p_{B(e_k)}^{1+\eta} \frac{dB(e_k)/de_k}{dp_{bjk}/de_k}}_{\text{entry/exit of buyers}} \right) d\ln p_{bjk}.$$

I expect $\frac{dB(e_k)}{de_k} < 0$ since a depreciation (increase in e_k) reduces the real fixed import cost, encouraging more buyers to enter. I also expect $\frac{dp(e_k)}{de_k} > 0$.⁷¹

⁷¹With a continuum of buyers the expression becomes:

$$\frac{d\ln S(b', j, k)}{d\ln p(b', j, k)} = (1 + \eta) \left(1 - \int_0^1 S(b, j, k) \frac{\partial \ln p(b, j, k)}{\partial \ln p(b', j, k)} db \right) - p^{1+\eta}(B(e_k), j, k) \frac{dB(e_k)}{d \ln p(b', j, k)}.$$

A.2 Appendix: Empirical Part

A.2.1 Data Sources

Table 7: Data Sources

Data	Data Source	Note
Colombian Exports	DANE (2007-2020)	Data accessed through Datamyne
Colombian Imports	DANE (2007-2020)	Data accessed through Datamyne
Rauch Classification	Rauch (1999)	4-digit SIC Rev. 3.1 classification
Exchange Rates Shocks	IMF (2007-2020)	Bilateral nominal exchange rates

Notes: This table shows a summary of the datasets I combine in the empirical section and their sources.

A.2.2 Buyer Size and Price Level

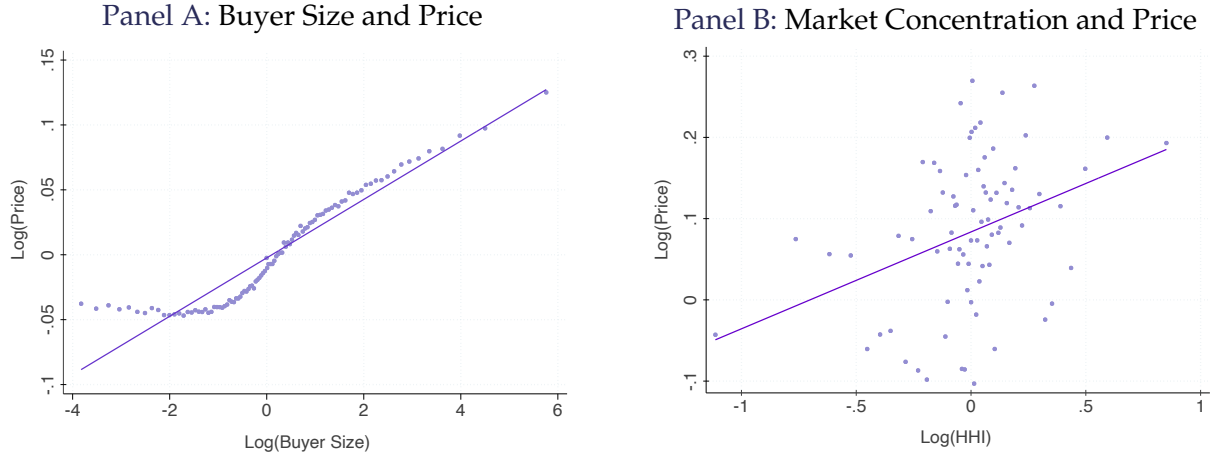
In this section, I explore the relationship in the data between the size of the buyer and the price. For doing so, I run the following regression:

$$\ln(\text{price}_{s,b,j,k,t}) = \xi BS_{b,j,k,t} + FE_{s,j,k,t} + X_{k,t} + \epsilon_{s,b,j,k,t} \quad (\text{A.17})$$

where $\ln(\text{price}_{s,b,j,k,t})$ is the price of Product j , Seller s charges to Buyer b at Destination k in period t and $X_{k,t}$ are control variables at the country and time level. Figure 7 Panel A shows that the price of the same product, sold to the same destination in a given year is increasing in the buyer's size. This is true, even controlling for destination and time specific variables. The reason for this is that even though the markdowns μ for firms with higher market shares are larger (meaning they pay lower prices relative to MRP), the marginal-revenue product for larger firms is also larger. Therefore, the net effect on the price level can go either way and empirically the estimated slope is positive.⁷²

⁷²Note that this result is analogous to [Berger, Herkenhoff and Mongey \(2022\)](#) where large firms pay higher wages.

Figure 7: Proposition I and II

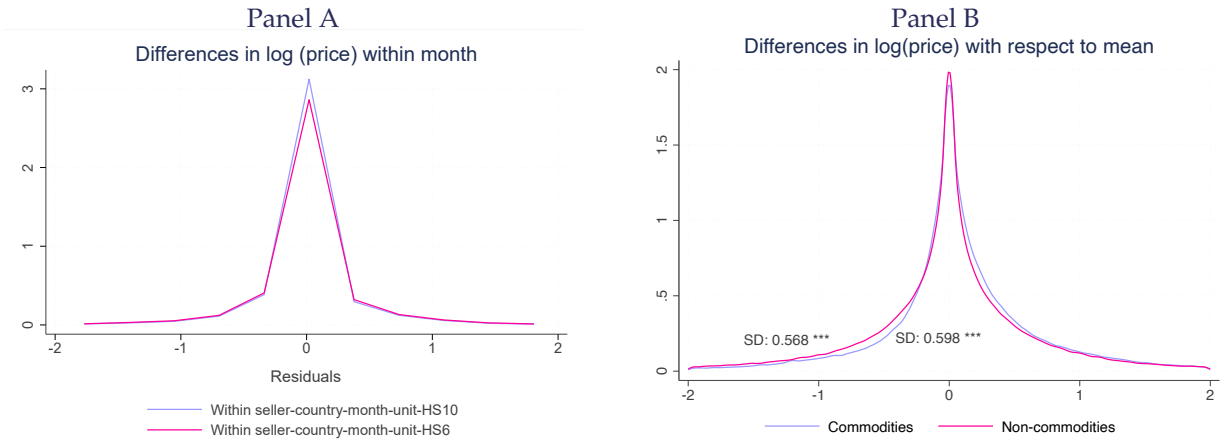


Notes: This figure illustrates Propositions I and II using Colombian data. Panel A examines the relationship between buyer market share and transaction-level log prices of the inputs, while Panel B explores the relationship between the average log price in a market (Year-Destination-HS10 combination) and buyer market concentration, measured by the Herfindahl-Hirschman Index of the market.

A.2.3 Robustness: Price Dispersion

In this section, I investigate whether price differences can be attributed to various factors, as opposed to the same seller discriminating among buyers. First, I examine the nature of the product being sold; differentiated products are more likely to exhibit price variations. Second, given the high volatility of exchange rates, differences in prices could potentially be explained by monthly fluctuations in the price of the goods. To test this hypothesis, I analyze price dispersion at the monthly level.

Figure 8: Price Dispersion



Notes: This figure illustrates the heterogeneity in prices in a different time period and for different types of products. Panel A shows the price dispersion for monthly prices, which represents the range of prices for a given HS10 product, destined for a specific location and offered by a particular seller within the same month. Panel B highlights the variations in price dispersion based on the type of product, distinguishing between commodities and non-commodities.

Table 8: Price Dispersion in different HS levels

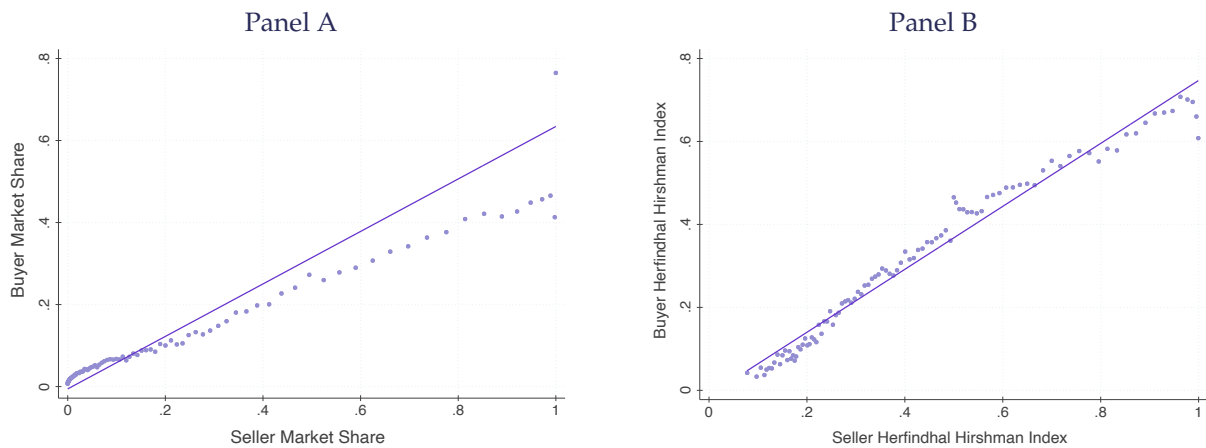
	Products	Products with different prices	%
HS6	3654	2299	62.9
HS10	4889	2811	57.5

Notes: The table shows how many products have variations in their prices depending on the HS level.

A.2.4 Assortative Matching

Figure 9 shows that large buyers are more likely to trade with larger sellers and that highly concentrated markets in terms of buyers are also highly concentrated in terms of sellers.

Figure 9: Assortative Matching



Notes: This figure illustrates the correlation between major buyers and sellers and market concentration in terms of both buyers and sellers. Panel A displays the correlation between buyer market share and seller market share within a transaction, while Panel B demonstrates the correlation between the HHI for buyer market concentration and the HHI for seller market concentration.

A.2.5 Robustness: Seller Market Power

In this section, I include a variable in the baseline regressions that isolates the buyer market power effect from the seller side. Table 9 shows that estimates are still significant and have the expected sign.

Table 9: Robustness with Seller Size

	(1)	(2)	(3)	(4)	(5)
	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$
$\Delta \ln ER$	0.106** (0.0488)	0.116** (0.0506)	-0.0178 (0.0408)	0.0218 (0.0428)	0.0160 (0.0433)
S_{t-1}	-0.0198*** (0.00571)	-0.0425*** (0.00613)			
$\Delta \ln ER \times S_{t-1}$	-0.118** (0.0485)	-0.124* (0.0654)			
X_{t-1}	-0.0146* (0.00805)		-0.0357*** (0.00695)	-0.0458*** (0.00777)	
$\Delta \ln ER \times X_{t-1}$	0.00952 (0.0364)		0.0673 (0.0497)	0.0722 (0.0510)	
$Xhsyear_{t-1}$		-0.000597 (0.0144)			
$\Delta \ln ER \times Xhsyear_{t-1}$		-0.00642 (0.0333)			0.126* (0.0682)
$Shsyear_{t-1}$			-0.0586*** (0.0196)	-0.0543*** (0.0193)	-0.0682*** (0.0194)
$\Delta \ln ER \times Shsyear_{t-1}$			-0.354*** (0.128)	-0.346*** (0.129)	-0.357*** (0.133)
HS - Year - Seller FE			✓	✓	✓
Country-HS-Seller FE	✓	✓			
Country FE				✓	✓
Buyer FE		✓			
N	499308	480223	525080	525080	525080

Standard errors in parentheses
* p<0.1, ** p<0.05, *** p<0.01

Notes: The table shows results for equation 3.3 that include a variable corresponding to the size of the seller (X_{t-1}) at the product-destination-time level, $Xhsyear_{t-1}$) at the product-time level and $Shsyear_{t-1}$ corresponding to the size of the buyer at the product-year level. One of the columns includes an alternative measure for buyer size (S_{t-1}). Standard errors are clustered at the country-time level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.2.6 Robustness: Import Intensity and Exchange Rate Pass-through

Table 10 includes information on the import intensity when evaluating the exchange rate pass-through.

Table 10: Import Intensity and Exchange Rate Pass-through

	(1)	(2)	(3)
	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$
$\Delta \ln ER$	0.142*** (0.0538)	0.157*** (0.0565)	
$S_{.t} - 1$	-0.0492*** (0.00647)	-0.0546*** (0.00704)	-0.0493*** (0.00909)
$\Delta \ln ER \times S_{.t} - 1$	-0.103* (0.0610)	-0.136* (0.0694)	-0.245*** (0.0654)
Import Intensity	0.000287 (0.000662)	-0.00106 (0.000678)	-0.000153*** (0.0000348)
$\Delta \ln ER \times \text{Import Intensity}$	0.00733*** (0.00192)	0.0134*** (0.00225)	0.00210*** (0.000446)
Country-HS FE	✓		
Period-Seller FE	✓	✓	
Country-Seller FE	✓		✓
Country-HS-Seller FE		✓	
Period-Country-HS FE			✓
N	604221	562960	524209

Notes: The table shows results for Equation 3.3 including as a control the interaction between log changes in the exchange rate and import intensity of the exporter, named *Import Intensity*. Standard errors are clustered at the country-time level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.2.7 Robustness: Other measures of Buyer Market Power

Table 11 includes other measures of buyer market share. Columns (1)-(2) include an interaction of the exchange rate with buyer market share defined as Buyer b 's share of the nominal value of all exports of Product j in Year t ($Shsyear_{t-1}$).

Table 11: Robustness with Other Measures of Buyer Market Power

	(1)	(2)	(3)	(4)
	$\Delta \text{Log(Prices)}$	$\Delta \text{Log(Prices)}$	$\Delta \text{Log(Prices)}$	$\Delta \text{Log(Prices)}$
$\Delta \ln ER$	0.0527 (0.0378)		0.0703* (0.0381)	
$Shsyear_{t-1}$	-0.0902*** (0.0263)	-0.0906*** (0.0260)		
$\Delta \ln ER \times Shsyear_{t-1}$	-0.363** (0.152)	-0.334** (0.153)		
$Xseller_{t-1}$			-0.0583*** (0.00690)	-0.0574*** (0.00693)
$\Delta \ln ER \times Xseller_{t-1}$			-0.0701 (0.0476)	-0.0947* (0.0515)
HS - Year - Seller FE	✓	✓	✓	✓
Country-HS-Seller FE	✓	✓		
Country-Year FE		✓		✓
Country FE			✓	
N	494237	494211	525080	525063

Notes: The table shows results for Equation 3.3 including alternative measures of buyer market power. Standard errors are clustered at the country-time level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.2.8 Quantity Discounts

Large buyers may receive lower prices or different pass-through because they place larger orders. To evaluate this explanation, the baseline equation is re-estimated after controlling for log quantity and its interaction with the exchange-rate shock. This specification asks whether the buyer-size gradient remains after conditioning directly on shipment scale.

Table 12: Quantity discounts and the buyer-size pass-through gradient

	(1)	(2)
	Paper baseline	Quantity controls
$\Delta \ln ER \times \text{buyer share}$	-0.164** (0.0789)	-0.156* (0.0939)
Log quantity		-0.0331*** (0.00293)
$\Delta \ln ER \times \text{log quantity}$		0.0359** (0.0153)
Quantity controls	No	Yes
Period-seller FE	Yes	Yes
Country-HS-seller FE	Yes	Yes
Observations	517100	528136

Notes: Standard errors in parentheses. The table reports the coefficients most directly related to the quantity-discount explanation. Column 1 reproduces the paper baseline. Column 2 controls for log shipment quantity and its interaction with the exchange-rate shock; lower-order terms are included in the specification. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 12 shows that the negative buyer-size gradient is not simply a quantity-discount result. After adding log quantity and its exchange-rate interaction, the buyer-share interaction remains negative, -0.156 . The quantity interaction is positive and statistically significant, indicating that shipment scale is empirically relevant, but it does not remove the negative buyer-size pass-through gradient. These checks support the interpretation that the baseline gradient is tied to foreign-buyer size rather than only to shipment scale.

A.2.9 Robustness: Length of Contracts and Volatility Unrelated to Exchange Rate Shocks

Barro (1977) and Carlton (1991) argue that buyer–seller prices could be less responsive to shocks due to the use of contracts which specify fixed prices for a period of time. Given the existence of long-term relationships might be more likely to use either implicit or explicit contracts, they could exhibit lower pass-through of shocks (Heise, 2019). Table 13 shows different specifications that aim to control for the length of the relationship in my baseline regression.

Table 13: Robustness with length of buyer-seller relationships

	(1)	(2)	(3)
	$\Delta \text{Log(Prices)}$	$\Delta \text{Log(Prices)}$	$\Delta \text{Log(PricesBS)}$
$\Delta \ln ER$	0.0237 (0.0564)	0.178** (0.0714)	0.204*** (0.0725)
$Tenure_t$	0.00552 (0.0109)	0.00718 (0.00473)	
$\Delta \ln ER \times Tenure_t$	0.0607* (0.0332)	-0.0152 (0.0248)	
S_{t-1}		-0.0567*** (0.00682)	-0.0563*** (0.00693)
$\Delta \ln ER \times S_{t-1}$		-0.134* (0.0686)	-0.129* (0.0682)
$TenureHS_t$			0.00576 (0.00529)
$\Delta \ln ER \times TenureHS_t$			-0.0394 (0.0290)
HS - Dest FE	✓	✓	✓
Dest - HS - Seller FE	✓		
Year FE			
Buyer FE	457488	562960	562960

Notes: The table presents results for Equation 3.3, which includes variables corresponding to the tenure of the relationship between a buyer and a seller. The variable $Tenure_t$ represents the number of years the buyer and the seller have been interacting prior to the transaction at time t . The variable $TenureHS_t$ indicates the number of years the buyer and the seller have interacted but within the same HS10 product category. Standard errors are clustered at the country-time level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

In all the cases, even though the fact that longer relationships have less change in prices in the buyer currency, they do not seem to be explaining the mechanism this paper proposes.

A.2.10 Robustness: Dominant Currency Paradigm

In this section, I replicate my findings following the data cleaning and specification of [Gopinath et al. \(2020\)](#). First, I restrict the data to the manufacturing sector, using the HS codes proposed in the paper. Second, I start as a benchmark specification with [Gopinath et al. \(2020\)](#)'s main regression. Table 14 shows that including time fixed effects reduces the coefficient in magnitude, though it remains statistically significant with the same sign.

Table 14: Dominant Currency Paradigm and Exchange Rate Pass-Through

	(1)	(2)	(3)
	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$
$\Delta \ln ER = \alpha$	0.887*** (0.284)	0.464*** (0.162)	0.108* (0.0601)
BS_{t-1}		-0.0170 (0.0103)	-0.0330*** (0.00934)
$\Delta \ln ER \times S_{t-1} = \beta$		-0.395** (0.196)	-0.149** (0.0721)
Country-HS-Seller	x	x	x
HS - Period FE			
HS - Period - Seller FE			x
Year FE		x	
N	165100	170796	163463

Notes: The table shows results for an equivalent version of equation 3.3 in [Gopinath et al. \(2020\)](#). Standard errors are clustered at the country-time level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.2.11 Robustness: Mechanical Endogeneity and Market Definition of Buyer Shares

The baseline analysis defines buyer size within a foreign market. This is the relevant object for the reduced-form pass-through exercise because exchange-rate shocks vary at the destination-period level and the customs data identify buyers, sellers, products, destinations, and years. This appendix disciplines the share measure in two ways. First, because buyer shares are constructed from transaction values, it verifies that the pass-through gradient is not driven by a mechanical link between the seller's own unit value and the share. Second, it clarifies that the destination-product-period share measures exposure to a foreign buyer in the market where the exchange-rate shock is observed, rather than the seller's total outside option.

For this check, the buyer share is reconstructed leave-one-seller-out. For seller s , buyer b , product i , destination d , and period t , define

$$S_{bid-1}^{-s} = \frac{\sum_{s' \neq s} X_{s'bid-1}}{\sum_{b'} \sum_{s' \neq s} X_{s'b'id-1}},$$

where X denotes export sales. This measure removes seller s 's own transactions from both the numerator and the denominator of the buyer share. The interaction is then identified from the buyer's importance among other Colombian sellers in the same destination-product-period market, not from the seller's own price or quantity.

The buyer-share measure should therefore be interpreted as foreign-buyer exposure where exchange-rate shocks and export prices are observed. This does not require assuming that sellers lack other outlets: the domestic-market evidence in [Appendix A.2.12](#) documents that Colombian manufacturers have a substantial domestic market outside the export transaction.

The table reports two alternative share definitions that discipline the interpretation of buyer size. The leave-one-seller-out share directly addresses mechanical endogeneity because it removes seller s 's

own transaction value from the share. The seller-buyer share is closer to a bilateral bargaining object and asks whether pass-through varies with the buyer's importance to the relationship.

Table 15: Buyer-share definitions and exchange-rate pass-through

	(1) Paper baseline	(2) Leave-one-out	(3) Seller-buyer
$\Delta \ln ER$	0.148*** (0.0547)	0.195** (0.0773)	0.175*** (0.0616)
Buyer share	-0.0481*** (0.00684)		
$\Delta \ln ER \times$ buyer share	-0.164** (0.0789)		
Leave-one-out share		0.0256* (0.0153)	
$\Delta \ln ER \times$ leave-one-out share		-0.351*** (0.135)	
Seller-buyer share			-0.0422*** (0.00790)
$\Delta \ln ER \times$ seller-buyer share			-0.184** (0.0860)
Period-seller FE	Yes	Yes	Yes
Country-HS-seller FE	Yes	Yes	Yes
Share definition	Main paper	Leave-one-seller-out	Seller-buyer
Observations	517100	139485	411090

Notes: Standard errors in parentheses. Column 1 reproduces the paper baseline. Columns 2–3 use the cleaned export-regression file. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table 15 shows that the negative pass-through gradient is not driven by the mechanical construction of the share. The baseline interaction is -0.164 , indicating that exchange-rate pass-through is lower in transactions involving larger buyers. When buyer size is measured leave-one-seller-out, the interaction becomes more negative, -0.351 , so the gradient strengthens after removing the seller's own sales from the construction of buyer size. The bilateral seller-buyer share also delivers a negative interaction, -0.184 , showing that the result is robust to a measure closer to the buyer's importance within the relationship. Taken together, these estimates support the reduced-form claim that pass-through is systematically lower for transactions with larger foreign buyers, while using share definitions that address the main mechanical-endogeneity and bilateral-outside-option issues.

A.2.12 Robustness: Domestic Sales and the Outside Option

A related question is how large the domestic market is relative to exports. This matters because the paper's buyer-share measure is defined within foreign markets, where exchange-rate shocks and export prices are observed. To document the scale of domestic absorption, this appendix assembles domestic-sales measures from DANE's Encuesta Anual Manufacturera (EAM) historical annexes. DANE reports, by industrial group and year, sales of products manufactured by the establishment in Colombia and abroad. It also reports product-level sales by CPC code, including total sales and sales abroad. These data allow a direct description of the size of domestic absorption relative to exports.

For industry g and year t , define

$$DomesticShare_{gt} = \frac{Sales_{gt}^{domestic}}{Sales_{gt}^{domestic} + Sales_{gt}^{export}}.$$

At the CPC-product level, the same object is computed using total product sales less sales abroad. In 2007, the CPC annex reports the percent sold abroad rather than the value sold abroad, so export sales are recovered as total sales times the reported export percent.

Table 16: Domestic sales in Colombian manufacturing, DANE EAM

Year	CIU	Industry shares		CPC product shares	
		Domestic	Export	Weighted domestic	Median domestic
2007	Rev. 3	83.8	16.2	83.7	100.0
2008	Rev. 3	82.2	17.8	82.1	99.2
2009	Rev. 3	82.9	17.1	82.8	99.3
2010	Rev. 3	81.4	18.6	81.3	99.6
2011	Rev. 3	84.2	15.8	84.1	99.7
2012	Rev. 3	86.2	13.8	86.1	99.6
2013	Rev. 4	88.1	11.9	88.1	99.7
2014	Rev. 4	85.5	14.5	85.5	99.7
2015	Rev. 4	83.8	16.2	83.8	99.7
2016	Rev. 4	83.9	16.1	83.9	99.6
2017	Rev. 4	84.7	15.3	84.7	99.5
2018	Rev. 4	84.7	15.3	84.7	99.5
2019	Rev. 4	84.1	15.9	84.1	99.4
2020	Rev. 4	83.9	16.1	83.9	99.6

Notes: Shares are percentages. Industry shares use DANE EAM table 5.15; CPC shares use table 6.2. Domestic CPC sales equal total sales less exports; in 2007 exports are recovered from the reported percent sold abroad.

Appendix Table 16 shows that domestic absorption is large. At the national manufacturing level, the domestic share of sales ranges from 81.4 percent to 88.1 percent between 2007 and 2020, with an average of 84.2 percent. The product-level CPC calculation gives nearly the same sales-weighted domestic share, averaging 84.2 percent. The median CPC product is even more domestic-oriented: the median domestic-sales share is about 99.6 percent.

The DANE data can also be used to incorporate the domestic market directly into the buyer-size measure. The analysis constructs two domestic-market adjustments. First, the baseline buyer share is scaled by the lagged DANE export-sales share of the matched manufacturing industry, so the same foreign-buyer share receives less weight in industries where most sales are domestic. Second, a broader HS6-year buyer share adds DANE domestic sales to Colombian exports in the denominator and identifies buyers in the upper tail of this domestic-inclusive exposure distribution.

Appendix Table 17 shows that the buyer-size pass-through gradient remains visible when the domestic market is incorporated into the exposure measure. The continuous domestic-adjusted share has a negative interaction with exchange-rate changes. Using the broader HS6-year denominator that includes domestic sales, transactions with buyers in the top quartile of domestic-inclusive exposure have 4.9 percentage points lower pass-through, and transactions with buyers in the top decile have 9.4 percentage points lower pass-through. The DANE exercise therefore strengthens the main interpretation: Colombian sellers have meaningful domestic outlets, but pass-through is still systematically lower in transactions with larger foreign-buyer exposure.

A.2.13 Robustness: Alternative Samples

Table 17: Domestic-market exposure and exchange-rate pass-through

	(1)	(2)	(3)
	Adj. share	Top quartile	Top decile
$\Delta \ln ER$	0.153*** (0.0428)	0.154*** (0.0396)	0.151*** (0.0380)
Domestic-adjusted buyer share	-0.209*** (0.0249)		
$\Delta \ln ER \times$ domestic-adjusted share	-0.346* (0.192)		
$\Delta \ln ER \times$ top quartile		-0.0494** (0.0225)	
$\Delta \ln ER \times$ top decile			-0.0940*** (0.0295)
Exposure measure	Continuous	Top quartile	Top decile
Domestic market	DANE export share	In denominator	In denominator
Main FE	Yes	Yes	Yes
Observations	460,710	493,770	493,770

Notes: The dependent variable is the change in log unit values. The domestic-adjusted buyer share equals the baseline buyer share multiplied by the lagged DANE export-sales share of the matched manufacturing industry. The domestic-inclusive buyer share adds DANE domestic sales to Colombian exports in the HS6-year denominator; columns 2 and 3 report interactions for buyers in the top quartile and top decile of that distribution. All regressions include destination-product-seller and seller-period fixed effects. Standard errors are clustered by destination-period. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 18: Exchange Rate Pass-through into prices

	OECD Destinations		All Destinations			
	(1)	(2)	(3)	(4)	(5)	(6)
	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$	$\Delta \text{Log(Price)}$
$\Delta \ln ER$	0.356*** (0.0836)	0.357*** (0.0862)	0.200*** (0.0542)	0.172*** (0.0577)	0.158*** (0.0603)	0.158*** (0.0479)
S_{t-1}	-0.0797*** (0.0104)	-0.0889*** (0.0119)	-0.0338*** (0.0110)	-0.0552*** (0.0145)	-0.0707*** (0.00947)	-0.0402*** (0.00752)
$\Delta \ln ER \times S_{t-1}$	-0.442*** (0.0837)	-0.475*** (0.0879)	-0.350*** (0.0890)	-0.258** (0.109)	-0.196** (0.0964)	-0.198*** (0.0639)
Period-Seller FE	✓	✓	✓	✓	✓	✓
Country-HS-Seller FE	✓	✓	✓	✓	✓	✓
N	235688	225054	196499	164783	474959	220482
Sample	A	D	C	H	D	R

Notes: Results from equation 3.3 for alternative samples. The sample includes all HS products (A) in Column (1), only commodities (C) in Column (3), only homogeneous goods (H) in Column (4), only differentiated goods (D) in Columns (2) and (5) and only referenced goods (R) in Column (6). Standard errors are clustered at the country-time level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.2.14 Robustness: Depreciations versus Appreciations

In this section I estimate the different effects for the case of a depreciation versus an appreciation. As shown in Table 19, I find the effects are stronger for depreciations. One caveat about this effect is that my sample contains few appreciation events of meaningful magnitude for COP. Potentially the reason why I find almost no effect for appreciation is the appreciation events are insignificant and reverted shortly after they occur.

Table 19: Positive or negative ΔER

	Depreciations		Appreciations	
	(1) $\Delta \text{Log(Price)}$	(2) $\Delta \text{Log(Price)}$	(3) $\Delta \text{Log(Price)}$	(4) $\Delta \text{Log(Price)}$
$\Delta \ln ER$	0.195** (0.0889)		0.146* (0.0744)	
S_{t-1}	-0.0252** (0.0107)	-0.0255** (0.0105)	0.00253 (0.0138)	0.00258 (0.0137)
$\Delta \ln ER \times S_{t-1}$	-0.207** (0.100)	-0.192* (0.0987)	0.00456 (0.128)	-0.00267 (0.128)
Period-Seller FE	✓		✓	
Country-HS-Seller FE	✓	✓	✓	✓
Country-Period FE		✓		✓
N	297342	297580	171620	174940

Notes: This table presents the results derived from equation 3.3 under two scenarios: first, when the exchange rate change signifies an appreciation of the Colombian currency, shown in Columns (1) and (2); and second, when it indicates a depreciation of the Colombian currency, displayed in Columns (3) and (4). Standard errors are clustered at the country-time level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.2.15 Robustness: Prices versus Quantities

Table 20: Robustness with Disaggregated Price and Quantity Effects

	(1) $\Delta \text{Log(ExportValue)}$	(2) $\Delta \text{Log(ExportValue)}$	(3) $\Delta \text{Log(Quantity)}$	(4) $\Delta \text{Log(Quantity)}$	(5) $\Delta \text{Log(Price)}$	(6) $\Delta \text{Log(Price)}$
$\Delta \ln ER$	0.275** (0.1000)	0.319** (0.0834)	0.133 (0.122)	0.231 (0.0921)	0.142** (0.0548)	0.129** (0.0474)
S_{t-1}	-1.211** (0.0183)		-1.158** (0.0183)		-0.0530** (0.00696)	
$\Delta \ln ER \times S_{t-1}$	-0.00597 (0.128)		0.169 (0.139)		-0.175 (0.0795)	
$Shsyear_{t-1}$		-1.392** (0.0334)		-1.365** (0.0327)		-0.0348** (0.0108)
$\Delta \ln ER \times Shsyear_{t-1}$		-0.489** (0.173)		-0.394 (0.186)		-0.220 (0.103)
Period-Seller FE	✓	✓	✓	✓	✓	✓
Country-HS FE		✓		✓		✓
Country-HS-Seller FE	✓		✓		✓	
N	536252	564720	536252	564720	536252	578890

Notes: The table displays results for equation 3.3 under different scenarios. Columns (1) and (2) present outcomes when the dependent variable is the exported value. Columns (3) and (4) showcase results when the dependent variable is the quantity exported, while Columns (5) and (6) depict results considering prices. Standard errors are clustered at the country-time level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.3 Appendix: Estimation

A.3.1 Indirect Inference

Set Up the Model The baseline equilibrium is simulated using an exchange rate of one for every currency.

1. Establish the size of the economy (number of countries, and products)
2. Define the parameters' (η , θ , and σ) values.

3. Simulate a random vector of productivity weights z_{bjk} from the log-normal distribution.
4. For each country guess an initial matrix of p_{bjk} .
5. For each country, find S_{bjk} , ϵ_{bjk} , μ_{bjk} , and $\frac{q_{bjk}}{Q_k}$.
6. Given the equilibrium values, update p_{bjk} .
7. Find the distance between p_{bjk}^r and p_{bjk}^{r-1} .
8. Repeat the whole process until p_{bjk} converges.

The equilibrium after the shock is computed as follows.

1. Simulate a random vector of exchange rates from a log-normal distribution.
2. Set the initial guesses to be the solution of the baseline model.
3. Repeat the same algorithm as for the baseline model.

Estimate for σ To estimate σ using the structure of the model, I use the ratio between prices of different products from the same buyer. I derive an expression that can be estimated through regression using the available data, and its coefficient becomes a function of σ , specifically $\frac{1}{\sigma}$.

$$p_{bjk} = \frac{\epsilon_{bjk}}{1 + \epsilon_{bjk}} z_{bjk} \left(\frac{Q_{bk}^{finalg}}{q_{bjk}} \right)^{\frac{1}{\sigma}} e_k$$

$$\frac{p_{bjk}}{p_{bj'k}} = \frac{\epsilon_{bjk} (1 + \epsilon_{bj'k})}{(1 + \epsilon_{bjk}) \epsilon_{bj'k}} \frac{z_{bjk}}{z_{bj'k}} \left(\frac{q_{bj'k}}{q_{bjk}} \right)^{\frac{1}{\sigma}}$$

$$\ln \left(\frac{p_{bjk}}{p_{bj'k}} \right) \approx \omega_0 + \omega_1 \ln \left(\frac{q_{bj'k}}{q_{bjk}} \right), \quad \omega_1 = \frac{1}{\sigma}$$

Standard Errors The standard errors are the diagonal's square root of the key parameters' varcov (V) given the following specification:

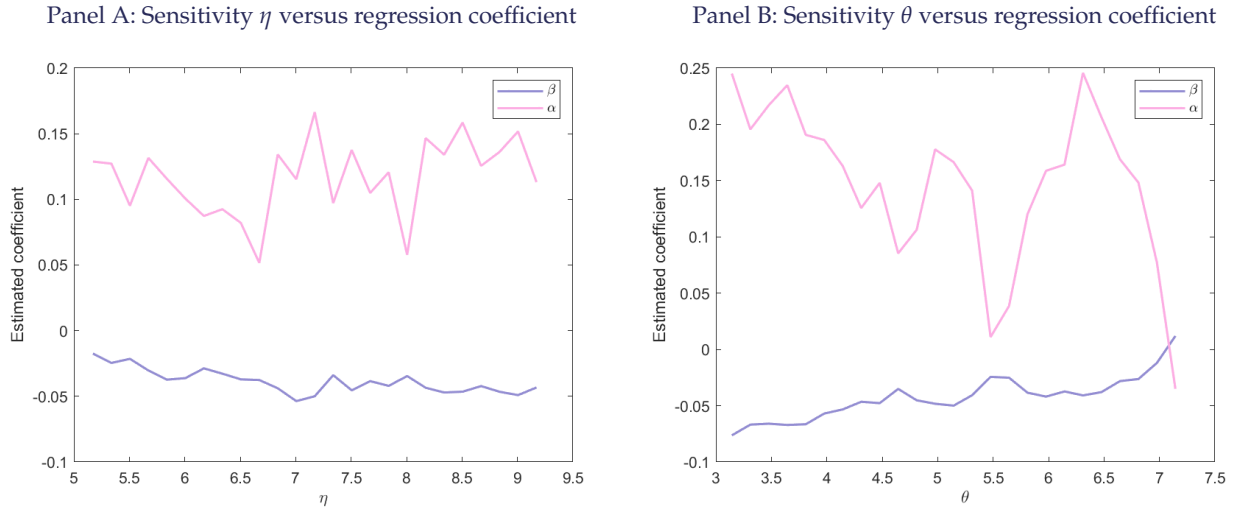
$$V = \left(1 + \frac{1}{s} \right) \left(G' \Omega^{-1} G \right)^{-1}$$

Where G is the matrix of partial derivatives of the model moments $m(\beta_1, \beta_2)$ with respect to (η, θ) . The term Ω denotes the varcov matrix of the moments, and S indicates the number of simulations used for estimation. These components are estimated through bootstrapping.

A.3.2 Sensitivity Analysis

To perform a sensitivity analysis, I observe how the equilibrium and regression coefficients change when the parameters of the model change.

Figure 10: Sensitivity Analysis

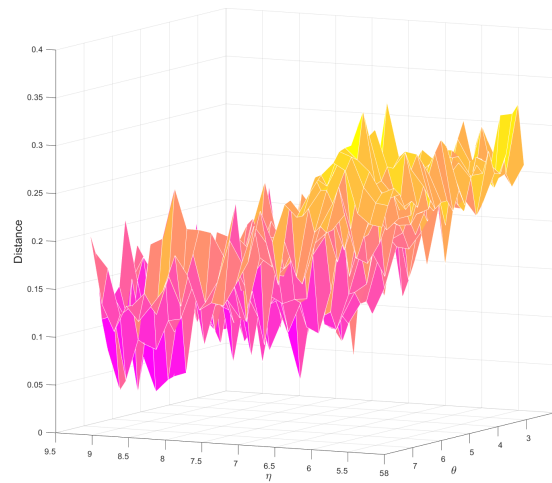


Notes: In these figures, I hold all parameters constant at the estimated values, except one parameter which varies along the horizontal axis. The vertical axis reports the values taken by a given target moment.

A.3.3 Uniqueness

In this section, I challenge the uniqueness of the estimated elasticities, η and θ . The estimators minimize the distance between the simulated data and the model's expressions while matching the ERPT coefficients. Considering that the estimated parameters are approximately $\eta = 6.98$ and $\theta = 4.01$, one can see a clear convexity around that point, and it seems like a feasible minimum of the objective function. Furthermore, given the observable trends, the estimated values are consistent with a global value and, therefore, with a unique set of equilibrium values.

Figure 11: Uniqueness of values



Notes: This figure illustrates the distance between the reduced form estimated values for α and β and the ones in the simulated data for pairs of η and θ .

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